

5G RAN

Multi-Operator Sharing Feature Parameter Description

Issue	Draft B
Date	2026-01-30



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1 Change History

The following describes the specific changes in different issues.

5G RAN10.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
None	None	Low-frequency TDD	None

- Editorial changes
Revised descriptions in this document, which do not involve changes in meanings and therefore are not detailed.

5G RAN10.1 Draft A (2025-12-31)

This issue introduces the following changes to 5G RAN9.1 05 (2025-11-07).

- Technical changes

Change Description	Parameter Change	RAT	Base Station Model
Added a mutually exclusive relationship between percentage-based dynamic RB sharing among operators and network slice resource management. For details, see 4.3.3.2.2 Mutually Exclusive Functions .	None	High-frequency TDD	5900 series base stations (macro base stations)
Added an impact relationship between frequency-based 5GC selection and network slice deployment. For details, see 4.8.3.2.3 Function Impacts .	None	High-frequency TDD	5900 series base stations (macro base stations)
Added a mutually exclusive relationship between operator-specific NCI configuration and X2-U IP address sending control. For details, see 4.6.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD High-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added the "prohibition of multi-operator sharing during Xn-U self-setup" function. For details, see 4.1.1.6 Interface Management and 4.1.4.1 Data Configuration .	Modified parameter: Added the XN_UP_SELF_SETUP_NO_MOCN_SW option to the gNBInterfaceParam.InterfaceSelfConfigOptSw parameter.	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added the "prohibition of multi-operator sharing during X2-U self-setup" function. For details, see 4.1.1.6 Interface Management and 4.1.4.1 Data Configuration .	Modified parameter: Added the X2_UP_SELF_SETUP_NO_MOCN_SW option to the gNBInterfaceParam.InterfaceSelfConfigOptSw parameter.	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added support for a shared 5GC by the gNodeB. For details, see 4.1.1.6 Interface Management and 4.1.4.1.2 Using MML Commands .	None	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between RAN sharing with common carrier and flexible experience differentiation. For details, see 4.1.3.2.3 Function Impacts .	None	Low-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite

Change Description	Parameter Change	RAT	Base Station Model
Added support for X2 interface configurations for multi-operator sharing in SA networking. For details, see 4.1.1.6 Interface Management .	None	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added a mutually exclusive relationship between SA X2 setup and operator-specific NCI configuration. For details, see 4.6.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD High-frequency TDD	3900 and 5900 series base stations (macro base stations) - DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between mmWave PUSCH MU-MIMO and percentage-based dynamic RB sharing among operators. For details, see 4.3.3.2.3 Function Impacts .	None	High-frequency TDD	5900 series base stations (macro base stations)
Added an impact relationship between inter-frequency MLB for mmWave PCells and multi-operator sharing. For details, see 4.1.3.2.3 Function Impacts and 5.1.3.2.3 Function Impacts .	None	High-frequency TDD	5900 series base stations (macro base stations)

Change Description	Parameter Change	RAT	Base Station Model
Added a mutually exclusive relationship between FSD transmission mode and operator-specific guarantee on the minimum number of available RBs. For details, see 4.3.3.2.2 Mutually Exclusive Functions .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite
Added an impact relationship between enhanced L4S congestion control based on rate adjustment and RAN sharing with common carrier. For details, see 4.1.3.2.3 Function Impacts .	None	Low-frequency TDD	• 3900 and 5900 series base stations (macro base stations) • DBS3900 LampSite and DBS5900 LampSite

- Editorial changes
Outlined the scope of operator-specific dedicated cell reselection priority management described in this document. For details, see [4.5.1 Principles](#).
Added descriptions of operators each using their own self-setup X2-U links in RAN sharing with common carrier. For details, see [4.1.1.6 Interface Management](#).
Revised other descriptions in this document, which do not involve changes in meanings and therefore are not detailed. Examples of such changes include changes in the names of reference documents.

2 About This Document

2.1 General Statements

Purpose

Feature Parameter Description documents are intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in Feature Parameter Description documents apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Features in This Document

This document describes the following features.

Feature ID	Feature Name	Chapter/Section
FOFD-021211	Multi-Operator Sharing	4.1 Basic Networking for RAN Sharing with Common Carrier 4.2 Operator-Specific License Management 5.2 Operator-Specific License Management 4.3 Operator-Specific RB Management 4.4 Operator-Specific Roaming Fairness 4.5 Operator-Specific Dedicated Cell Reselection Priority Management 4.6 Operator-Specific NCI Configuration 5.3 Operator-Specific NCI Configuration 4.7 Operator-Specific Function Configuration 4.8 Frequency-based 5GC Selection 5.4 Frequency-based 5GC Selection 4.9 Operator-Specific Paging Resource Management 5.5 Operator-Specific Paging Resource Management 5.1 Basic Networking for RAN Sharing with Dedicated Carrier

2.3 Differences

Differences Between NSA and SA

Function Name	Difference	Chapter/Section
Basic networking for RAN sharing with common carrier	None	4.1 Basic Networking for RAN Sharing with Common Carrier

Function Name	Difference	Chapter/Section
Operator-specific license management	None	4.2 Operator-Specific License Management 5.2 Operator-Specific License Management
Operator-specific RB management	None	4.3 Operator-Specific RB Management
Operator-specific roaming fairness	Supported only in SA networking	4.4 Operator-Specific Roaming Fairness
Operator-specific dedicated cell reselection priority management	Supported only in SA networking	4.5 Operator-Specific Dedicated Cell Reselection Priority Management
Operator-specific NCI configuration	None	4.6 Operator-Specific NCI Configuration 5.3 Operator-Specific NCI Configuration
Operator-specific function configuration	Supported only in SA networking	4.7 Operator-Specific Function Configuration
Frequency-based 5GC selection	Supported only in SA networking	4.8 Frequency-based 5GC Selection 5.4 Frequency-based 5GC Selection
Operator-specific paging resource management	None	4.9 Operator-Specific Paging Resource Management 5.5 Operator-Specific Paging Resource Management
Basic networking for RAN sharing with dedicated carrier	None	5.1 Basic Networking for RAN Sharing with Dedicated Carrier

Differences Between High Frequency Bands and Low Frequency Bands

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

Function Name	Difference	Chapter/Section
Basic networking for RAN sharing with common carrier	Supported in both low and high frequency bands, with the following differences: Both SA networking and NSA networking are supported in low frequency bands, but only SA networking in FWA scenarios and NSA networking are supported in high frequency bands.	4.1 Basic Networking for RAN Sharing with Common Carrier
Operator-specific license management	Supported in both low and high frequency bands, with the following differences: Both SA networking and NSA networking are supported in low frequency bands, but only SA networking in FWA scenarios and NSA networking are supported in high frequency bands.	4.2 Operator-Specific License Management 5.2 Operator-Specific License Management
Operator-specific RB management	Supported in both low and high frequency bands, with the following differences: All functions are supported in low frequency bands. In contrast, only the percentage-based dynamic RB sharing among operators function is supported in high frequency bands.	4.3 Operator-Specific RB Management
Operator-specific roaming fairness	None	4.4 Operator-Specific Roaming Fairness
Operator-specific dedicated cell reselection priority management	None	4.5 Operator-Specific Dedicated Cell Reselection Priority Management

Function Name	Difference	Chapter/Section
Operator-specific NCI configuration	Supported in both low and high frequency bands, with the following differences: Both SA networking and NSA networking are supported in low frequency bands, but only SA networking in FWA scenarios and NSA networking are supported in high frequency bands.	4.6 Operator-Specific NCI Configuration 5.3 Operator-Specific NCI Configuration
Operator-specific function configuration	None	4.7 Operator-Specific Function Configuration
Frequency-based 5GC selection	None	4.8 Frequency-based 5GC Selection 5.4 Frequency-based 5GC Selection
Operator-specific paging resource management	None	4.9 Operator-Specific Paging Resource Management 5.5 Operator-Specific Paging Resource Management
Basic networking for RAN sharing with dedicated carrier	Supported in both low and high frequency bands, with the following differences: Both SA networking and NSA networking are supported in low frequency bands, but only SA networking in FWA scenarios and NSA networking are supported in high frequency bands.	5.1 Basic Networking for RAN Sharing with Dedicated Carrier

3 Introduction

Multi-operator sharing is introduced to address challenges that arise from independent network construction. These challenges include high spectrum license fees and network deployment costs, provisioning of relatively wide network coverage within a short time, and deployment difficulties. Multi-operator sharing allows for sharing or cooperation on telecom infrastructure, network equipment, and spectrum resources to any extent among operators (or with third parties).

- The telecom infrastructure includes site locations, equipment room facilities, towers, and power supplies.
- The network equipment includes base stations, transmission equipment, and core network equipment.

This document focuses on the sharing of network equipment and spectrum resources. Sharing modes are decoupled from RATs. For example, when the gNodeB works in multi-operator sharing, the eNodeB can operate in independent operator mode or multi-operator sharing mode. This document describes multi-operator sharing on the gNodeB side. For details about multi-operator sharing on the eNodeB side, see *RAN Sharing of eRAN Feature Documentation*.

This document focuses on the sharing of network equipment and spectrum resources. Multi-operator sharing is categorized into RAN sharing with common carrier and RAN sharing with dedicated carrier, as described in [Table 3-1](#).

Table 3-1 Multi-operator sharing modes

Multi-Operator Sharing Mode	Mode Description	Chapter/Section
RAN sharing with common carrier	Multiple operators share network equipment and spectrum resources, and each NR cell is shared among operators. Currently, an NR cell can be shared among up to 12 operators.	4 RAN Sharing with Common Carrier

Multi-Operat or Sharin g Mode	Mode Description	Chapter/Section
RAN sharing with dedicat ed carrier	Multiple operators share network equipment but do not share spectrum resources. That is, services from multiple operators run on the same network equipment whereas each NR cell belongs only to one operator. Currently, a piece of network equipment can be shared among up to 12 operators.	5 RAN Sharing with Dedicated Carrier

NOTE

According to section 6.2.2 "Message definitions" in 3GPP TS 38.331 V17.6.0, a maximum of 12 PLMN IDs can be broadcast in a single cell. Huawei gNodeBs support network sharing among up to 12 operators. Note that there are special requirements if the value of an operator ID is greater than 5 (that is, the value of **gNBOperator.OperatorId** is greater than 5) in network sharing scenarios. For details, see [4.1.1.2 Network Sharing Among More Operators](#) and [5.1.1.2 Network Sharing Among More Operators](#).

In network sharing scenarios, if a base station can only be deployed with the public key infrastructure (PKI) server of one operator (the primary operator), IPsec tunnels of secondary operators must be authenticated using the certificate issued by the PKI server of the primary operator. This means that the IPsec tunnel reliability of secondary operators depends on the PKI server of the primary operator. To address this issue, the Base Station Supporting Multi-operator PKI feature is introduced. For details, see *Base Station Supporting Multi-operator PKI*.

4 RAN Sharing with Common Carrier

4.1 Basic Networking for RAN Sharing with Common Carrier

4.1.1 Principles

4.1.1.1 Overview

RAN sharing with common carrier can be used when:

- One operator shares its licensed radio spectrum with other operators.
- Multiple operators share their licensed radio spectrum to serve as a resource pool, which can be used by themselves as well as operators that have no radio spectrum.

Figure 4-1 shows spectrum assignment for operators in RAN sharing with common carrier, which supports the following two implementations:

- Implementation 1 (typical scenario 1): All cells under a gNodeB are shared by the primary and secondary operators.
- Implementation 2 (typical scenarios 2 and 3): One or more cells under a gNodeB are exclusive to a secondary operator.

Table 4-1 provides further details.

Figure 4-1 Spectrum assignment in RAN sharing with common carrier

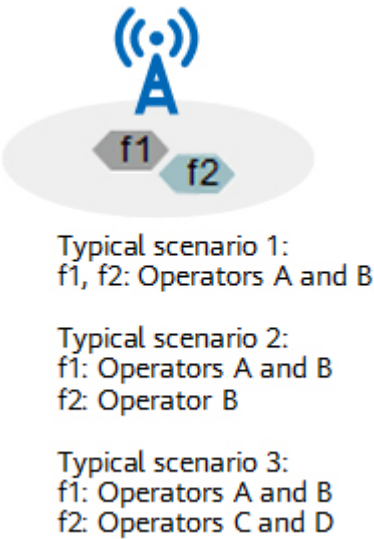


Table 4-1 Frequency use in RAN sharing with common carrier

Scenario	Frequency f1	Frequency f2
Typical scenario 1	Operators A and B share this frequency. A cell working on this frequency broadcasts the PLMN IDs of operators A and B.	Operators A and B share this frequency. A cell working on this frequency broadcasts the PLMN IDs of operators A and B.
Typical scenario 2		Operator B exclusively uses this frequency. A cell working on this frequency broadcasts the PLMN IDs of operators A and B, and the PLMN ID of operator A is automatically configured as reserved .
Typical scenario 3		Operators C and D share this frequency. A cell working on this frequency broadcasts the PLMN IDs of operators A, C, and D, and the PLMN ID of operator A is automatically configured as reserved .
Note: Operator A is the primary operator, and operators B, C, and D are secondary operators.		

The operator information of a cell is collectively configured using the **NRDUCellOp.NrDuCellId** and **NRDUCellOp.OperatorId** parameters in the **NRDUCellOp** MO. If a cell is shared among operators, all those operators' information needs to be configured. However, if a cell is dedicated to an operator, only this operator's information needs to be configured. RAN sharing with

common carrier is enabled by setting the **gNBSharingMode.gNBMultiOpSharingMode** parameter to **SHARED_FREQ**. The **gNBOperator.NrNetworkingOption** parameter specifies the networking option of an operator.

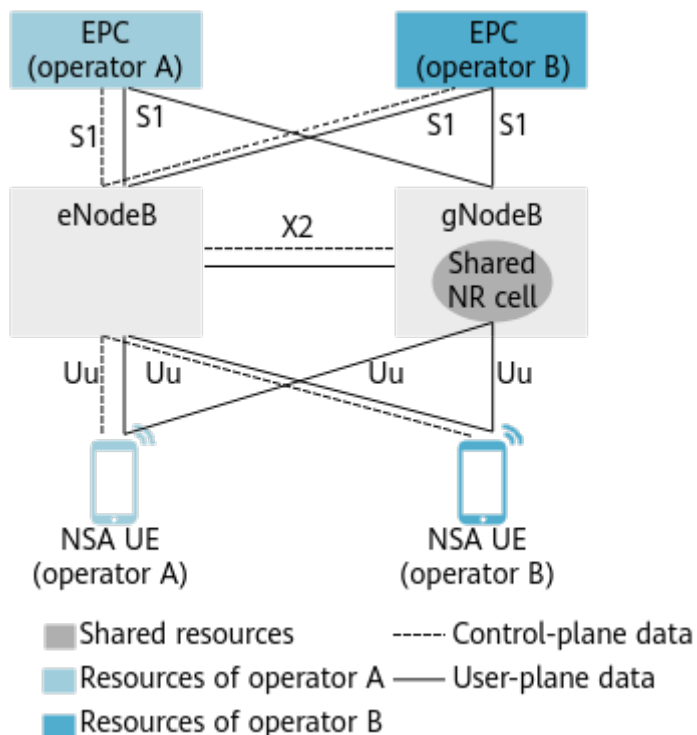
The evolution of networking modes has created a need for multi-operator sharing in the following networking scenarios: NSA networking, NSA and SA hybrid networking, and SA networking. [Table 4-2](#) provides details.

Table 4-2 Networking modes in multi-operator sharing

Networking Mode	Networking of Operators	Network Architecture
NSA networking	All operators use NSA networking and only NSA UEs can access the gNodeB.	Figure 4-2 shows the network architecture.
NSA and SA hybrid networking	Both NSA and SA UEs can access the gNodeB. The following two scenarios are involved: - Scenario 1: At least one operator uses NSA and SA hybrid networking. - Scenario 2: Some operators use SA networking, while others use NSA networking.	- Scenario 1: Figure 4-3 shows the network architecture. - Scenario 2: Figure 4-4 shows the network architecture.
SA networking	All operators use SA networking and only SA UEs can access the gNodeB.	Figure 4-5 shows the network architecture.

- RAN sharing with common carrier in NSA networking
The **gNBOperator.NrNetworkingOption** parameter is set to **NSA** for all operators. In this networking mode, each cell served by the gNodeB belongs to both operators A and B.

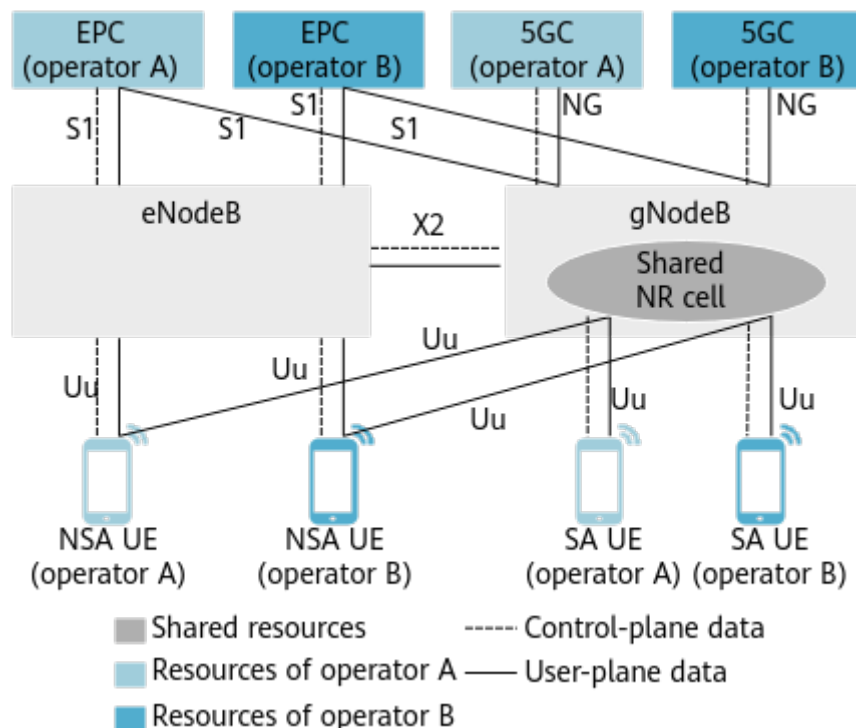
Figure 4-2 RAN sharing with common carrier in NSA networking (using Option 3x as an example)



- RAN sharing with common carrier in NSA and SA hybrid networking

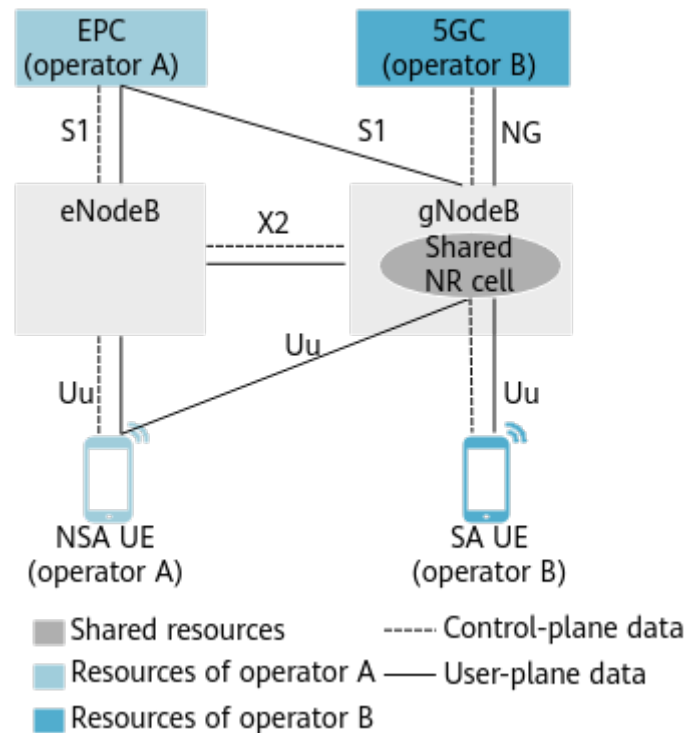
The **gNBOperator.NrNetworkingOption** parameter can be set to **SA_NSA** for all operators. Alternatively, this parameter can be set to **SA_NSA** for some operators and to **SA** or **NSA** for others. The following figure shows an example of the network architecture when the **gNBOperator.NrNetworkingOption** parameter is set to **SA_NSA** for all operators (operators A and B). In this scenario, each cell served by the gNodeB belongs to both operators A and B.

Figure 4-3 Scenario 1: RAN sharing with common carrier in NSA and SA hybrid networking (using Option 3x and Option 2 as an example)



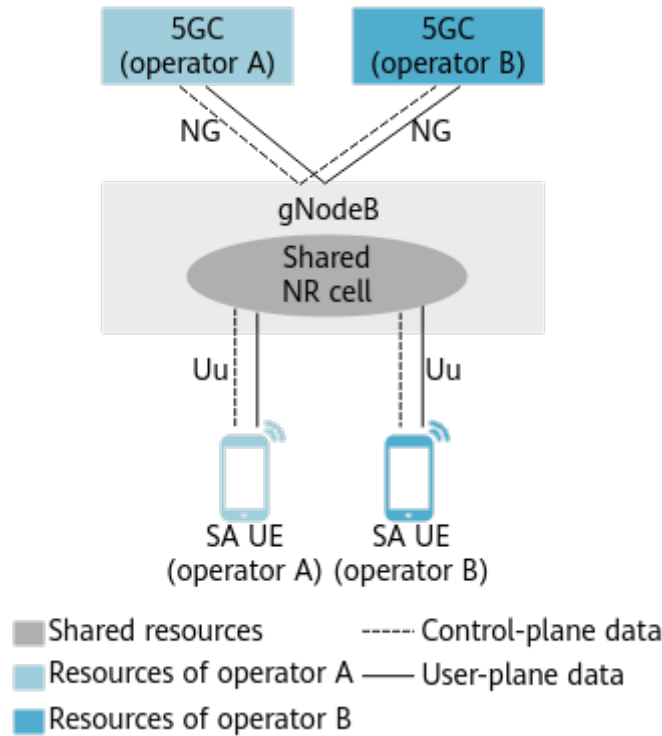
In this networking mode, the **gNBOperator.NrNetworkingOption** parameter can also be set to **SA** for some operators and to **NSA** for others. The following figure shows an example of the network architecture when this parameter is set to **NSA** for operator A and **SA** for operator B. In this scenario, each cell served by the gNodeB belongs to both operators A and B.

Figure 4-4 Scenario 2: RAN sharing with common carrier in NSA and SA hybrid networking (using Option 3x and Option 2 as an example)



- RAN sharing with common carrier in SA networking
The **gNBOperator.NrNetworkingOption** parameter is set to **SA** for all operators. In this networking mode, each cell served by the gNodeB belongs to both operators A and B.

Figure 4-5 RAN sharing with common carrier in SA networking (using Option 2 as an example)



NOTE

Sharing on the gNodeB side does not necessarily require sharing on the eNodeB side, meaning that sharing on the eNodeB side can be enabled or not. In the preceding examples of RAN sharing with common carrier in NSA networking and RAN sharing with common carrier in NSA and SA hybrid networking, sharing is enabled on both the eNodeB and gNodeB. For details about multi-operator sharing on the eNodeB side, see *RAN Sharing of eRAN Feature Documentation*.

RAN sharing with common carrier differs from the independent operator mode as follows:

- Mobility management. For details, see [4.1.1.3 Mobility Management in Idle Mode](#) and [4.1.1.4 Mobility Management in Connected Mode](#).
- Interface management. For details, see [4.1.1.6 Interface Management](#).
- License management. For details, see [4.1.1.7 License Management](#).
- OSS management. For details, see [4.1.1.8 OSS Management](#).

4.1.1.2 Network Sharing Among More Operators

If the value of an operator ID is greater than 5 (that is, the value of **gNBOperator.OperatorId** is greater than 5) in network sharing scenarios, the following special requirements apply:

- Main control boards are required to be UMPTg series boards or later, and baseband processing units are required to be boards other than UBBPfw series.
- The OSS is required to be upgraded to a version later than V100R024C10 and the NE to V100R020C10 or later. Otherwise, there is a possibility that the

configured operator information is incomplete during neighboring cell configuration.

- All of the involved base stations are required to be of V100R020C10 or later and use UMPTg series boards or later. This is because a base station of an earlier version or a board earlier than UMPTg series can only store and parse information about a maximum of six operators. Failure to comply with this requirement would entail incomplete configuration of operator information as well as the following impacts:
 - UE mobility and access are affected, which may increase the service drop rate.
 - The functions that require the configuration of neighbor relationships among multiple base stations fail to take effect for the neighboring cells of a cell with incomplete operator information. This is because the external-cell PLMN list (configured by the **NRExternalNCellPlmn** MO) is incomplete due to missing information of certain operators. Such functions include inter-base-station carrier aggregation (CA), inter-base-station super uplink, and inter-base-station coordinated multipoint transmission/reception (CoMP).

4.1.1.3 Mobility Management in Idle Mode

4.1.1.3.1 NSA Mobility Management in Idle Mode

In NSA networking, for an operator with the **gNBOperator.NrNetworkingOption** parameter set to **NSA**, the *PLMN-IdentityInfo* IE contained in the system information does not contain tracking area code (TAC) information. For details, see section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331 V15.5.0. For the operator, TAC planning is not needed, the **gNBTrackingArea.Tac** parameter must be set to the invalid TAC, and the **NRDUCell.TrackingAreald** parameter must be set to the tracking area ID corresponding to the invalid TAC for the cell of the operator.

4.1.1.3.2 SA Mobility Management in Idle Mode

Cell Search and Roaming Fairness

According to section 4.4 "PLMN selection process" in 3GPP TS 23.122 V15.6.0, a roaming UE randomly selects a PLMN ID given the same conditions, such as the same signal quality. With roaming, a UE still has access to mobile communication services in a non-ownership place after leaving the ownership place. As a cell broadcasts a list of PLMN IDs in a fixed sequence, a third-party UE is more likely to select the top PLMN ID in the list during roaming to a common-carrier sharing cell, particularly when it is at an airport or harbor. To achieve roaming fairness in RAN sharing with common carrier, the duration for which a PLMN ID is placed at the top of the list can be operator-specifically configured. For details, see [4.4 Operator-Specific Roaming Fairness](#).

Cell Reselection

After a UE transitions from RRC_CONNECTED mode to RRC_IDLE mode or selects a PLMN ID, the UE needs to select a suitable cell to camp on, after which it listens

to system information. The UE then measures the signal quality of the serving and neighboring cells to check for a cell with better signal quality to camp on.

During cell reselection, the UE takes into account the cell reselection priorities of neighboring frequencies. In the case of intra-frequency cell reselection, intra-frequency cells have the same frequency priority; therefore, the UE ignores this frequency priority.

The priorities of NR frequencies and E-UTRAN frequencies are classified into the following types:

- Common priority. A cell broadcasts the common priority to all UEs in the cell. The common priority is not operator-specific.
- Dedicated priority. The dedicated priority is generated by a dedicated priority algorithm during an RRC connection release procedure. The dedicated priority is signaled to a specific UE in the *CellReselectionPriorities* IE in the RRCRelease message. However, the RRCRelease message does not contain the *CellReselectionPriorities* IE in the following scenarios:

- The UE has not sent the Msg5 message.

The Msg5 message carries the serving PLMN ID of a UE. Before the UE sends the Msg5 message to the gNodeB, the gNodeB does not know the serving PLMN ID of the UE and cannot determine the frequencies to be delivered and frequency priority configurations based on the serving PLMN ID of the UE. In this case, the RRCRelease message does not carry *CellReselectionPriorities*.

- The UE is in the RRC_INACTIVE state.

In this scenario, the source gNodeB processes the RRCRelease message when the UE moves to another gNodeB. The source gNodeB cannot obtain the frequency and frequency priority configurations of the target gNodeB. In this case, the RRCRelease message does not carry *CellReselectionPriorities*.

In RAN sharing with common carrier, dedicated cell reselection priorities of frequencies can be configured for operators based on the PLMN IDs of the serving cell, enabling camping policies for RRC_IDLE UEs to be operator-specific. For details, see [4.5 Operator-Specific Dedicated Cell Reselection Priority Management](#).

System Information Broadcast

In SA networking, for an operator with the **gNBOperator.NrNetworkingOption** parameter set to **SA_NSA** or **SA**, the *PLMN-IdentityInfo* IE contained in the system information contains TAC information, as shown in [Figure 4-6](#). Additionally, UEs signal the selected PLMN IDs to the gNodeB in the *RRCSetupComplete* IE. Based on the PLMN IDs selected by UEs, the gNodeB determines the core networks corresponding to the PLMN IDs. For details, see the definition of "RRCSetupComplete" in section 6.2.2 "Message definitions" of 3GPP TS 38.331 V15.5.0. For such an operator, TAC planning is required. The **gNBTrackingArea.Tac** parameter must be set to a valid TAC for the operator, and the **NRDUCell.TrackingAreaId** parameter must be set to the tracking area ID corresponding to the valid TAC for the cell of the operator. In this version, TAC configuration can be operator-specific. For details, see [4.7 Operator-Specific Function Configuration](#).

NOTE

If a UE includes an incorrectly selected PLMN ID from those of a common-carrier sharing cell in the *RRCSetupComplete* IE, the UE will be rejected by the core network corresponding to the PLMN ID. In this case, it is recommended that reject cause values (for example, **PLMN not allowed**) be configured on the core network to avoid the signaling impact on common-carrier sharing cells caused by repeated UE attempts.

Figure 4-6 TAC contained in the *PLMN-IdentityInfo* IE

-- ASN1START	
-- TAG-PLMN-IDENTITYINFOLIST-START	
PLMN-IdentityInfoList ::=	SEQUENCE (SIZE (1..maxPLMN)) OF PLMN-IdentityInfo
PLMN-IdentityInfo ::=	SEQUENCE {
plmn-IdentityList	SEQUENCE (SIZE (1..maxPLMN)) OF PLMN-Identity,
trackingAreaCode	TrackingAreaCode
ranac	RAN-AreaCode
cellIdentity	CellIdentity,
cellReservedForOperatorUse	ENUMERATED {reserved, notReserved},
...	
}	
-- TAG-PLMN-IDENTITYINFOLIST-STOP	
-- ASN1STOP	

Cell state information is signaled to UEs in SIB1. Upon an NG link fault or setup failure, the **reserved** (related to operators' PLMN IDs) and **barred** (related to cells) states can be adaptively changed in SIB1.

- If all NG links of the primary operator are faulty or not set up, the gNodeB automatically sets the **reserved** state for the PLMN ID of the primary operator.
- If all NG links of a secondary operator are faulty or not set up, the gNodeB does not include the PLMN ID of the secondary operator in the **PLMN-Identity** field.
- When the **gNBOperator.NrNetworkingOption** parameter is set to **SA** for the primary and secondary operators, and all NG links of these operators fail to be established or are faulty, the gNodeB deactivates the cell. When the **gNBOperator.NrNetworkingOption** parameter is set to **SA_NSA** or **NSA** for at least one of the primary and secondary operators, and all NG links of these operators fail to be established or are faulty, the gNodeB signals the barred state of the cell in a broadcast message.

For details about SA mobility management in idle mode, see *SA Mobility Management in Idle Mode*.

4.1.1.4 Mobility Management in Connected Mode

Table 4-3 provides the triggering scenarios and thresholds for coverage-based inter-frequency handover.

In terms of the protection of UEs under weak uplink coverage function in RAN sharing with common carrier, threshold 2 for event A5 related to uplink-coverage-based inter-frequency handover is equal to the sum of **NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld2** and **gNBOperatorQciParam.InterFreqA5RsrpThld2Ofs**. For details about the protection of UEs under weak uplink coverage function, see *Multi-Frequency Smart Selection*.

Table 4-3 Triggering scenarios of coverage-based inter-frequency handover

Triggering Scenario	Threshold for Coverage-based Measurement	Calculated Value
Event A5 for triggering handover execution	Thresh for event A1	Sum of NRCellInterFHoMeaGrp.CovInterFreqA1RsrpThld and gNBOperatorQciParam.InterFreqA1RsrpThldOfs
	Thresh for event A2	Sum of NRCellInterFHoMeaGrp.CovInterFreqA2RsrpThld and gNBOperatorQciParam.InterFreqA2RsrpThldOfs
	Thresh2 for event A5	Sum of NRCellInterFHoMeaGrp.CovInterFreqA5RsrpThld2 and gNBOperatorQciParam.InterFreqA5RsrpThld2Ofs
Event A3 for triggering handover execution	Thresh for event A1	Sum of NRCellInterFHoMeaGrp.A3InterFreqHoA1RsrpThld and gNBOperatorQciParam.InterFreqA1RsrpThldOfs
	Thresh for event A2	Sum of NRCellInterFHoMeaGrp.A3InterFreqHoA2RsrpThld and gNBOperatorQciParam.InterFreqA2RsrpThldOfs

4.1.1.4.1 NSA Mobility Management in Connected Mode

In NSA networking, NSA DC UEs connect to both an eNodeB and a gNodeB to use the radio resources of both base stations for transmission. The eNodeB serves as the master eNodeB (MeNB) and connects to the EPC, and the gNodeB serves as the secondary gNodeB (SgNB) and connects to the eNodeB over an X2 interface. The UEs connect to both the primary cell (PCell) served by the MeNB and the primary SCG cell (PSCell) served by the SgNB. For details, see *EPC-based NSA Basics*, *EPC-based NSA Performance Enhancement*, and *EPC-based NSA Experience Improvement*.

In regard to NSA mobility management in connected mode, RAN sharing with common carrier mainly differs from the independent operator mode as follows (refer to section 9.1.4 "Messages for E-UTRAN-NR Dual Connectivity Procedures" of 3GPP TS 36.423 V15.4.0 for details).

Configuration of Shared Neighboring Cells

On a gNodeB with a common-carrier sharing cell configured as a neighboring cell, the information about the shared neighboring cell must be configured for the eNodeB to add the PSCell or for the gNodeB to change the PSCell.

Intra-RAT cases

- gNodeB: The following configuration is required for the gNodeB to change the PSCell.
 - Use the **NRExternalNCellPlmn** MO to configure the NR external neighboring cell PLMN list.
 - **NRExternalNCellPlmn.Mcc** and **NRExternalNCellPlmn.Mnc** are set to the MCC and MNC of the primary operator that an NR external neighboring cell belongs to, respectively.
 - **NRExternalNCellPlmn.SharedMcc** and **NRExternalNCellPlmn.SharedMnc** are set to the MCC and MNC of a secondary operator, respectively.
 - Use the **NRCellRelation** MO to configure a new neighbor relationship.
- eNodeB: The following configuration is required for the eNodeB to add the PSCell.
 - Use the **NrExternalCellPlmn** MO to configure the NR external-cell PLMN list.
 - **NrExternalCellPlmn.Mcc** and **NrExternalCellPlmn.Mnc** are set to the MCC and MNC of the primary operator that an external NR cell for LTE belongs to, respectively.
 - **NrExternalCellPlmn.SharedMcc** and **NrExternalCellPlmn.SharedMnc** are set to the MCC and MNC of a secondary operator, respectively.
 - Use the **NrNRelationship** MO to configure a new neighbor relationship.

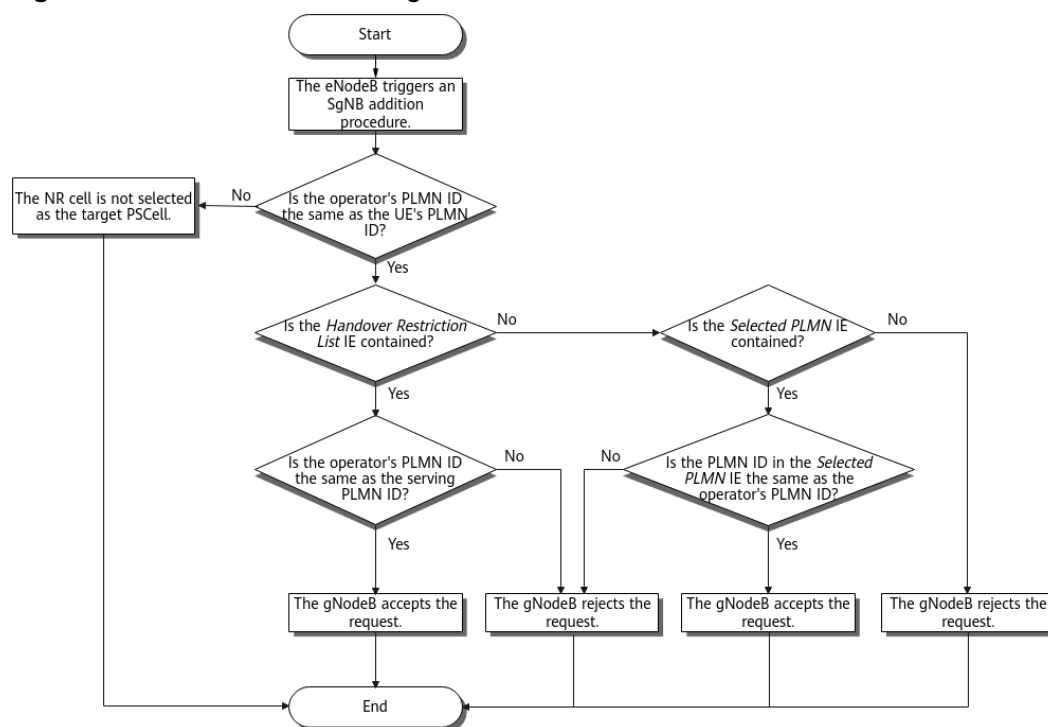
Inter-RAT cases

Management of mobility from NR cells to E-UTRAN cells is not required, and there is no need to perform related configuration.

Adding the PSCell

[Figure 4-7](#) shows the procedure for adding the PSCell.

Figure 4-7 Procedure for adding the PSCell



In an eNodeB-triggered SgNB addition procedure, the eNodeB checks configurations of a neighboring NR cell to determine whether this cell's primary-operator PLMN ID or secondary-operator PLMN ID is the same as the PLMN ID or equivalent public land mobile network (EPLMN) ID of the UE.

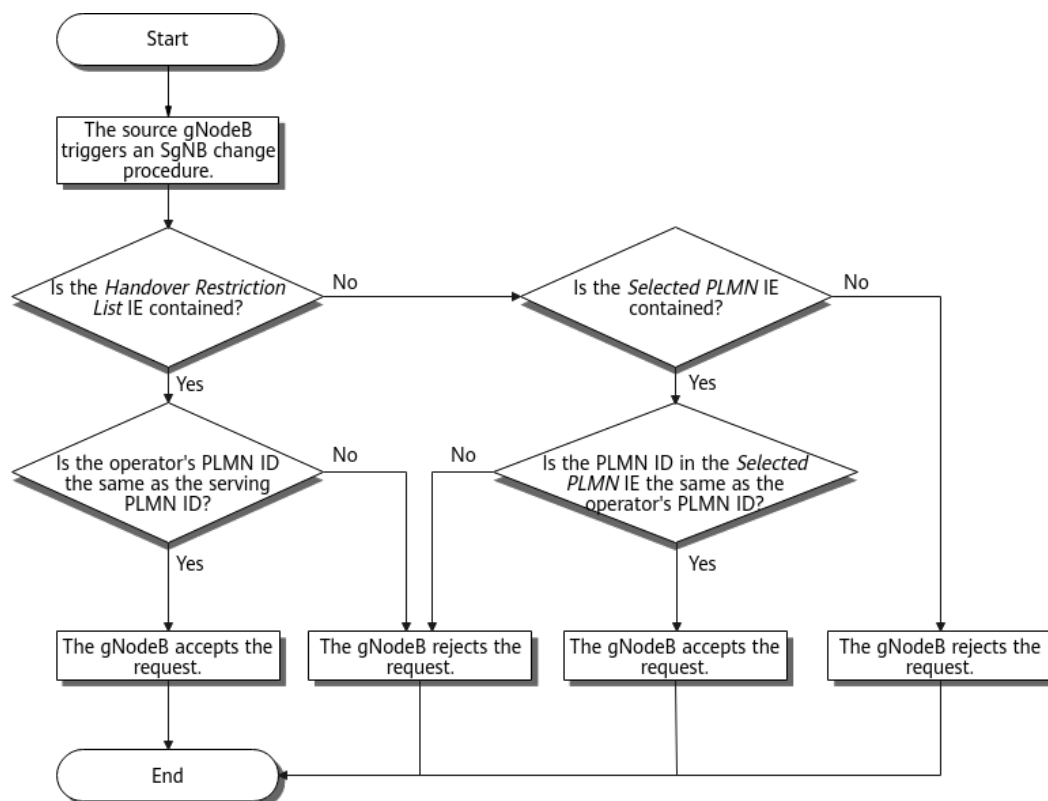
- If they are the same, the eNodeB selects the NR cell as the target PSCell and sends a signaling message to the gNodeB.
The gNodeB checks whether the *Handover Restriction List* IE is contained in the SGNB ADDITION REQUEST message sent by the eNodeB.
 - If it is contained, the serving PLMN ID indicated in the *Handover Restriction List* IE is compared with the primary-operator PLMN ID or secondary-operator PLMN ID of the target NR cell.
 - If the value is the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the gNodeB accepts the request.
 - If the value is not the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the gNodeB rejects the request.
 - If it is not contained, the gNodeB checks whether the *Selected PLMN* IE is contained in the SGNB ADDITION REQUEST message sent by the eNodeB.
 - If it is contained, the PLMN ID indicated in the *Selected PLMN* IE is compared with the primary-operator PLMN ID or secondary-operator PLMN ID of the target NR cell.
 - If the value is the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the gNodeB accepts the request.
 - If the value is not the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the gNodeB rejects the request.

- If it is not contained, the gNodeB rejects the request.
- If they are different, the eNodeB does not select the NR cell as the target PSCell.

Changing the PSCell

Figure 4-8 shows the procedure for changing the PSCell.

Figure 4-8 Procedure for changing the PSCell



In an SgNB change procedure triggered by the source gNodeB, the source gNodeB selects an NR cell as the target PSCell based on the neighbor relationship, and sends a signaling message to the target gNodeB through the eNodeB. The target gNodeB checks whether the *Handover Restriction List* IE is contained in the SGNB ADDITION REQUEST message sent by the eNodeB.

- If it is contained, the serving PLMN ID indicated in the *Handover Restriction List* IE is compared with the primary-operator PLMN ID or secondary-operator PLMN ID of the target NR cell.
 - If the value is the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the target gNodeB accepts the request.
 - If the value is not the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the target gNodeB rejects the request.
- If it is not contained, the target gNodeB checks whether the *Selected PLMN* IE is contained in the SGNB ADDITION REQUEST message sent by the eNodeB.

- If it is contained, the PLMN ID indicated in the *Selected PLMN* IE is compared with the primary-operator PLMN ID or secondary-operator PLMN ID of the target NR cell.
 - If the value is the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the target gNodeB accepts the request.
 - If the value is not the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the target gNodeB rejects the request.
- If it is not contained, the target gNodeB rejects the request.

4.1.1.4.2 SA Mobility Management in Connected Mode

SA mobility management in connected mode ensures intra-RAT handovers (including intra-frequency handovers and inter-frequency handovers) and inter-RAT handovers. In this process, appropriate cells are selected for UEs. For details about the handover procedure, see *SA Mobility Management in Connected Mode*.

In regard to SA mobility management in connected mode, RAN sharing with common carrier mainly differs from the independent operator mode as follows.

Configuration of Shared Neighboring Cells

On a gNodeB with a common-carrier sharing cell configured as a neighboring cell, the information about the shared neighboring cell must be configured to help determine the target cell for a handover.

Intra-RAT cases

- gNodeB: The following configuration is required for the source gNodeB to select the target NR cell during an intra-RAT handover.
 - Use the **NRExternalNCellPlmn** MO to configure the NR external neighboring cell PLMN list.
 - Use the **NRCeCellRelation** MO to configure a new neighbor relationship.
- eNodeB: No configuration is required.

Inter-RAT cases

- gNodeB: The following configuration is required for the source gNodeB to select the target E-UTRAN cell during an inter-RAT handover.
 - Use the **gNBExtCellPlmn** MO to configure the E-UTRAN external-cell PLMN list.
 - Use the **NRCeCellRelation** MO to configure a new neighbor relationship.
- eNodeB: The following configuration is required for the source eNodeB to select the target NR cell during an inter-RAT handover.
 - Use the **NrExtCellPlmn** MO to configure the NR external-cell PLMN list.
 - Use the **NrNRelationship** MO to configure a new neighbor relationship.

Outgoing Handover

Based on the UE's serving PLMN ID, the source gNodeB selects a target handover cell from the neighboring cells that have the PLMN IDs allowed for handovers for

a UE. If inter-PLMN handover is enabled, the UE's serving PLMN ID can be changed after a handover. If inter-PLMN handover is disabled, the UE's serving PLMN ID cannot be changed after a handover. For details about inter-PLMN handover, see *SA Mobility Management in Connected Mode*.

When all the following conditions are met, the gNodeB instructs the UE to perform an Xn-based handover. Otherwise, the gNodeB instructs the UE to perform an NG-based handover.

- An available Xn link is present. That is, the PLMN ID contained in the *Global NG-RAN Node ID* IE of the XN SETUP RESPONSE message sent from the target gNodeB to the source gNodeB is present in the neighboring cell list (NCL) (configured using the **NRExternalNCell** MO) of the source gNodeB.
- The UE's serving PLMN ID is present in the *TAI Support List* IE of the XN SETUP RESPONSE message sent from the target gNodeB to the source gNodeB.
- The UE's serving PLMN ID in the target handover cell selected by the source gNodeB is same as that in the source gNodeB.

If the source gNodeB works in RAN sharing with common carrier:

- Intra-RAT handover: If the target gNodeB works in RAN sharing with dedicated carrier and the target cell belongs to a secondary operator, the system information broadcast by the cell of the secondary operator contains the PLMN ID (**Reserved** state) of the primary operator. To prevent service drops caused by incorrect handovers of UEs served by the primary operator from the source gNodeB to the target gNodeB, set the **NRExternalNCell.PlmnReservedFlag** parameter to **TRUE** when adding a neighboring cell.
- Inter-RAT handover: If the target eNodeB works in RAN sharing with special dedicated carrier and the target cell belongs to a secondary operator, the system information broadcast by the cell of the secondary operator contains the PLMN ID (**Reserved** state) of the primary operator. To prevent service drops caused by incorrect handovers of UEs served by the primary operator from the source gNodeB to the target eNodeB, select the **MASTER_PLMN_RESERVED_FLAG** option of the **gNB EutraExternalCell.AggregationAttribute** parameter when adding a neighboring cell. For details, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

In regard to inter-RAT mobility from NG-RAN to E-UTRAN, the event A2 threshold offset can be set using the **gNB OperatorQciParam.EutranA2B2RsrpThldOffset** parameter. This setting facilitates the configuration of the operator-specific event A2 threshold. For details about the event A2 threshold, see *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*.

When the **HO_TARGET_CELL_ID_PRIMARYPLMN** option of the **NRExternalGNB.CompatibilityConfig** parameter is selected:

- In the case of NG-based handovers, the PLMN ID of the primary operator to which the neighboring cell belongs is specified in the **PLMN Identity** field (**CHOICE Source NG-RAN Node to Target NG-RAN Node Transparent Container > Target Cell ID > NR CGI**) in a HANDOVER REQUIRED message.

- In the case of Xn-based handovers, the PLMN ID of the primary operator to which the neighboring cell belongs is specified in the **PLMN Identity** field (**Target Cell Global ID > NR CGI**) in a HANDOVER REQUEST message.

When the **HO_TARGET_CELL_ID_PRIMARYPLMN** option of the **NREXTERNALGNB.CompatibilityConfig** parameter is deselected, the UE's serving PLMN ID is specified in the **PLMN Identity** field.

Incoming Handover

In an incoming handover, the target gNodeB must correctly identify the serving PLMN ID of a UE in the target cell. Otherwise, there is a possibility that an NG- or Xn-based handover fails.

In an NG-based handover:

- If the handover request message sent by the access and mobility management function (AMF) to the target gNodeB contains the *Mobility Restriction List* IE, the target gNodeB uses the serving PLMN ID indicated in the *Mobility Restriction List* IE as the serving PLMN ID of a UE in the target cell.
- If the handover request message sent by the AMF to the target gNodeB does not contain the *Mobility Restriction List* IE, the target gNodeB uses the PLMN ID indicated in the *GUAMI* IE of the handover request as the serving PLMN ID of a UE in the target cell.

In an Xn-based handover:

- If the handover request message sent by the source gNodeB to the target gNodeB contains the *Mobility Restriction List* IE, the target gNodeB uses the serving PLMN ID indicated in the *Mobility Restriction List* IE as the serving PLMN ID of a UE in the target cell.
- If the handover request message sent by the source gNodeB to the target gNodeB does not contain the *Mobility Restriction List* IE, the target gNodeB uses the PLMN ID indicated in the *GUAMI* IE of the handover request as the serving PLMN ID of a UE in the target cell.

4.1.1.5 Admission Control

In RAN sharing with common carrier, UE-number-based preemption, which is incorporated in admission control, allows a preempting UE to preempt resources of other UEs, regardless of whether the preempting UE and other UEs are served by the same operator or different operators. The preemption range is specified by the **NRDUCellRac.UeNumPreemptionRange** parameter. For details, see *Admission Control*.

4.1.1.6 Interface Management

Interface management for multi-operator sharing varies with networking options:

- In NSA networking, interface management for multi-operator sharing includes S1 interface management and X2 interface management.
- In NSA and SA hybrid networking, interface management for multi-operator sharing includes S1 interface management, X2 interface management, NG interface management, and Xn interface management.

- In SA networking, interface management for multi-operator sharing includes X2 interface management, NG interface management, and Xn interface management.

Interface configurations are described as follows.

S1 Interface

In the Option 3x architecture, the S1-C interface is not present between the gNodeB and the EPC, and only the S1-U interface needs to be established for user-plane data transmission between the gNodeB and the EPC. The S1-U link of each operator to the EPC requires an S1 object to be configured using the **gNBCUS1** MO, and the operator information to be configured for the S1 object using the **gNBCUS1Op** MO.

In this version, independent EPC and EPC sharing are supported. [Table 4-4](#) describes the configuration requirements.

Table 4-4 Configuration requirements

Scenario	User-Plane and Control-Plane IP Address Planning and Configuration	Endpoint Group Configuration	Interface Configuration
Independent EPC	Different host S1 IP addresses are planned, and different user-plane hosts are configured (using the USERPLANEHOST MO) for different operators.	Different endpoint groups are configured (using the EPGROUP MO).	Different operators are associated with different S1 objects using the gNBCUS1Op.gNBCuS1Id and gNBCUS1Op.OperatorId parameters.
EPC sharing	One host S1 IP address is planned, and one user-plane host is configured (using the USERPLANEHOST MO).	Operators share an endpoint group (configured using the EPGROUP MO).	Different operators are associated with the same S1 object using the gNBCUS1Op.gNBCuS1Id and gNBCUS1Op.OperatorId parameters.
Note: Support for EPC sharing depends on the implementation of the EPC.			

NOTE

In the Option 3 architecture, S1 interface configuration is not required on the gNodeB side.

X2 Interface

In gNodeB sharing mode (RAN sharing with common carrier or RAN sharing with dedicated carrier), the requirements for X2 configuration on the eNodeB and gNodeB are consistent, as described in [Table 4-5](#).

Table 4-5 Configuration requirements

eNodeB Sharing Mode	gNodeB NCI Configuration	eNodeB X2 Configuration	gNodeB X2 Configuration
RAN sharing with common carrier or RAN sharing with special dedicated carrier	Non-operator-specific NCI configuration	Operators share a single host X2 IP address.	Operators share a single host X2 IP address.
	Operator-specific NCI configuration	Operators share a single host X2 IP address.	Operators do not share host X2 IP addresses.
RAN sharing with dedicated carrier	Operator-specific NCI configuration/Non-operator-specific NCI configuration	Operators do not share host X2 IP addresses.	Operators do not share host X2 IP addresses.
Independent operator	Non-operator-specific NCI configuration	A single host X2 IP address is configured.	Operators share a single host X2 IP address.
	Operator-specific NCI configuration	A single host X2 IP address is configured.	Operators do not share host X2 IP addresses.

eNodeB Sharing Mode	gNodeB NCI Configuration	eNodeB X2 Configuration	gNodeB X2 Configuration
Note: - On the eNodeB side: - Host X2 IP address sharing among operators: X2 objects are operator-specifically configured with unique X2.CpEpGroupId values and the same X2.UpEpGroupId value. The X2.CpEpGroupId values are associated with the same SCTPHOST MO. - Separate host X2 IP addresses for operators: X2 objects are operator-specifically configured with unique X2.CpEpGroupId and X2.UpEpGroupId values, and the values of X2.CpEpGroupId and X2.UpEpGroupId are associated with different SCTPHOST and USERPLANEHOST MOs, respectively. - On the gNodeB side: - Host X2 IP address sharing among operators: Operators are associated with the same X2 object using the gNBCUX2Op.gNBCuX2Id and gNBCUX2Op.OperatorId parameters. - Separate host X2 IP addresses for operators: Operators are associated with different X2 objects using the gNBCUX2Op.gNBCuX2Id and gNBCUX2Op.OperatorId parameters, with unique values of gNBCUX2.CpEpGroupId and gNBCUX2.UpEpGroupId for these X2 objects. The values of gNBCUX2.CpEpGroupId and gNBCUX2.UpEpGroupId are associated with different SCTPHOST and USERPLANEHOST MOs, respectively. For details about operator-specific NCI configuration, see 4.6 Operator-Specific NCI Configuration. In RAN sharing with common carrier, operator-specific virtual routing and forwarding (VRF) must be planned if operators need to isolate transmission. When the eNodeB works in RAN sharing with dedicated carrier, transport network configurations must be operator-specific. In addition, if the X2 peer is manually configured on the eNodeB and gNodeB, the configured peer X2 addresses must belong to the same operator. For details about the X2 interface, see <i>X2 and S1 Self-Management in NSA Networking</i> and <i>X2 Self-Management in SA Networking</i>.			

NOTE

When operators each use their own self-setup X2-U links in RAN sharing with common carrier:

- X2 user plane optimization in RAN sharing with common carrier needs to be enabled (by selecting the **SHARE_FREQ_X2_UP_OPT_SW** option of the **EnodebAlgoExtSwitch.NetworkPrfmOptSwitch** parameter) on the eNodeB.
- X2-U self-removal optimization needs to be enabled (by selecting the **X2SON_X2U_AUTO_DEL_OPT_SWITCH** option of the **gNBX2SonConfig.X2SonConfigSwitch** parameter) on the gNodeB.
- Prohibition of multi-operator sharing during X2-U self-setup needs to be enabled (by selecting the **X2_UP_SELF_SETUP_NO_MOCN_SW** option of the **gNBInterfaceParam.InterfaceSelfCfgOptSw** parameter) on the gNodeB.
- When the base station is configured with two main control boards and the IPv4/IPv6 dual-stack network is applied, both IPv4 and IPv6 addresses must be configured for the X2 and Xn interfaces planned for dual-stack transmission on a single main control board. It is not allowed that one main control board is configured only with IPv4 addresses and the other main control board is configured only with IPv6 addresses.
- If X2-U self-removal optimization is enabled on the gNodeB and the **gNBSharingMode.MultipleNCISwitch** parameter is set to **OFF**, the following mechanism applies:
 - When removing a redundant faulty X2-C interface, the gNodeB does not correspondingly remove an available X2-U interface.
 - When removing an available X2-C interface or the last faulty X2-C interface, the gNodeB also removes the X2-U interfaces of all operators of the peer eNodeB.
- Penalty upon handover preparation failures needs to be enabled on the gNodeB. When this function is enabled, you are advised to set both of the parameters **gNBMobilityCommParam.NecHoPrepFailPunishTimer** and **gNBMobilityCommParam.OptHoPrepFailPunishTimer** to 3s.

With different design models, the eNodeB and gNodeB differ in X2 link status indication.

- For the eNodeB, the PLMN ID of an X2 link peer is the primary-operator PLMN ID of the gNodeB. The **DSP X2INTERFACE** command can be used to query the link status. In this case, a link is identified by the **CN Operator ID** value configured on the local end.
- For the gNodeB, the PLMN ID of an X2 link peer is the PLMN ID of the operator to which the X2 link belongs. The **DSP GNBCUX2INTERFACE** command can be used to query the link status. In this case, a link is identified by the **Neighboring eNodeB PLMN ID** value configured on the peer end.

NG Interface

For details about NG interface configuration, see *NG and Xn Self-Management*.

For multi-operator sharing in SA networking, the NG link of each operator to the 5GC requires an NG object to be configured using the **gNBCUNg** MO and operator information to be configured for the NG object using the **gNBCUNgOp** MO.

For multi-operator sharing in NSA and SA hybrid networking:

- If an NG link to 5GC is set up for only one operator; that is, the **gNBOperator.NrNetworkingOption** parameter is set to **SA** or **SA_NSA** for only one operator, only an NG object needs to be configured using the **gNBCUNg** MO, and operator information does not need to be configured for the NG object using the **gNBCUNgOp** MO.

- If NG links to 5GC are set up for multiple operators; that is, the **gNBOperator.NrNetworkingOption** parameter is set to **SA** or **SA_NSA** for multiple operators, NG objects need to be configured using the **gNBCUNg** MO, and operator information needs to be configured for the NG objects using the **gNBCUNgOp** MO.

The gNodeB can work with an independent or shared 5GC. [Table 4-6](#) describes the configuration requirements in different 5GC scenarios.

Table 4-6 Configuration requirements

Scenario	User-Plane and Control-Plane IP Address Planning and Configuration	Endpoint Group Configuration	Interface Configuration
Independent 5GC	Either of the following control-plane configurations is used: • Different host NG IP addresses are planned, and different control-plane hosts are configured (using the SCTPHOST MO) for operators. • The same host NG IP address (configured using the SCTPHOST MO) is planned. Either of the following user-plane configurations is used: • Different host NG IP addresses are planned, and different user-plane hosts are configured (using the USERPLANEHOST MO) for operators. • The same host NG IP address (configured using the USERPLANEHOST MO) is planned.	Different control-plane endpoint groups and SCTP peer IDs are configured (using the EPGROUP and SCTPPEER2EPGRP MOs, respectively). Either of the following user-plane configurations is used: • When different host NG IP addresses are planned, different user-plane endpoint groups are configured (using the EPGROUP MO). • When the same host NG IP address is planned, the same user-plane endpoint group is configured (using the EPGROUP MO).	Different operators are associated with different NG objects using the gNBCUNgOp.gNB CuNgId and gNBCUNgOp.OperatorId parameters.
5GC sharing	One host NG IP address is planned, and one control-plane host and one user-plane host are configured (using the SCTPHOST and USERPLANEHOST MOs, respectively).	Operators share an endpoint group (configured using the EPGROUP MO).	Different operators are associated with the same NG object using the gNBCUNgOp.gNB CuNgId and gNBCUNgOp.OperatorId parameters.

Xn Interface

For details about Xn interface configuration, see *NG and Xn Self-Management*.

The gNodeB allows operators to share an Xn interface or use their own Xn interfaces. [Table 4-7](#) lists the configuration requirements in different scenarios.

Table 4-7 Configuration requirements

Scenario	User-Plane and Control-Plane IP Address Planning and Configuration	Endpoint Group Configuration	Interface Configuration
Independent Xn	Different host Xn IP addresses are planned, and different control-plane hosts and user-plane hosts are configured (using the SCTPHOST and USERPLANEHOST MOs, respectively) for different operators.	Different endpoint groups are configured (using the EPGROUP MO).	Different operators are associated with different Xn objects using the gNBCUXnOp.gNBCuXnId and gNBCUXnOp.OperatorId parameters.
Xn sharing	One host Xn IP address is planned, and one control-plane host and one user-plane host are configured (using the SCTPHOST and USERPLANEHOST MOs, respectively).	Operators share an endpoint group (configured using the EPGROUP MO).	Different operators are associated with the same Xn object using the gNBCUXnOp.gNBCuXnId and gNBCUXnOp.OperatorId parameters.

Note: When the operator-specific NCI configuration function is enabled, the Xn interface can only work in independent mode. For details, see [4.6 Operator-Specific NCI Configuration](#).

The "prohibition of multi-operator sharing during Xn-U self-setup" function is designed to enable separated Xn-U links for different operators to achieve service isolation. This function is controlled by the **XN_UP_SELF_SETUP_NO_MOCN_SW** option of the **gNBInterfaceParam.InterfaceSelfCfgOptSw** parameter. After this function is enabled, multiple operators each deploy their own self-setup Xn-U links, and these operators do not share user-plane transmission resources.

 NOTE

It is recommended that only one Xn link be configured for a single operator.

4.1.1.7 License Management

A license is an authorization agreement between the vendor and the customer in terms of the application scope and service lifetime of products.

In RAN sharing with common carrier, a license purchase order is placed by only one party, which can either be the primary operator or a joint venture with multiple operators. Each NE has only one license file, which does not distinguish between operators.

License management includes feature and capacity license management.

4.1.1.7.1 Feature License Management

Feature licenses are shared among operators, and their sales are not operator-specific.

4.1.1.7.2 Capacity License Management

Capacity licenses are shared among operators, and their sales are not operator-specific. Percentages of the licensed limits defined by some capacity licenses, including the RRC Connected User License and RE(Resource Element) License, can be operator-specific. For details, see [4.2 Operator-Specific License Management](#).

4.1.1.8 OSS Management

The OSS mainly provides configuration management and performance management for both shared and non-shared gNodeBs.

Configuration Management

OSS configuration management in RAN sharing with common carrier is the same as that in independent operator mode. Cell-level and base-station-level parameters are configured by the O&M team to the values determined through negotiation among operators.

Cell-level common configurations and base-station-level common configurations are reported over the northbound interface (Itf-N) of the OSS. The reported configurations are the same as those in the independent operator mode.

The scope of operator information query on the OSS depends on operator-specific account permissions.

- An account of a secondary operator has the permission to query only the information about the secondary operator itself.
- An account of the primary operator has the permission to query the information about all operators.

Performance Management

On the gNodeB, statistics collection on some key performance counters is operator-specific. For details, see [4.1.4.3 Network Monitoring](#).

Performance counter reporting over the Itf-N of the OSS is operator-specific, meaning the OSS reports counters specific to each operator.

4.1.2 Network Analysis

4.1.2.1 Benefits

- Capital expenditure (CAPEX) and operating expense (OPEX) of operators reduce.
- Network deployments are accelerated.
- Network coverage of operators expands.

4.1.2.2 Impacts

The impacts between this function and other functions are described in [4.1.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.1.3 Requirements

4.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021211	Multi-Operator Sharing	NR0S00MULS00	per Cell
For the license control item NR0S00MULS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.1.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with common carrier	gNBSharingMode.gNBMultiOpSharingMode set to SHARED_FREQ	<i>Multi-Operator Sharing</i>	Prohibition of multi-operator sharing during Xn-U self-setup requires RAN sharing with common carrier to be deployed on the live network.

4.1.3.2.2 Mutually Exclusive Functions

None

4.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	NSA networking based on EPC	NRCellAlgoSwitch.NsaDcSwitch	<i>EPC-based NSA Basics</i>	When both "dual connectivity (DC) in NSA networking based on EPC" and "basic networking for RAN sharing with common carrier" are enabled, data split is only supported between an E-UTRAN cell and an NR cell that belong to the same operator.
Low-frequency TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	The PLMN list of a cooperating cell for a UE must contain the ID of the PLMN with which the UE has registered.
High-frequency TDD	Inter-frequency MLB for mmWave PCells	MMW_PCELL_INTER_FREQ_MLB_SW option of the NRCellMlb.MlbAlgoEnhSwitch parameter	<i>Mobility Load Balancing in Connected Mode</i>	If a UE only supports certain operators in multi-operator sharing scenarios, the UE may fail to be handed over blindly during load redistribution.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Flexible experience differentiation	FLEX_EXP_DIFF_SW option of the NRDUCellFeatureSw.QciBasedDiffServExpSw parameter	<i>Service-differentiated Experience Guarantee</i>	After "flexible experience differentiation" is enabled in RAN sharing with common carrier scenarios, the base station estimates the uplink or downlink peak data rate based on the available resource range of the UE's operator and performs further data rate control. As a result, the difference between the estimated data rate and the actually measured data rate may increase.
Low-frequency TDD	Enhanced L4S congestion control based on rate adjustment	• UL_L4SPLUS_FOR_FAIR_SW option of the NRDUCellULSchAlgoSwitch parameter • DL_L4SPLUS_FOR_FAIR_SW option of the NRDUCellDISchAlgoEnhSwitch parameter	<i>Adaptive Managed Latency</i>	If enhanced L4S congestion control based on rate adjustment is enabled in RAN sharing with common carrier scenarios, the gNodeB restricts the available RBs for L4S UEs based on the available resource range of the UEs' operator, which may cause the calculated maximum rate and fair rate to be inaccurate.

4.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.1.3.4 Others

None

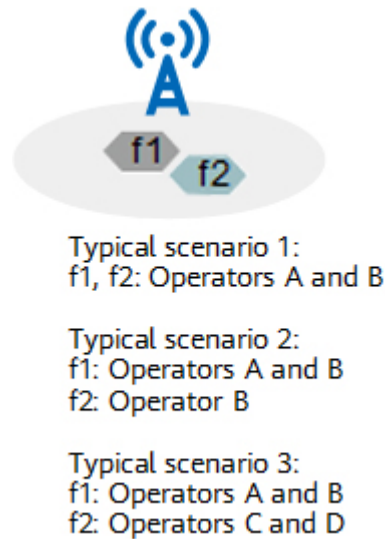
4.1.4 Operation and Maintenance

4.1.4.1 Data Configuration

This section uses typical scenario 1 (as shown in [Figure 4-9](#)) as an example to describe how to configure data.

In typical scenarios 2 and 3, the cells on frequency f1 are configured similarly to those on frequency f1 or f2 in typical scenario 1, and the cells on frequency f2 are configured similarly to those in RAN sharing with dedicated carrier. For detailed configuration, see [5.1.4.1 Data Configuration](#).

Figure 4-9 Spectrum assignment in RAN sharing with common carrier



Interface configuration: [4.1.1.6 Interface Management](#) presents a brief setting note on transmission configuration.

- For details about S1 and X2 interface configuration on the eNodeB side in endpoint mode, see *S1 and X2 Self-Management of eRAN Feature Documentation*. For details about S1 and X2 interface configuration on the eNodeB side in link mode, see *Transmission Resource Management of eRAN Feature Documentation*.
- For details about NG and Xn interface configuration on the gNodeB side, see *NG and Xn Self-Management*.

Neighboring cell configuration: If ANR is enabled, neighboring cells do not need to be manually configured. For details, see *ANR*. This section describes only the neighboring cell configuration on the gNodeB side. For the neighboring cell

configuration on the eNodeB side, see *EPC-based NSA Basics*, *EPC-based NSA Performance Enhancement*, and *EPC-based NSA Experience Improvement*.

In RAN sharing with common carrier, reconfiguration is required in any of the five scenarios described below.

NOTICE

A change to the sharing mode of the gNodeB does not take effect until the application referenced by the gNodeB function is manually restarted by running the **RST APP** command. The ID of the application referenced by the gNodeB function can be queried using the **LST GNODEBFUNCTION** command.

Scenario 1: Changing the Sharing Mode of the gNodeB from the Independent Operator Mode to RAN Sharing with Common Carrier

In this scenario, the sharing mode of the gNodeB needs to be changed and cell-level operator information needs to be added (one primary operator and one secondary operator). The required MO reconfiguration is as follows:

- NSA networking
 - a. Configure the sharing mode of the gNodeB (using the **gNBSharingMode** MO).
 - b. Configure gNodeB-level operator information (using the **gNBOperator** MO).
 - c. Configure cell-level operator information (using the **NRDUCellOp** MO).
 - d. Configure the operator information of an S1 object (using the **gNBCUS1Op** MO).
 - e. Configure the operator information of an X2 object (using the **gNBCUX2Op** MO).
 - f. Configure neighboring cell information.
 - On the gNodeB side
 - 1) (Optional) Based on the network plan, configure the NR external neighboring cells of the primary operator (using the **NRExternalNCell** MO), and the NR external neighboring cells of the secondary operator and the external neighboring cell PLMN list (using the **NRExternalNCell** and **NRExternalNCellPlmn** MOs, respectively) on the gNodeB.
 - 2) (Optional) Based on the network plan, configure the neighbor relationships with NR cells of the primary and secondary operators (using the **NRCeCellRelation** MO) on the gNodeB.
 - On the eNodeB side
 - 1) (Optional) Based on the network plan, configure the external NR cells of the primary operator (using the **NrExternalCell** MO), and the external NR cells of the secondary operator and the external-cell PLMN list (using the **NrExternalCell** and **NrExternalCellPlmn** MOs, respectively) on the eNodeB.

- 2) (Optional) Based on the network plan, configure the neighbor relationships with NR cells of the primary and secondary operators (using the **NrNRRelationship** MO) on the eNodeB.
- NSA and SA hybrid networking
In addition to configuration in NSA networking, the following configuration is required:
 - a. (Optional) Based on the network plan, configure the external E-UTRAN cells of the primary operator (using the **gNBEutraExternalCell** MO), and the external E-UTRAN cells of the secondary operator and the external-cell PLMN list (using the **gNBEutraExternalCell** and **gNBEutraExtCellPlmn** MOs, respectively) on the gNodeB.
 - b. (Optional) Based on the network plan, configure the neighbor relationships with E-UTRAN cells of the primary and secondary operators (using the **NRCeutraNRRelation** MO) on the gNodeB.
 - c. (Optional) If the NR networking option is set to **SA** or **SA_NSA** for an operator, configure the operator information of an NG object (using the **gNBCUNgOp** MO).
 - d. Configure a valid TAC for an operator for which the NR networking option is set to **SA** or **SA_NSA** (using the **gNBTrackingArea** MO).
 - e. Configure a valid TAC for the cells of an operator for which the NR networking option is set to **SA** or **SA_NSA** (using the **NRDUCell** MO).
 - SA networking
 - a. Configure the sharing mode of the gNodeB (using the **gNBSharingMode** MO).
 - b. Configure gNodeB-level operator information (using the **gNBOperator** MO).
 - c. Configure cell-level operator information (using the **NRDUCellOp** MO).
 - d. Configure the operator information of an NG object (using the **gNBCUNgOp** MO).
 - e. Configure the operator information of an Xn object (using the **gNBCUXnOp** MO).
 - f. Configure neighboring cell information.
 - i. (Optional) Based on the network plan, configure the NR external neighboring cells of the primary operator (using the **NRExternalNCell** MO), and the NR external neighboring cells of the secondary operator and the external neighboring cell PLMN list (using the **NRExternalNCell** and **NRExternalNCellPlmn** MOs, respectively) on the gNodeB.
 - ii. (Optional) Based on the network plan, configure the neighbor relationships with NR cells of the primary and secondary operators (using the **NRCeutraNRRelation** MO) on the gNodeB.
 - iii. (Optional) Based on the network plan, configure the external E-UTRAN cells of the primary operator (using the **gNBEutraExternalCell** MO), and the external E-UTRAN cells of the secondary operator and the external-cell PLMN list (using the **gNBEutraExternalCell** and **gNBEutraExtCellPlmn** MOs, respectively) on the gNodeB.

- iv. (Optional) Based on the network plan, configure the neighbor relationships with E-UTRAN cells of the primary and secondary operators (using the **NRCellEutraNRelation** MO) on the gNodeB.
- g. Configure a valid TAC for an operator for which the NR networking option is set to **SA** or **SA_NSA** (using the **gNBTrackingArea** MO).
- h. Configure a valid TAC for the cells of an operator for which the NR networking option is set to **SA** or **SA_NSA** (using the **NRDUCell** MO).

Scenario 2: Changing the Sharing Mode of the gNodeB from RAN Sharing with Common Carrier to the Independent Operator Mode

In this scenario, cell-level operator information needs to be removed and the sharing mode of the gNodeB needs to be changed. The required MO reconfiguration is as follows:

- NSA networking
 - a. Remove neighboring cell information.
 - On the gNodeB side
 - (Optional) Based on the network plan, remove the neighbor relationships with NR cells of the secondary operator (using the **NRCellRelation** MO) from the gNodeB.
 - (Optional) Based on the network plan, remove the NR external neighboring cells of the secondary operator and the external neighboring cell PLMN list (using the **NRExternalNCell** and **NRExternalNCellPlmn** MOs, respectively) from the gNodeB.
 - On the eNodeB side
 - (Optional) Based on the network plan, remove the neighbor relationships with NR cells of the secondary operator (using the **NrNRRelationship** MO) from the eNodeB.
 - (Optional) Based on the network plan, remove the external NR cells of the secondary operator and the external-cell PLMN list (using the **NrExternalCell** and **NrExternalCellPlmn** MOs, respectively) from the eNodeB.
 - b. Deactivate cells (using the **NRCell** MO).
 - c. Remove the operator information of an S1 object (using the **gNBCUS1Op** MO).
 - d. Remove the operator information of an X2 object (using the **gNBCUX2Op** MO).
 - e. Remove cell-level operator information (using the **NRDUCellOp** MO).
 - f. Remove gNodeB-level operator information (using the **gNBOperator** MO).
 - g. Change the sharing mode of the gNodeB to the independent operator mode (using the **gNBSharingMode** MO).
 - h. Activate cells (using the **NRCell** MO).
- NSA and SA hybrid networking

In addition to configuration in NSA networking, the following configuration prior to deactivating cells is required:

- a. (Optional) Based on the network plan, remove the neighbor relationships with E-UTRAN cells of the secondary operator (using the **NRCeutraNRelation** MO) from the gNodeB.
- b. (Optional) Based on the network plan, remove the external E-UTRAN cells of the secondary operator and the external-cell PLMN list (using the **gNBEutraExternalCell** and **gNBEutraExtCellPlmn** MOs, respectively) from the gNodeB.
- c. (Optional) If the NR networking option is set to **SA** or **SA_NSA** for the secondary operator to be removed, remove the operator information of an NG object (using the **gNBCUNgOp** MO).
- SA networking
 - a. Remove neighboring cell information.
 - i. (Optional) Based on the network plan, remove the neighbor relationships with NR cells of the secondary operator (using the **NRCeutralCell** MO) from the gNodeB.
 - ii. (Optional) Based on the network plan, remove the NR external neighboring cells of the secondary operator and the external neighboring cell PLMN list (using the **NRExternalNCell** and **NRExternalNCellPlmn** MOs, respectively) from the gNodeB.
 - iii. (Optional) Based on the network plan, remove the neighbor relationships with E-UTRAN cells of the secondary operator (using the **NRCeutraNRelation** MO) from the gNodeB.
 - iv. (Optional) Based on the network plan, remove the external E-UTRAN cells of the secondary operator and the external-cell PLMN list (using the **gNBEutraExternalCell** and **gNBEutraExtCellPlmn** MOs, respectively) from the gNodeB.
 - b. Deactivate cells (using the **NRCeutralCell** MO).
 - c. Remove the operator information of an NG object (using the **gNBCUNgOp** MO).
 - d. Remove the operator information of an Xn object (using the **gNBCUXnOp** MO).
 - e. Remove cell-level operator information (using the **NRDUCellOp** MO).
 - f. Remove gNodeB-level operator information (using the **gNBOperator** MO).
 - g. Change the sharing mode of the gNodeB to the independent operator mode (using the **gNBSharingMode** MO).
 - h. Activate cells (using the **NRCeutralCell** MO).

Scenario 3: Changing the Sharing Mode of the gNodeB from RAN Sharing with Common Carrier to RAN Sharing with Dedicated Carrier

In this scenario, cell-level operator information and the sharing mode of the gNodeB need to be changed. The required MO reconfiguration is as follows:

- NSA networking
 - a. Deactivate cells (using the **NRCeutralCell** MO).
 - b. Remove cell-level operator information (using the **NRDUCellOp** MO).

- c. (Optional) If no cells are available for a secondary operator, add a cell for the secondary operator and configure operator information for the cell (using the **NRDUCell** and **NRDUCellOp** MOs, respectively).
- d. Change the sharing mode of the gNodeB to RAN sharing with dedicated carrier (using the **gNBSharingMode** MO).
- e. Activate cells (using the **NRCell** MO).
- NSA and SA hybrid networking
The configuration is the same as that in NSA networking.
- SA networking
The configuration is the same as that in NSA networking.

Scenario 4: Adding a Secondary Operator

In this scenario, the configuration of a secondary operator needs to be added. The required MO reconfiguration is as follows:

- NSA networking
 - a. Configure gNodeB-level operator information (using the **gNBOperator** MO).
 - b. Configure cell-level operator information (using the **NRDUCellOp** MO).
 - c. Configure the operator information of an S1 object (using the **gNBCUS1Op** MO).
 - d. Configure the operator information of an X2 object (using the **gNBCUX2Op** MO).
 - e. Configure neighboring cell information.
 - On the gNodeB side
 - 1) (Optional) Based on the network plan, configure the NR external neighboring cells of the secondary operator and the external neighboring cell PLMN list (using the **NRExternalNCell** and **NRExternalNCellPlmn** MOs, respectively) on the gNodeB.
 - 2) (Optional) Based on the network plan, configure the neighbor relationships with NR cells of the secondary operator (using the **NRCellRelation** MO) on the gNodeB.
 - On the eNodeB side
 - 1) (Optional) Based on the network plan, configure the external NR cells of the secondary operator and the external-cell PLMN list (using the **NrExternalCell** and **NrExternalCellPlmn** MOs, respectively) on the eNodeB.
 - 2) (Optional) Based on the network plan, configure the neighbor relationships with NR cells of the secondary operator (using the **NrNRRelationship** MO) on the eNodeB.
- NSA and SA hybrid networking
In addition to configuration in NSA networking, the following configuration is required:
 - a. (Optional) Based on the network plan, configure the external E-UTRAN cells of the secondary operator and the external-cell PLMN list (using the

- gNBEutraExternalCell** and **gNBEutraExtCellPlmn** MOs, respectively) on the gNodeB.
- b. (Optional) Based on the network plan, configure the neighbor relationships with E-UTRAN cells of the secondary operator (using the **NRCeutraNRelation** MO) on the gNodeB.
 - c. (Optional) If the NR networking option is set to **SA** or **SA_NSA** for the newly added secondary operator, configure the operator information of an NG object (using the **gNBCUNgOp** MO).
- SA networking
 - a. Configure gNodeB-level operator information (using the **gNBOperator** MO).
 - b. Configure cell-level operator information (using the **NRDUCellOp** MO).
 - c. Configure the operator information of an NG object (using the **gNBCUNgOp** MO).
 - d. Configure the operator information of an Xn object (using the **gNBCUXnOp** MO).
 - e. Configure neighboring cell information.
 - i. (Optional) Based on the network plan, configure the NR external neighboring cells of the secondary operator and the external neighboring cell PLMN list (using the **NRExternalNCell** and **NRExternalNCellPlmn** MOs, respectively) on the gNodeB.
 - ii. (Optional) Based on the network plan, configure the neighbor relationships with NR cells of the secondary operator (using the **NRCeutraNRelation** MO) on the gNodeB.
 - iii. (Optional) Based on the network plan, configure the external E-UTRAN cells of the secondary operator and the external-cell PLMN list (using the **gNBEutraExternalCell** and **gNBEutraExtCellPlmn** MOs, respectively) on the gNodeB.
 - iv. (Optional) Based on the network plan, configure the neighbor relationships with E-UTRAN cells of the secondary operator (using the **NRCeutraNRelation** MO) on the gNodeB.

Scenario 5: Removing a Secondary Operator

In this scenario, the configuration of a secondary operator needs to be deleted. The required MO reconfiguration is as follows:

- NSA networking
 - a. Remove neighboring cell information.
 - On the gNodeB side
 - (Optional) Based on the network plan, remove the neighbor relationships with NR cells of the secondary operator (using the **NRCeutraNRelation** MO) from the gNodeB.
 - (Optional) Based on the network plan, remove the NR external neighboring cells of the secondary operator and the external neighboring cell PLMN list (using the **NRExternalNCell** and **NRExternalNCellPlmn** MOs, respectively) from the gNodeB.

- On the eNodeB side
 - 1) (Optional) Based on the network plan, remove the neighbor relationships with NR cells of the secondary operator (using the **NrNRRelationship** MO) from the eNodeB.
 - 2) (Optional) Based on the network plan, remove the external NR cells of the secondary operator and the external-cell PLMN list (using the **NrExternalCell** and **NrExternalCellPlmn** MOs, respectively) from the eNodeB.
- b. Remove the operator information of an S1 object (using the **gNBCUS1Op** MO).
- c. Remove the operator information of an X2 object (using the **gNBCUX2Op** MO).
- d. Remove cell-level operator information (using the **NRDUCellOp** MO).
- e. Remove gNodeB-level operator information (using the **gNBOperator** MO).
- NSA and SA hybrid networking

In addition to configuration in NSA networking, the following configuration prior to removing the operator information of an S1 object is required:

 - a. (Optional) Based on the network plan, remove the neighbor relationships with E-UTRAN cells of the secondary operator (using the **NRCellEutraNRelation** MO) from the gNodeB.
 - b. (Optional) Based on the network plan, remove the external E-UTRAN cells of the secondary operator and the external-cell PLMN list (using the **gNBEutraExternalCell** and **gNBEutraExtCellPlmn** MOs, respectively) from the gNodeB.
 - c. (Optional) If the NR networking option is set to **SA** or **SA_NSA** for the secondary operator to be removed, remove the operator information of an NG object (using the **gNBCUNgOp** MO).
- SA networking
 - a. Remove neighboring cell information.
 - i. (Optional) Based on the network plan, remove the neighbor relationships with NR cells of the secondary operator (using the **NRCellRelation** MO) from the gNodeB.
 - ii. (Optional) Based on the network plan, remove the NR external neighboring cells of the secondary operator and the external neighboring cell PLMN list (using the **NRExternalNCell** and **NRExternalNCellPlmn** MOs, respectively) from the gNodeB.
 - iii. (Optional) Based on the network plan, remove the neighbor relationships with E-UTRAN cells of the secondary operator (using the **NRCellEutraNRelation** MO) from the gNodeB.
 - iv. (Optional) Based on the network plan, remove the external E-UTRAN cells of the secondary operator and the external-cell PLMN list (using the **gNBEutraExternalCell** and **gNBEutraExtCellPlmn** MOs, respectively) from the gNodeB.
 - b. Remove the operator information of an NG object (using the **gNBCUNgOp** MO).

- c. Remove the operator information of an Xn object (using the **gNBCUXnOp** MO).
- d. Remove cell-level operator information (using the **NRDUCellOp** MO).
- e. Remove gNodeB-level operator information (using the **gNBOperator** MO).

4.1.4.1.1 Data Preparation

The parameters for initial configuration related to different networking options are described as follows:

- NSA networking
 - **Table 4-8**, **Table 4-9**, and **Table 4-10** describe the parameters used for function activation.
 - The parameters used for neighboring cell configuration related to this function are as follows:
 - gNodeB: **Table 4-11** to **Table 4-13**
 - eNodeB: **Table 4-18** to **Table 4-20**
 - No parameters are required for function optimization.
- NSA and SA hybrid networking
 - **Table 4-8**, **Table 4-9**, and **Table 4-10** describe the parameters used for function activation.
 - The parameters used for neighboring cell configuration related to this function are as follows:
 - gNodeB: **Table 4-11** to **Table 4-16**
 - eNodeB: **Table 4-18** to **Table 4-20**
 - No parameters are required for function optimization.
- SA networking
 - **Table 4-8**, **Table 4-9**, and **Table 4-10** describe the parameters used for function activation.
 - **Table 4-11** to **Table 4-16** describe the parameters used for neighboring cell configuration related to this function.
 - No parameters are required for function optimization.

Configuring Parameters on the gNodeB

Table 4-8 Parameters for configuring the sharing mode of the gNodeB and operator information

Parameter Name	Parameter ID	Setting Notes
gNodeB Multi Operator Sharing Mode	gNBSharingMode.gNB MultiOpSharingMode	In RAN sharing with common carrier, set this parameter to SHARED_FREQ .

Parameter Name	Parameter ID	Setting Notes
Operator ID	gNBOperator.OperatorID	Set this parameter based on the network plan.
Mobile Country Code	gNBOperator.Mcc	Set this parameter to the operator's MCC.
Mobile Network Code	gNBOperator.Mnc	Set this parameter to the operator's MNC.
Operator Name	gNBOperator.OperatorName	Set this parameter based on the network plan.
Operator Type	gNBOperator.OperatorType	Set this parameter based on the network plan.
NR Networking Option	gNBOperator.NrNetworkingOption	• In NSA networking, set this parameter to NSA for all operators. • In NSA and SA hybrid networking, the following settings are supported: – Set this parameter to SA_NSA for all operators. – Set this parameter to SA_NSA for some operators and to SA or NSA for other operators. – Set this parameter to SA for some operators and to NSA for other operators. • In SA networking, set this parameter to SA for all operators.
NR DU Cell ID ^a	NRDUCellOp.NrDuCellID	Set this parameter based on the network plan.
Operator ID	NRDUCellOp.OperatorID	Set this parameter based on the network plan.
a: If the NRDUCell.DuplexMode parameter is set to CELL_SUL , the NRDUCellOp MO does not need to be configured.		

Table 4-9 Parameters for configuring a tracking area

Parameter Name	Parameter ID	Setting Notes
Tracking Area Code ^a	gNBTrackingArea.Tac	Set this parameter to the invalid TAC (4294967295) for an operator for which the gNBOperator.NrNetworkingOption parameter is set to NSA . Set this parameter to a valid TAC (ranging from 0 to 16777215) for an operator for which the gNBOperator.NrNetworkingOption parameter is set to SA or SA_NSA .
Tracking Area ID	gNBTrackingArea.TrackingAreaId	Set this parameter based on the network plan.
NR DU Cell ID	NRDUCell.NrDuCellId	Set this parameter based on the network plan.
Tracking Area ID	NRDUCell.TrackingAreaId	Set this parameter to the tracking area ID that corresponds to the invalid TAC for the cell of an operator for which the gNBOperator.NrNetworkingOption parameter is set to NSA . Set this parameter to the tracking area ID that corresponds to a valid TAC for the cell of an operator for which the gNBOperator.NrNetworkingOption parameter is set to SA or SA_NSA .
a: For multi-operator sharing in SA networking, different TACs can be configured for different SA operators in a shared cell. For details, see 4.7.4 Operation and Maintenance .		

Table 4-10 Parameters for configuring interfaces

Parameter Name	Parameter ID	Option	Setting Notes
gNodeB CU S1 ID	gNBCUS1Op.gNBCuS1Id	None	Configure these parameters only when NSA Option 3x is used on live networks.
Operator ID	gNBCUS1Op.OperatorId	None	
gNodeB CU NG ID	gNBCUNgOp.gNBCuNgId	None	These parameters need to be configured when the gNBOperator.NrNetworkingOption parameter is set to SA or SA_NSA for an operator.
Operator ID	gNBCUNgOp.OperatorId	None	
gNodeB CU Xn ID	gNBCUXnOp.gNBCuXnId	None	Set these parameters based on the network plan. It is recommended that one operator be configured with only one Xn object.
Operator ID	gNBCUXnOp.OperatorId	None	
gNodeB CU X2 ID	gNBCUX2Op.gNBCuX2Id	None	Set these parameters based on the network plan.
Operator ID	gNBCUX2Op.OperatorId	None	
Interface Self-Config Optimization Switch	gNBInterfaceParam.InterfaceSelfConfigOptSw	X2_UP_SELF_SETUP_NO_MOCN_SW	Set this parameter based on the network plan.
Interface Self-Config Optimization Switch	gNBInterfaceParam.InterfaceSelfConfigOptSw	XN_UP_SELF_SETUP_NO_MOCN_SW	Set this parameter based on the network plan.
Note: The preceding table lists only the interface configuration parameters that must be additionally prepared for multi-operator sharing. For details about other related interface configuration parameters, see <i>X2 and S1 Self-Management in NSA Networking</i> , <i>NG and Xn Self-Management</i> , and <i>S1 and X2 Self-Management of eRAN Feature Documentation</i> .			

Table 4-11 Parameters for configuring an NR external neighboring cell of the gNodeB

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	NRExternalNCell.Mcc	- If the gNodeB serving an NR external neighboring cell works in multi-operator sharing, set these parameters to the MCC and MNC of the primary operator to which the external NR cell belongs, separately. - If the gNodeB serving an NR external neighboring cell does not work in multi-operator sharing, set these parameters to the MCC and MNC of the operator to which the external NR cell belongs, separately.
Mobile Network Code	NRExternalNCell.Mnc	
gNodeB ID	NRExternalNCell.gNBId	Set this parameter based on the network plan. gNodeB IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NRExternalNCell.CellId	Set this parameter based on the network plan.
Physical Cell ID	NRExternalNCell.PhysicalCellId	Set this parameter based on the network plan.
SSB Frequency Position	NRExternalNCell.SsbFrequencyPos	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Tracking Area Code	NRExternalNCell.Tac	Set this parameter to the invalid TAC (4294967295) for an external neighboring cell for which the NRExternalNCell.NrNetworkingOption parameter is set to NSA . Set this parameter to a valid TAC (ranging from 0 to 16777215) for an external neighboring cell for which the NRExternalNCell.NrNetworkingOption parameter is set to SA or SA_NSA .
NR Networking Option	NRExternalNCell.NrNetworkingOption	Set this parameter based on the network plan.
PLMN Reserved Flag	NRExternalNCell.PlmnReservedFlag	Set this parameter to TRUE if the gNodeB serving an NR external neighboring cell works in RAN sharing with dedicated carrier, and the external neighboring cell belongs to a secondary operator.

Table 4-12 Parameters for configuring the NR external neighboring cell PLMN list of the gNodeB

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NRExternalNCellPlmn.gNBId	Set this parameter based on the network plan. gNodeB IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NRExternalNCellPlmn.CellId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	NRExternalNCellPlmn.<i>Mcc</i>	Set these parameters to the MCC and MNC of the primary operator to which the NR external neighboring cell belongs, separately.
Mobile Network Code	NRExternalNCellPlmn.<i>Mnc</i>	
Shared Mobile Country Code	NRExternalNCellPlmn.<i>SharedMcc</i>	Set these parameters to the MCC and MNC of the secondary operator to which the NR external neighboring cell belongs, separately.
Shared Mobile Network Code	NRExternalNCellPlmn.<i>SharedMnc</i>	
Tracking Area Code	NRExternalNCellPlmn.<i>Tac</i>	Set this parameter to the secondary-operator TAC of an NR external neighboring cell. Set this parameter to the invalid TAC (4294967295) for an external neighboring cell for which the NRExternalNCellPlmn.<i>NrNetworkingOption</i> parameter is set to NSA. Set this parameter to a valid TAC (ranging from 0 to 16777215) for an external neighboring cell for which the NRExternalNCellPlmn.<i>NrNetworkingOption</i> parameter is set to SA or SA_NSA.
NR Networking Option	NRExternalNCellPlmn.<i>NrNetworkingOption</i>	Set this parameter to the secondary-operator NR networking option of an NR external neighboring cell.
Note: The preceding parameters need to be configured only when the gNodeB serving an NR external neighboring cell works in multi-operator sharing.		

Table 4-13 Parameters for configuring a relationship between NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellRelation.NrCellId	Set this parameter based on the network plan.
Mobile Country Code	NRCellRelation.Mcc	Set these parameters to the MCC and MNC of the primary operator to which the neighboring NR cell belongs, separately.
Mobile Network Code	NRCellRelation.Mnc	
gNodeB ID	NRCellRelation.gNBId	Set this parameter based on the network plan. gNodeB IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NRCellRelation.CellId	Set this parameter based on the network plan.

Table 4-14 Parameters for configuring an external E-UTRAN cell of the gNodeB

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	gNBEutraExternal-Cell.Mcc	- If the eNodeB serving an external E-UTRAN cell works in RAN sharing with common carrier, RAN sharing with special dedicated carrier, or hybrid RAN sharing, set these parameters to the MCC and MNC of the primary operator to which the external E-UTRAN cell belongs, separately. - If the eNodeB serving an external E-UTRAN cell works in RAN sharing with dedicated carrier or does not work in RAN sharing, set these parameters to the MCC and MNC of the operator to which the external E-UTRAN cell belongs, separately.
Mobile Network Code	gNBEutraExternal-Cell.Mnc	
eNodeB ID	gNBEutraExternal-Cell.EnodebId	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	gNBEutraExternal-Cell.CelId	Set this parameter based on the network plan.
Downlink EARFCN	gNBEutraExternal-Cell.DlEarfcn	Set this parameter based on the network plan.
Physical Cell ID	gNBEutraExternal-Cell.PhysicalCellId	Set this parameter based on the network plan.
Tracking Area Code	gNBEutraExternal-Cell.Tac	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	gNBEutraExternal-Cell.AggregationAttribute	Select the MASTER_PLMN_RESERVED_FLAG option of the gNBEutraExternal-Cell.AggregationAttribute parameter if both of the following conditions are met: (1) The eNodeB serving an external E-UTRAN cell works in RAN sharing with special dedicated carrier. (2) The external cell belongs to a secondary operator.

Table 4-15 Parameters for configuring the E-UTRAN external-cell PLMN list of the gNodeB

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	gNBEutraExtCellPlmn.Mcc	Set these parameters to the MCC and MNC of the primary operator to which the external E-UTRAN cell belongs, separately.
Mobile Network Code	gNBEutraExtCellPlmn.Mnc	
eNodeB ID	gNBEutraExtCellPlmn.EnodebId	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	gNBEutraExtCellPlmn.CellId	Set this parameter based on the network plan.
Shared Mobile Country Code	gNBEutraExtCellPlmn.SharedMcc	Set these parameters to the MCC and MNC of the secondary operator to which the external E-UTRAN cell belongs, separately.
Shared Mobile Network Code	gNBEutraExtCellPlmn.SharedMnc	
Note: The preceding parameters need to be configured only when the eNodeB serving an external E-UTRAN cell works in RAN sharing with common carrier or RAN sharing with special dedicated carrier.		

Table 4-16 Parameters for configuring the relationship of an NR cell with an E-UTRAN cell

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellEutraNRelation.NrCellId	Set this parameter based on the network plan.
Mobile Country Code	NRCellEutraNRelation.Mcc	- If the eNodeB serving a neighboring E-UTRAN cell works in RAN sharing with common carrier, RAN sharing with special dedicated carrier, or hybrid RAN sharing, set these parameters to the MCC and MNC of the primary operator to which the neighboring E-UTRAN cell belongs, separately. - If the eNodeB serving a neighboring E-UTRAN cell works in RAN sharing with dedicated carrier or does not work in RAN sharing, set these parameters to the MCC and MNC of the operator to which the neighboring E-UTRAN cell belongs, separately.
Mobile Network Code	NRCellEutraNRelation.Mnc	
eNodeB ID	NRCellEutraNRelation.EnodebId	Set this parameter based on the network plan.
Cell ID	NRCellEutraNRelation.CellId	Set this parameter based on the network plan.

Table 4-17 Operator-specific QCI parameters of the gNodeB

Parameter Name	Parameter ID	Setting Notes
Inter-freq A1 RSRP Threshold Offset	gNBOperatorQciParam.InterFreqA1RsrpThresholdOfs	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Inter-freq A2 RSRP Threshold Offset	gNBOperatorQciParam.InterFreqA2RsrpThresholdOfs	Set this parameter based on the network plan.
Inter-freq A5 RSRP Threshold2 Offset	gNBOperatorQciParam.InterFreqA5RsrpThreshold2Ofs	Set this parameter based on the network plan.

Configuring Parameters on the eNodeB

Table 4-18 Parameters for configuring an external NR cell of the eNodeB

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NrExternalCell.GnodebId	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NrExternalCell.CellId	Set this parameter based on the network plan.
Mobile Country Code	NrExternalCell.Mcc	- If the gNodeB serving an external NR cell works in multi-operator sharing, set these parameters to the MCC and MNC of the primary operator to which the external NR cell belongs, separately. - If the gNodeB serving an external NR cell does not work in multi-operator sharing, set these parameters to the MCC and MNC of the operator to which the external NR cell belongs, separately.
Mobile Network Code	NrExternalCell.Mnc	
Physical Cell ID	NrExternalCell.PhyCellId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Tracking Area Code	NrExternalCell.Tac	Set this parameter based on the network plan.
Downlink ARFCN	NrExternalCell.DlArfcn	Set this parameter based on the network plan.
NR Networking Option	NrExternalCell.NrNetworkingOption	Set this parameter based on the network plan.

Table 4-19 Parameters for configuring the NR external-cell PLMN list of the eNodeB

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NrExternalCellPlmn.GnodebId	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NrExternalCellPlmn.CellId	Set this parameter based on the network plan.
Mobile Country Code	NrExternalCellPlmn.Mcc	Set these parameters to the MCC and MNC of the primary operator to which the external NR cell belongs, separately.
Mobile Network Code	NrExternalCellPlmn.Mnc	
Shared Mobile Country Code	NrExternalCellPlmn.SharedMcc	Set these parameters to the MCC and MNC of the secondary operator to which the external NR cell belongs, separately.
Shared Mobile Network Code	NrExternalCellPlmn.SharedMnc	
Tracking Area Code	NrExternalCellPlmn.Tac	This parameter indicates the ID of base station serving the external NR cell for a RAN sharing operator. The ID uniquely identifies the base station for the operator.
Note: The preceding parameters need to be configured only when the gNodeB serving an external NR cell works in multi-operator sharing.		

Table 4-20 Parameters for configuring a relationship of an E-UTRAN cell with an NR cell

Parameter Name	Parameter ID	Setting Notes
Local cell ID	NrNRRelationship.LocalCellId	Set this parameter based on the network plan.
Mobile Country Code	NrNRRelationship.Mcc	Set these parameters to the MCC and MNC of the primary operator to which the neighboring NR cell belongs, separately.
Mobile Network Code	NrNRRelationship.Mnc	
gNodeB ID	NrNRRelationship.GnodebId	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NrExternalCellPlmn.CellId	Set this parameter based on the network plan.

4.1.4.1.2 Using MML Commands

Before using MML commands, refer to [4.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this feature, you may need to retrieve basic operator information—for example, by running the **LST GNBOPERATOR** command. For details about basic MML commands used for multi-operator sharing, see **Man-Machine Interface Reference > 3900 & 5900 Series Base Station MML Command Reference > MML Command Reference > gNodeBFunction MML Commands > gNodeBFunction Management > gNBCU Management > Sharing Management** in *3900 & 5900 Series Base Station Product Documentation*.

In NSA and SA hybrid networking, the NG and Xn interfaces must be configured for the gNodeB serving SA UEs. For details about NG and Xn interface configurations, see *NG and Xn Self-Management*. Other configurations are as follows:

Example 1: Changing the sharing mode of the gNodeB from the independent operator mode to RAN sharing with common carrier for new deployments in NSA and SA hybrid networking

```
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Configuring the sharing mode, gNodeB-level operator information, and cell-level operator
information
//Configuring the sharing mode of the gNodeB
MOD GNBSHARINGMODE: gNBMultiOpSharingMode=SHARED_FREQ;
//Manually restarting the application referenced by the gNodeB function
RST APP: AID=1;
//Configuring the primary-operator information of the gNodeB
ADD GNBOPERATOR: OperatorId=0, OperatorName="unicom", OperatorType=PRIMARY_OPERATOR,
Mcc="460", Mnc="01", NrNetworkingOption=SA_NSA;
//Configuring the secondary-operator information of the gNodeB
ADD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,
Mcc="460", Mnc="03", NrNetworkingOption=SA_NSA;
//Configuring the TAC (not required in NSA networking)
//Configuring a valid TAC for the operator with the NR networking option set to SA or SA_NSA
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;
//Configuring a valid TAC for the cells of the operator with the NR networking option set to SA or SA_NSA
ADD NRDUCELL: NrDuCellId=1, NrDuCellName="1", DuplexMode=CELL_TDD, CellId=1, PhysicalCellId=1,
FrequencyBand=N77, Dlnarfcn=620000, UlBandwidth=CELL_BW_40M, DlBandwidth=CELL_BW_40M,
SlotAssignment=4_1_DDDSU, SlotStructure=SS2, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7712, LogicalRootSequenceIndex=0;
ADD NRDUCELL: NrDuCellId=2, NrDuCellName="2", DuplexMode=CELL_TDD, CellId=2, PhysicalCellId=1,
FrequencyBand=N77, Dlnarfcn=620000, UlBandwidth=CELL_BW_40M, DlBandwidth=CELL_BW_40M,
SlotAssignment=4_1_DDDSU, SlotStructure=SS2, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7712, LogicalRootSequenceIndex=0;
//Configuring operator information for cell 1
ADD NRDUCELLOP: NrDuCellId=1, OperatorId=0;
ADD NRDUCELLOP: NrDuCellId=1, OperatorId=1;
//Configuring operator information for cell 2
ADD NRDUCELLOP: NrDuCellId=2, OperatorId=0;
ADD NRDUCELLOP: NrDuCellId=2, OperatorId=1;
//Configuring interfaces
//Configuring the S1 and X2 interfaces in NSA networking
//Configuring the S1 interface
//Configuring the primary-operator information of the S1 object if Option 3x is used (given an assumption
that the S1 object of the primary operator has been added)
ADD GNBCUS1OP: gNBCuS1Id=0, OperatorId=0;
//Adding an S1 object of the secondary operator and an IP address, and configuring the secondary-operator
information of the S1 object if Option 3x is used
ADD EPGROUP: EPGROUPID=60;
ADD USERPLANEHOST: UPHOSTID=60, IPVERSION=IPv4, LOCIPV4="5.5.3.30", IPSECSWITCH=DISABLE;
ADD UPHOST2EPGRP: EPGROUPID=60, UPHOSTID=60;
ADD GNBCUS1: gNBCuS1Id=1, UpEpGroupId=60;
ADD GNBCUS1OP: gNBCuS1Id=1, OperatorId=1;
//Configuring the X2 interface
//Configuring the X2 interface—either shared or independent—based on the operator's policy
//(Case 1) Configuring a shared X2 interface (given an assumption that the IP address and related
configurations of the shared X2 interface have been added)
//Specifying the operators to be bound to the X2 object
ADD GNBCUX2OP: gNBCuX2Id=0, OperatorId=0;
ADD GNBCUX2OP: gNBCuX2Id=0, OperatorId=1;
//(Case 2) Configuring independent X2 interfaces (configuring different X2 addresses and related
configurations for different operators)
//Adding endpoint groups for the X2 interfaces
ADD EPGROUP: EPGROUPID=3;
ADD EPGROUP: EPGROUPID=4;
//Adding X2 control-plane hosts (IPv4 used as an example) with different operators using different local
addresses
ADD SCTPHOST: SCTPHOSTID=3, IPVERSION=IPv4, SIGIPV4="5.5.3.40", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=38422, SCTPTEMPLATEID=0;
ADD SCTPHOST: SCTPHOSTID=4, IPVERSION=IPv4, SIGIPV4="5.5.3.41", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=38422, SCTPTEMPLATEID=0;
//Adding X2 user-plane hosts with different operators using different local addresses
```



```

ADD USERPLANEHOST: UPHOSTID=3, IPVERSION=IPv4, LOCIPV4="5.5.3.40", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=4, IPVERSION=IPv4, LOCIPV4="5.5.3.41", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane hosts to the endpoint groups
ADD UPHOST2EPGRP: EPGROUPID=3, UPHOSTID=3;
ADD UPHOST2EPGRP: EPGROUPID=4, UPHOSTID=4;
//Adding the X2 control-plane hosts to the endpoint groups
ADD SCTPHOST2EPGRP: EPGROUPID=3, SCTPHOSTID=3;
ADD SCTPHOST2EPGRP: EPGROUPID=4, SCTPHOSTID=4;
//Adding X2 objects on the gNodeB side
ADD GNBCUX2: gNBCuX2Id=0, CpEpGroupId=3, UpEpGroupId=3;
ADD GNBCUX2: gNBCuX2Id=1, CpEpGroupId=4, UpEpGroupId=4;
//Specifying the operators to be bound to the X2 object
ADD GNBCUX2OP: gNBCuX2Id=0, OperatorId=0;
ADD GNBCUX2OP: gNBCuX2Id=1, OperatorId=1;
//Enabling prohibition of multi-operator sharing during X2-U self-setup
MOD GNBINTERFACEPARAM: InterfaceSelfCfgOptSw=X2_UP_SELF_SETUP_NO_MOCN_SW-1;
//Configuring the NG and Xn interfaces in SA networking
//Configuring the NG interface—either shared or independent—based on the operator's policy
//(Case 1) Configuring a shared NG interface (given an assumption that the IP address and related
configurations of the shared NG interface have been added)
//Configuring the primary-operator information of the NG object
ADD GNBCUNGOP: gNBCuNgId=0, OperatorId=0;
//Configuring the secondary-operator information of the NG object
ADD GNBCUNGOP: gNBCuNgId=0, OperatorId=1;
//(Case 2) Configuring independent NG interfaces (configuring different NG addresses and related
configurations for different operators)
//Configuring the primary-operation information of the NG object (given an assumption that the NG object
of the primary operator has been added)
ADD GNBCUNGOP: gNBCuNgId=0, OperatorId=0;
//Adding the NG object of the secondary operator and the secondary-operator information of the NG object
ADD GNBCUNG: gNBCuNgId=1, CpEpGroupId=60, UpEpGroupId=60;
ADD GNBCUNGOP: gNBCuNgId=1, OperatorId=1;
//Configuring the Xn interface—either shared or independent—based on the operator's policy
//(Case 1) Configuring a shared Xn interface (given an assumption that the IP address and related
configurations of the shared Xn interface have been added)
//Specifying the operators to be bound to the Xn object
ADD GNBCUXNOP: gNBCuXnId=0, OperatorId=0;
ADD GNBCUXNOP: gNBCuXnId=0, OperatorId=1;
//(Case 2) Configuring independent Xn interfaces (configuring different Xn addresses and related
configurations for different operators)
//Adding endpoint groups for the Xn interfaces
ADD EPGROUP: EPGROUPID=3;
ADD EPGROUP: EPGROUPID=4;
//Adding Xn control-plane hosts (IPv4 used as an example) with different operators using different local
addresses
ADD SCTPHOST: SCTPHOSTID=3, IPVERSION=IPv4, SIGIPV4="5.5.3.38", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=38422, SCTPTEMPLATEID=0;
ADD SCTPHOST: SCTPHOSTID=4, IPVERSION=IPv4, SIGIPV4="5.5.3.39", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=38422, SCTPTEMPLATEID=0;
//Adding Xn user-plane hosts with different operators using different local addresses
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="5.5.3.38", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv4, LOCIPV4="5.5.3.39", IPSECSWITCH=DISABLE;
//Adding the Xn user-plane hosts to the endpoint groups
ADD UPHOST2EPGRP: EPGROUPID=3, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=4, UPHOSTID=2;
//Adding the Xn control-plane hosts to the endpoint groups
ADD SCTPHOST2EPGRP: EPGROUPID=3, SCTPHOSTID=3;
ADD SCTPHOST2EPGRP: EPGROUPID=4, SCTPHOSTID=4;
//Adding Xn objects on the gNodeB side
ADD GNBCUXN: gNBCuXnId=0, CpEpGroupId=3, UpEpGroupId=3;
ADD GNBCUXN: gNBCuXnId=1, CpEpGroupId=4, UpEpGroupId=4;
//Specifying the operators to be bound to the Xn object
ADD GNBCUXNOP: gNBCuXnId=0, OperatorId=0;
ADD GNBCUXNOP: gNBCuXnId=1, OperatorId=1;
//Enabling prohibition of multi-operator sharing during Xn-U self-setup
MOD GNBINTERFACEPARAM: InterfaceSelfCfgOptSw=XN_UP_SELF_SETUP_NO_MOCN_SW-1;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;

```

//Configuring neighboring cells

```
//Assuming that the neighboring base station works in RAN sharing with common carrier
//Adding NR external neighboring cells, with Mcc and Mnc respectively set to the MCC and MNC of the
primary operator
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=2, CellId=3, PhysicalCellId=3, Tac=1,
SsbFreqPos=26639, NrNetworkingOption=SA_NSA;
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=2, CellId=4, PhysicalCellId=4, Tac=1,
SsbFreqPos=26639, NrNetworkingOption=SA_NSA;
//Adding external E-UTRAN cells, with Mcc and Mnc respectively set to the MCC and MNC of the primary
operator (not required in NSA networking)
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="01", EnodebId=0, CellId=3, DLEarfcn=2200,
PhysicalCellId=3, Tac=0;
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="01", EnodebId=0, CellId=4, DLEarfcn=2200,
PhysicalCellId=4, Tac=0;
//Adding the secondary-operator information to the neighboring-cell PLMN list, with Mcc and Mnc
respectively set to the MCC and MNC of the primary operator
ADD NREXTERNALNCELLPLMN: Mcc="460", Mnc="01", gNBId=2, CellId=3, SharedMcc="460",
SharedMnc="03", Tac=1, NrNetworkingOption=SA_NSA;
ADD NREXTERNALNCELLPLMN: Mcc="460", Mnc="01", gNBId=2, CellId=4, SharedMcc="460",
SharedMnc="03", Tac=1, NrNetworkingOption=SA_NSA;
ADD GNBEUTRAEXTCELLPLMN: Mcc="460", Mnc="01", EnodebId=0, CellId=3, SharedMcc="460",
SharedMnc="03";
ADD GNBEUTRAEXTCELLPLMN: Mcc="460", Mnc="01", EnodebId=0, CellId=4, SharedMcc="460",
SharedMnc="03";
//Adding the neighbor relationships with NR cells served by the neighboring base station (assuming that
NrCellId of the two cells of the local base station is 1 and 2, separately), with Mcc and Mnc respectively set
to the MCC and MNC of the primary operator
ADD NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=2, CellId=3;
ADD NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=2, CellId=4;
ADD NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=2, CellId=3;
ADD NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=2, CellId=4;
//Adding the neighbor relationships with E-UTRAN cells served by the neighboring base station (assuming
that NrCellId of the two cells of the local base station is 1 and 2, separately), with Mcc and Mnc
respectively set to the MCC and MNC of the primary operator (not required in NSA networking)
ADD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=0, CellId=3;
ADD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=0, CellId=4;
ADD NRCELLEUTRANRELATION: NrCellId=2, Mcc="460", Mnc="01", EnodebId=0, CellId=3;
ADD NRCELLEUTRANRELATION: NrCellId=2, Mcc="460", Mnc="01", EnodebId=0, CellId=4;
//Configuring the PLMN reserved flag (required when the neighboring base station serving an added
neighboring NR cell works in RAN sharing with dedicated carrier and the neighboring NR cell belongs to a
secondary operator)
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=2, CellId=3, PhysicalCellId=3, Tac=1,
SsbFreqPos=26639, NrNetworkingOption=SA_NSA, PlmnReservedFlag=TRUE;
//Configurations on the eNodeB (not required in SA networking)
//Adding external NR cells
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=1, CellId=1, DLEarfcn=620000, PhyCellId=1, Tac=1,
NrNetworkingOption=SA_NSA;
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=1, CellId=2, DLEarfcn=620000, PhyCellId=2, Tac=1,
NrNetworkingOption=SA_NSA;
//Adding the secondary-operator information to the NR external-cell PLMN list
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="01", GnodebId=1, CellId=1, SharedMcc="460",
SharedMnc="03", NrNetworkingOption=SA_NSA;
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="01", GnodebId=1, CellId=2, SharedMcc="460",
SharedMnc="03", NrNetworkingOption=SA_NSA;
//Adding the neighbor relationships with NR cells served by the neighboring base station (assuming that
LocalCellId of the two cells of the local base station is 1 and 2, separately), with Mcc and Mnc respectively
set to the MCC and MNC of the primary operator
ADD NRNRELATIONSHIP: LocalCellId=1, Mcc="460", Mnc="01", GnodebId=1, CellId=1;
ADD NRNRELATIONSHIP: LocalCellId=1, Mcc="460", Mnc="01", GnodebId=1, CellId=2;
ADD NRNRELATIONSHIP: LocalCellId=2, Mcc="460", Mnc="01", GnodebId=1, CellId=1;
ADD NRNRELATIONSHIP: LocalCellId=2, Mcc="460", Mnc="01", GnodebId=1, CellId=2;
//Configuring the operator-specific QCI parameters and event-specific threshold offsets of the gNodeB
//Adding QCI 1-specific threshold offsets for events A1, A2, and A5 related to coverage-based inter-
frequency measurements for operator 0
ADD GNBOPERATORQCIPARAM: OperatorId=0, Qci=1, InterFreqA1RsrpThldOfs=1,
InterFreqA2RsrpThldOfs=1, InterFreqA5RsrpThld2Ofs=1;
//Adding QCI 1-specific threshold offsets for events A1, A2, and A5 related to coverage-based inter-
frequency measurements for operator 1
```

```
ADD GNBOPERATORQCIPARAM: OperatorId=1, Qci=1, InterFreqA1RsrpThldOfs=0,
InterFreqA2RsrpThldOfs=2, InterFreqA5RsrpThld2Ofs=0;
```

Example 2: Changing the sharing mode of the gNodeB that originally belonged to one operator adopting NSA networking from the independent operator mode to RAN sharing with common carrier after an operator adopting SA networking is added

```
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Configuring the sharing mode, gNodeB-level operator information, and cell-level operator
information
//Configuring the sharing mode of the gNodeB
MOD GNBSHARINGMODE: gNBMultiOpSharingMode=SHARED_FREQ;
//Manually restarting the application referenced by the gNodeB function
RST APP: AID=1;
//Configuring the secondary-operator information of the gNodeB
ADD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,
Mcc="460", Mnc="03", NrNetworkingOption=SA;
//Configuring operator information for cell 1
ADD NRDUCELL: NrDuCellId=1, OperatorId=0;
ADD NRDUCELL: NrDuCellId=1, OperatorId=1;
//Configuring operator information for cell 2
ADD NRDUCELL: NrDuCellId=2, OperatorId=0;
ADD NRDUCELL: NrDuCellId=2, OperatorId=1;
//Configuring neighboring cells
//Assuming that the neighboring base station works in RAN sharing with common carrier
//Adding the secondary-operator information to the external neighboring cell PLMN list, with Mcc and Mnc
respectively set to the MCC and MNC of the primary operator
ADD NREXTERNALNCELLPLMN: Mcc="460", Mnc="01", gNBId=2, CellId=3, SharedMcc="460",
SharedMnc="03", Tac=1, NrNetworkingOption=SA;
ADD NREXTERNALNCELLPLMN: Mcc="460", Mnc="01", gNBId=2, CellId=4, SharedMcc="460",
SharedMnc="03", Tac=1, NrNetworkingOption=SA;
ADD GNBEUTRAEXTCELLPLMN: Mcc="460", Mnc="01", EnodebId=1, CellId=3, SharedMcc="460",
SharedMnc="03";
ADD GNBEUTRAEXTCELLPLMN: Mcc="460", Mnc="01", EnodebId=1, CellId=4, SharedMcc="460",
SharedMnc="03";
//Configuring the TAC by referring to Example 1
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
//Configuring the operator-specific QCI parameters and event-specific threshold offsets of the
gNodeB by referring to Example 1
```

Optimization Command Examples

N/A

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

Example 1: Changing the sharing mode of the gNodeB from RAN sharing with common carrier to the independent operator mode

```
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Removing neighboring cell configurations
//Removing the secondary-operator information from the external neighboring cell PLMN list
RMV NREXTERNALNCELLPLMN: CellId=3, gNBId=2, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03";
```

```

RMV NREXTERNALNCELLPLMN: CellId=4, gNBId=2, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03";
//Deleting interface configurations
//Configuring the S1 and X2 interfaces in NSA networking
//Configuring the S1 interface
//Removing the primary-operator information from the S1 object if Option 3x is used
RMV GNBCUS1OP: gNBCuS1Id=0, OperatorId=0;
//Removing the S1 object of the secondary operator and the secondary-operator information of the S1
object if Option 3x is used
RMV GNBCUS1OP: gNBCuS1Id=1, OperatorId=1;
RMV GNBCUS1: gNBCuS1Id=1;
//Removing a single S1-U link
RMV UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
RMV USERPLANEPEER: UPPEERID=0;
//Configuring the X2 interface
//Disabling prohibition of multi-operator sharing during X2-U self-setup
MOD GNBINTERFACEPARAM: InterfaceSelfCfgOptSw=X2_UP_SELF_SETUP_NO_MOCN_SW-0;
//Changing the X2 interface mode from the independent mode to the shared mode
//Removing the X2 interface binding relationship and the corresponding X2 object for the primary operator
RMV GNBCUX2OP: gNBCuX2Id=0, OperatorId=0;
RMV GNBCUX2: gNBCuX2Id=0;
//Binding the primary operator to the configured X2 interface of operator 1
ADD GNBCUX2OP: gNBCuX2Id=1, OperatorId=1;
//Configuring the NG and Xn interfaces in SA networking
//Configuring the NG interface
//Removing the NG object of the secondary operator and the secondary-operator information of the NG
object
RMV GNBCUNG: gNBCuNgId=1;
RMV GNBCUNGOP: gNBCuNgId=1, OperatorId=1;
//Removing a single NG-C link
RMV SCTPPEER2EPGRP: EPGROUPID=0, SCTPPEERID=0;
RMV SCTPPEER: SCTPPEERID=0;
//Removing a single NG-U link
RMV UPPEER2EPGRP: EPGROUPID=0, UPPEERID=0;
RMV USERPLANEPEER: UPPEERID=0;
//Configuring the Xn interface
//Disabling prohibition of multi-operator sharing during Xn-U self-setup
MOD GNBINTERFACEPARAM: InterfaceSelfCfgOptSw=XN_UP_SELF_SETUP_NO_MOCN_SW-0;
//Changing the Xn interface mode from the independent mode to the shared mode
//Removing the Xn interface binding relationship and the corresponding Xn object for the primary operator
RMV GNBCUXNOP: gNBCuXnId=0, OperatorId=0;
RMV GNBCUXN: gNBCuXnId=0;
//Binding the primary operator to the configured Xn interface of operator 1
ADD GNBCUXNOP: gNBCuXnId=1, OperatorId=1;
//Removing cell-level operator information and gNodeB-level operator information, and changing the
sharing mode
//Removing cell-level operator information
RMV NRDUCELLOP: NrDuCellId=1, OperatorId=0;
RMV NRDUCELLOP: NrDuCellId=1, OperatorId=1;
RMV NRDUCELLOP: NrDuCellId=2, OperatorId=0;
RMV NRDUCELLOP: NrDuCellId=2, OperatorId=1;
//Removing gNodeB-level secondary-operator information
RMV GNBOPERATOR: OperatorId=1;
//Changing the sharing mode of the gNodeB to the independent operator mode
MOD GNBSHARINGMODE: gNBMultiOpSharingMode=INDEPENDENT;
//Manually restarting the application referenced by the gNodeB function
RST APP: AID=1;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;

```

Example 2: Changing the sharing mode of the gNodeB from RAN sharing with common carrier to RAN sharing with dedicated carrier

```

//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Removing cell-level operator information and changing the sharing mode
//Removing cell-level operator information
RMV NRDUCELLOP: NrDuCellId=1, OperatorId=1;

```

```
RMV NRDUCELL: NrDuCellId=2, OperatorId=0;  
//Changing the sharing mode of the gNodeB to RAN sharing with dedicated carrier  
MOD GNBSHARINGMODE: gNBMultiOpSharingMode=SEPARATED_FREQ;  
//Manually restarting the application referenced by the gNodeB function  
RST APP: AID=1;  
//Activating cells  
ACT NRCELL: NrCellId=1;  
ACT NRCELL: NrCellId=2;
```

4.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.1.4.2 Activation Verification

Prerequisites

- The gNodeB is shared based on the network plan by two operators in RAN sharing with common carrier.
- The transmission links between the RAN and the core network are established and functioning normally.
- UE 1 and UE 2 separately belonging to two operators are ready.

Procedure

Step 1 Query the cell status.

Run the MML command **DSP NRCELL** to check the cell status.

Expected result: **NR Cell State** is **Normal**.

Step 2 Check SIB1. Specifically, log in to the MAE-Access, and choose **Monitor > Signaling Trace > Signaling Trace Management > Trace Type > NR > Application Layer > Uu Interface Trace**.

Expected result: SIB1 contains the correct PLMN ID of each operator.

In SIB1 of cell 1 and cell 2, **PLMN-Identity** indicates **46001** for the primary operator and **46003** for the secondary operator. In SIB1 of cell 1 for example, the MCC and MNC indicated by **PLMN-Identity** are the same as those in the PLMN ID configured for the corresponding operator, showing that the PLMN IDs of the primary and secondary operators are correctly delivered in SIB1 of cell 1.

```
systemInformationBlockType1  
  cellSelectionInfo  
    q-RxLevMin: ---- 0xfffffc0 (-64)  
  cellAccessRelatedInfo  
    plmn-IdentityList  
      PLMN-IdentityInfo  
        plmn-IdentityList  
          PLMN-Identity  
            mcc  
              MCC-MNC-Digit:0x4 (4)  
              MCC-MNC-Digit:0x6 (6)  
              MCC-MNC-Digit:0x0 (0)  
            mnc
```

```

MCC-MNC-Digit:0x6 (0)
MCC-MNC-Digit:0x1 (1)
trackingAreaCode: ---- 000000000000000000000000 (00 00 00)
cellIdentity: ---- 0000000000000000000000000000010001 (00 00 00 01 01)
cellReservedForOperatorUse: ---- notReserved (1)
PLMN-IdentityInfo
plmn-IdentityList
PLMN-Identity
mcc
MCC-MNC-Digit:0x4 (4)
MCC-MNC-Digit:0x6 (6)
MCC-MNC-Digit:0x0 (0)
mnc
MCC-MNC-Digit:0x0 (0)
MCC-MNC-Digit:0x3 (3)
trackingAreaCode: ---- 00000000000000000000000000 (00 00 00)
cellIdentity: ---- 0000000000000000000000000000010001 (00 00 00 01 01)
cellReservedForOperatorUse: ---- notReserved (1)

```

Step 3 Power on UE 1 and UE 2 and enable them to access cells.

Expected result: Service data is sent to respective core networks.

-----End

 NOTE

In SIB1, the *PLMN-IdentityInfo* IE contains **trackingAreaCode** for the operators with the **gNBOperator.NrNetworkingOption** parameter set to **SA** or **SA_NSA**, but not for the operators with the **gNBOperator.NrNetworkingOption** parameter set to **NSA**.

4.1.4.3 Network Monitoring

In networks where RAN sharing with common carrier is enabled, some cell-level performance counters can be monitored on a per operator basis. These counters are used to indicate whether the live network meets the key performance indicator (KPI) requirements of operators. Operators can use these counters for preliminary fault demarcation and locating.

For details about operator-specific performance counters, see the function sets in *3900 & 5900 Series Base Station Performance Counter Reference* in *3900 & 5900 Series Base Station Product Documentation*. These function sets include:

- Measurement of Operator-Specific NR DU Cell Performance (NRDUCell.Operator)
- Measurement of Operator-Specific NR CU Cell Performance (NRCell.Operator)
- Measurement of Operator-Specific DU Cell 5QI Performance (NRDUCELL.5QI.Operator)
- Measurement of Operator-Specific Neighboring NR Cell Performance (NRCellRelation.Operator)
- Measurement of Operator-Specific Cell 5QI Performance (NRCELL.5QI.Operator)

4.2 Operator-Specific License Management

4.2.1 Principles

On the gNodeB, specific percentages of the licensed limits defined by the RRC Connected User License and RE(Resource Element) License can be configured for operators. For eMBB services, each RRC_CONNECTED UE consumes one unit of both the RRC Connected User License and RE(Resource Element) License.

The RRC Connected User License and RE(Resource Element) License can be fully shared or statically shared in percentages, differentiating in the resource sharing mode. [Figure 4-10](#) illustrates license sharing between two operators, and [Table 4-21](#) provides further details.

Figure 4-10 License sharing

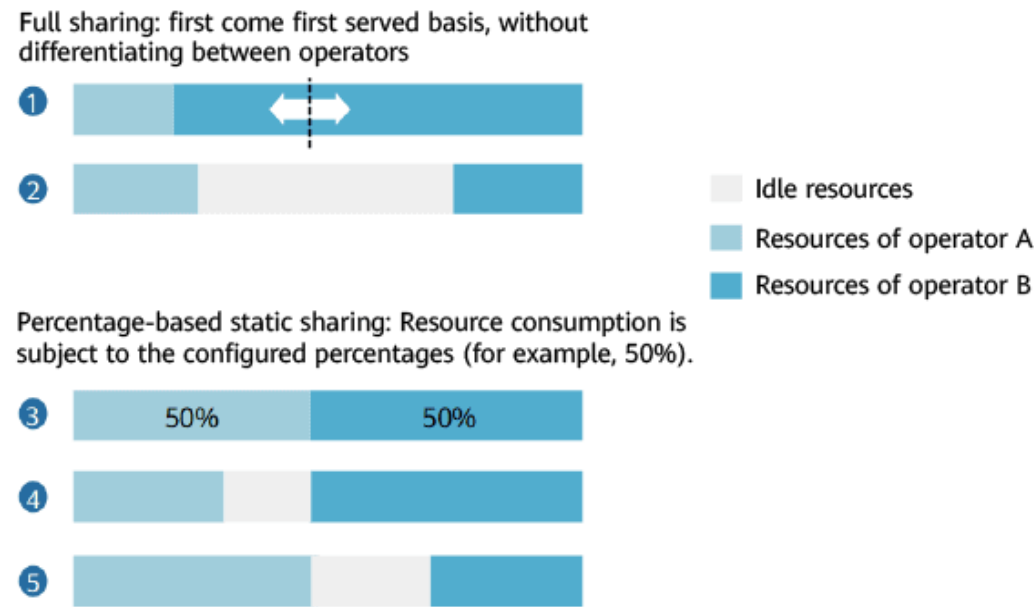


Table 4-21 License sharing description

No.	Description
1	During busy hours: Operators A and B contend for resources.
2	During off-peak hours: Operators A and B do not contend for resources.
3	During busy hours: Operators A and B use resources based on the configured percentages.
4	During off-peak hours for operator A: Operator A uses resources as required, and operator B uses resources based on the configured percentage.
5	During off-peak hours for operator B: Operator B uses resources as required, and operator A uses resources based on the configured percentage.

 NOTE

- If the resource usage of an operator exceeds the configured percentage, the operator is in busy hours.
- If the resource usage of an operator does not exceed the configured percentage, the operator is in off-peak hours.

RRC Connected User License

The **gNBOperator.RrcConnUserLicPercentage** parameter specifies a percentage of the licensed limit defined by the RRC Connected User License for each operator. The licensed resources available for each operator are subject to the specified percentage. The gNodeB calculates the licensed number of RRC_CONNECTED UEs for each operator based on the value of the **gNBOperator.RrcConnUserLicPercentage** parameter and performs access control based on the result.

- Full sharing
In this scenario, the **gNBOperator.RrcConnUserLicPercentage** parameter is either not set or set to **255** for an operator.
- Percentage-based static sharing
In this scenario, the **gNBOperator.RrcConnUserLicPercentage** parameter is set for all operators. To prevent the licensed resources from being wasted, it is recommended that the sum of the values of the **gNBOperator.RrcConnUserLicPercentage** parameter configured for all operators equal 100% (the maximum value).

RE(Resource Element) License

The **gNBOperator.ReLicPercentage** parameter specifies a percentage of the licensed limit defined by the RE(Resource Element) License for each operator. The licensed resources available for each operator are subject to the specified percentage. The gNodeB calculates the licensed number of REs for each operator based on the value of the **gNBOperator.ReLicPercentage** parameter and performs access control based on the result.

- Full sharing
In this scenario, the **gNBOperator.ReLicPercentage** parameter is either not set or set to **255** for an operator.
- Percentage-based static sharing
In this scenario, the **gNBOperator.ReLicPercentage** parameter is set for all operators. To prevent the licensed resources from being wasted, it is recommended that the sum of the values of the **gNBOperator.ReLicPercentage** parameter configured for all operators equal 100% (the maximum value).

4.2.2 Network Analysis

4.2.2.1 Benefits

Specific percentages of the licensed limits defined by the RRC Connected User License and RE(Resource Element) License can be configured for each operator.

With the setting, a certain amount of resources is always available to a specific operator.

4.2.2.2 Impacts

The impacts between this function and other functions are described in [4.2.3.2.3 Function Impacts](#).

When percentage-based static sharing is enabled and the configured percentage of licensed resources for an operator is smaller than the actual percentage required for the operator, UEs of the operator may fail to access the network due to insufficient licensed resources.

4.2.3 Requirements

4.2.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021211	Multi-Operator Sharing	NR0S00MULS00	per Cell
For the license control item NR0S00MULS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

NOTE

If the preceding license is unavailable, UEs of the secondary operator cannot access the network. In regard to the primary operator, UEs can access the network, and the RRC Connected User License and RE(Resource Element) License are consumed according to the configured percentages.

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.2.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.2.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with common carrier	gNBSharingMode .gNBMultiOpSharingMode set to SHARED_FREQ	<i>Multi-Operator Sharing</i>	Operator-specific license management requires RAN sharing with common carrier or RAN sharing with dedicated carrier to be deployed on the live network.
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with dedicated carrier	gNBSharingMode .gNBMultiOpSharingMode set to SEPARATED_FREQ	<i>Multi-Operator Sharing</i>	

4.2.3.2.2 Mutually Exclusive Functions

None

4.2.3.2.3 Function Impacts

None

4.2.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

If different carrier frequencies are configured on an RRU or AAU, the RF module must support multiple carriers.

For details about the carrier specifications of RF modules, see the RRU/AAU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

4.2.3.4 Others

None

4.2.4 Operation and Maintenance

4.2.4.1 Data Configuration

4.2.4.1.1 Data Preparation

Table 4-22 describes the parameters used for function activation. No parameters are required for function optimization.

Table 4-22 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
RRC Connected User License Percentage	gNBOperator.RrcConnU serLicPercentage	These parameters are configured based on negotiation among operators. One UE consumes one unit of both the RRC Connected User License and RE(Resource Element) License. Therefore, it is recommended that the two parameters be set to the same value for the same operator.
RE License Percentage	gNBOperator.ReLicPerc entage	

4.2.4.1.2 Using MML Commands

Before using MML commands, refer to **4.2.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

//Using NSA networking as an example (Set **gNBOperator.NrNetworkingOption** to the corresponding NR networking option in other networking scenarios.)

```
//Configuring specific percentages of the licensed limits defined by the RRC Connected User License and  
RE(Resource Element) License for the primary operator  
MOD GNBOPERATOR: OperatorId=0, OperatorName="unicom", OperatorType=PRIMARY_OPERATOR,  
Mcc="460", Mnc="01", NrNetworkingOption=NSA, ReLicPercentage=50, RrcConnUserLicPercentage=50;  
//Configuring specific percentages of the licensed limits defined by the RRC Connected User License and  
RE(Resource Element) License for the secondary operator  
MOD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,  
Mcc="460", Mnc="03", NrNetworkingOption=NSA, ReLicPercentage=50, RrcConnUserLicPercentage=50;
```

Optimization Command Examples

N/A

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Using NSA networking as an example (Set gNBOperator.NrNetworkingOption to the corresponding NR  
networking option in other networking scenarios.)  
//Removing the percentages of the licensed limits defined by the RRC Connected User License and  
RE(Resource Element) License for the primary operator  
MOD GNBOPERATOR: OperatorId=0, OperatorName="unicom", OperatorType=PRIMARY_OPERATOR,  
Mcc="460", Mnc="01", NrNetworkingOption=NSA, ReLicPercentage=255, RrcConnUserLicPercentage=255;  
//Removing the percentages of the licensed limits defined by the RRC Connected User License and  
RE(Resource Element) License for the secondary operator  
MOD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,  
Mcc="460", Mnc="03", NrNetworkingOption=NSA, ReLicPercentage=255, RrcConnUserLicPercentage=255;
```

4.2.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.2.4.2 Activation Verification

Prerequisites

- The gNodeB is shared between two operators (operator A and operator B) in RAN sharing with common carrier. The basic networking for RAN sharing with common carrier has been configured.
- The licensed limit defined by the RRC Connected User License is 100. The **gNBOperator.RrcConnUserLicPercentage** parameter is set to **50** for operator A and **50** for operator B. Additionally, 35 UEs of operator A and 80 UEs of operator B need to establish RRC connections.
- The licensed limit defined by the RE(Resource Element) License is 100. The **gNBOperator.ReLicPercentage** parameter is set to **50** for operator A and **50** for operator B. Additionally, 35 UEs of operator A and 80 UEs of operator B need to consume the RE(Resource Element) License.

Expected Result

All UEs of operator A are admitted successfully and can initiate services properly. However, some UEs of operator B fail to be admitted due to insufficient licensed

resources defined by the RRC Connected User License and RE(Resource Element) License.

4.2.4.3 Network Monitoring

Check the value of the **N.RRCConn.UserLic.Limit.Num.PLMN** counter, which indicates the number of times the licensed number of RRC_CONNECTED UEs is limited for a specific operator when consumption of licensed resources is controlled by pre-configured percentages. This counter applies only in network sharing scenarios.

4.3 Operator-Specific RB Management

4.3.1 Principles

In RAN sharing with common carrier, three functions are available: percentage-based dynamic RB sharing among operators, operator-specific limitation on the maximum number of available RBs, and operator-specific guarantee on the minimum number of available RBs. The RBs available for an operator are determined based on the association between **NRDUCellOp.CellResId** and **NRDUCellRes.CellResId**. For details, see [Table 4-23](#).

Table 4-23 Functions with their control switches and parameters specifying the percentages of uplink and downlink RBs

Function	Control Switch	Parameter Specifying the Uplink/Downlink RB Percentage
Percentage-based dynamic RB sharing among operators	RB_DYNAMIC_SHARING_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	Average percentage of uplink RBs: NRDUCellRes.UlRbAverageRatio
		Average percentage of downlink RBs: NRDUCellRes.DlRbAverageRatio
Operator-specific limitation on the maximum number of available RBs	MAX_RB_LIMIT_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	Maximum percentage of uplink RBs: NRDUCellRes.UlRbMaxRatio
		Maximum percentage of downlink RBs: NRDUCellRes.DlRbMaxRatio

Function	Control Switch	Parameter Specifying the Uplink/Downlink RB Percentage
Operator-specific guarantee on the minimum number of available RBs	MIN_RB_GUARANTEE_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	Minimum percentage of uplink RBs: NRDUCellRes.UIRbMinRatio
		Minimum percentage of downlink RBs: NRDUCellRes.DIRbMinRatio

NOTE

If three or more operators share a cell, some of these operators may form an alliance, which is regarded as an operator group. The alliance is made by setting **NRDUCellOp.CellResId** to the same value as **NRDUCellRes.CellResId** for all member operators in the group. In RB management, operators that belong to the same operator group are considered as a single operator; air interface resources are preferentially shared among operators in the group and can be shared with operators outside the group only when there are remaining resources.

Percentage-based Dynamic RB Sharing Among Operators

The **RB_DYNAMIC_SHARING_SW** option of the **NRDUCellAlgoSwitch.RanSharingAlgoSwitch** parameter determines whether to enable this function.

If this option is deselected, scheduling is not distinguished between operators, and all RBs are shared among operators as a result.

If this option is selected, however, operator-specific scheduling is performed. In this case, RBs are dynamically shared among operators. Further details are as follows:

- When all operators' RBs are occupied, RB consumption is subject to the specified percentage. Scheduling priority adjustment enables the percentage of RBs used by each operator during a specific period of time to reach or approach the specified average percentage of RBs. The **NRDUCellRes.DIRbAverageRatio** and **NRDUCellRes.UIRbAverageRatio** parameters specify the average percentages of downlink and uplink RBs for an operator, respectively.
- When an operator has idle RBs, they can be shared with other operators.

Uplink and downlink RBs have the same allocation rules. The following uses the uplink case as an example. It is recommended that the summed average percentages of uplink RBs configured for all operators reach the maximum allowed value of 100%. When the sum is less than 100%, the remaining RBs are treated as common resources. These common resources can be allocated to or preempted by operators, depending on their weights of average percentages of uplink RBs configured for them. In this situation, the actual RB usage by an operator may exceed the configured value.

Figure 4-11 illustrates RB sharing between two operators, and **Table 4-24** provides further details.

Figure 4-11 RB sharing

Full sharing: first come first served basis,
without differentiating between operators



Dynamic sharing: Resource consumption is subject to
the configured percentage (for example, 50%) in busy
hours, and idle resources can be shared in idle hours.



Table 4-24 RB sharing description

No.	Description
1	During busy hours: Operators A and B contend for resources.
2	During off-peak hours: Operators A and B do not contend for resources.
3	During busy hours: Operators A and B use resources based on the configured percentages.
4	During off-peak hours for operator A: Resources of operator A are dynamically shared with operator B.
5	During off-peak hours for operator B: Resources of operator B are dynamically shared with operator A.

 NOTE

- If the operator resource groups configured for a cell outnumber the operators in the cell and the function of percentage-based dynamic RB sharing among operators is enabled, some of the operator resource groups will not be associated with operators. In this situation, such operator resource groups still take effect, but they affect the resource configuration for operators associated with resource groups. As such, you are advised to configure operator resource groups that are associated with operators.
- In low-frequency scenarios, percentage-based dynamic RB sharing takes effect among operator groups, and full sharing of RBs takes effect within an operator group.
- For normal performance of percentage-based dynamic RB sharing among operators in high-frequency scenarios, it is recommended that all operators involved in a sector be configured with the same number of cells and the parameter settings related to the function be the same for all of the involved cells, including the settings of **NRDUCellRes.DIRbAverageRatio**, **NRDUCellRes.UIRbAverageRatio**, and the **RB_DYNAMIC_SHARING_SW** option of **NRDUCellAlgoSwitch.RanSharingAlgoSwitch**.

Operator-specific Limitation on the Maximum Number of Available RBs

The **MAX_RB_LIMIT_SW** option of the **NRDUCellAlgoSwitch.RanSharingAlgoSwitch** parameter determines whether to enable this function. This function is mutually exclusive with **Operator-specific Guarantee on the Minimum Number of Available RBs**.

If this option is deselected, the gNodeB does not limit the maximum number of RBs that can be used by an operator during scheduling.

If this option is selected, the gNodeB limits the maximum number of RBs that can be used by each operator during scheduling. The **NRDUCellRes.DIRbMaxRatio** and **NRDUCellRes.UIRbMaxRatio** parameters specify the maximum percentages of downlink and uplink RBs, respectively. To prevent user experience from being impacted, it is recommended the maximum percentages of downlink and uplink RBs both be greater than or equal to 10%.

If the number of RBs available for the data channel multiplied by the value of **NRDUCellRes.DIRbMaxRatio** or **NRDUCellRes.UIRbMaxRatio** is not an integer, the result is rounded down. If the rounddown result is not an integer multiple of the RBG size, RBs are wasted.

 NOTE

- If the RB quota specified by **NRDUCellRes.DIRbMaxRatio** is less than one RBG, the gNodeB allocates the minimum resources to ensure service continuity based on the resource scheduling unit that takes effect on UEs.
- If the RB quota specified by **NRDUCellRes.UIRbMaxRatio** includes fewer than four RBs, the gNodeB allocates the minimum resources to ensure service continuity based on the resource scheduling unit that takes effect on UEs.
- Resources scheduled for DTX retransmissions are not restricted by the **NRDUCellRes.DIRbMaxRatio** or **NRDUCellRes.UIRbMaxRatio** parameter.
- The number of RBs in an RBG varies with bandwidths. For details, see Table 5.3.2-1: Transmission bandwidth configuration N_{RB} for FR1 in 3GPP TS 38.104 V15.5.0, and Table 5.1.2.2.1-1: Nominal RBG size P in 3GPP TS 38.214 V15.4.0.
- Operator-specific limitation on the maximum number of available RBs takes effect based on RB percentages among operator groups, and full sharing of RBs takes effect within an operator group.

Operator-specific Guarantee on the Minimum Number of Available RBs

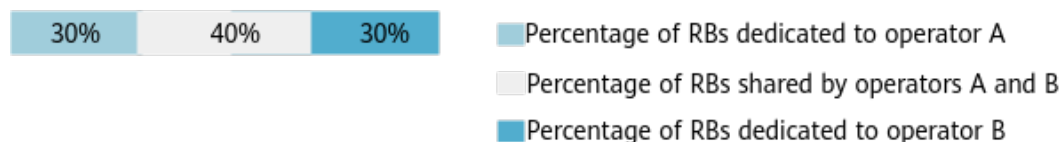
The **MIN_RB_GUARANTEE_SW** option of the **NRDUCellAlgoSwitch.RanSharingAlgoSwitch** parameter determines whether to enable this function. This function is mutually exclusive with **Operator-specific Limitation on the Maximum Number of Available RBs**.

If the option is deselected, the gNodeB does not guarantee the minimum number of RBs available for operators during scheduling.

If the option is selected, the gNodeB guarantees the minimum number of RBs available for operators during scheduling. The minimum percentages of downlink and uplink RBs are specified by the **NRDUCellRes.DlRbMinRatio** and **NRDUCellRes.UlRbMinRatio** parameters, respectively. To ensure user experience, it is recommended the minimum percentages of downlink and uplink RBs both be less than or equal to 90%. Currently, the function of operator-specific guarantee on the minimum number of available RBs supports only the mean-type control policy. With this control policy, RBs are scheduled based on the average PRB usage of UEs in an operator resource group; in this case, the resources available for these UEs do not need to meet the requirements in each TTI. This control policy ensures that the minimum number of available RBs for an operator is greater than or equal to the configured minimum number of RBs within a period of time.

Figure 4-12 shows how RBs are shared by two operators when the minimum percentages of downlink and uplink RBs are both set to 30%.

Figure 4-12 Operator-specific guarantee on the minimum number of available RBs



NOTE

- If the operator resource groups configured for a cell outnumber the operators in the cell and the function of operator-specific guarantee on the minimum number of available RBs is enabled, some of the operator resource groups will not be associated with operators. In this situation, such operator resource groups still take effect, but they affect the resources available for operators associated with resource groups. As such, you are advised to configure operator resource groups that are associated with operators.
- Operator-specific guarantee on the minimum number of available RBs takes effect based on RB percentages among operator groups, and full sharing of RBs takes effect within an operator group.

4.3.2 Network Analysis

4.3.2.1 Benefits

Percentage-based RB sharing among operators (or resource groups) ensures that a certain amount of resources are always available for a specific operator (or resource group).

4.3.2.2 Impacts

The impacts between this function and other functions are described in [4.3.3.2.3 Function Impacts](#).

The following describes the impacts on the uplink network performance after percentage-based dynamic RB sharing among operators is enabled. If (1) the sum of the average uplink RB percentages configured for all operators is equal to 100%, and (2) the RB percentages differ greatly among operators (for example, 90% for one operator and 10% for the other), the scheduling delay will slightly increase and the user-perceived rate will slightly decrease for the operator with a lower RB percentage, compared with those in scenarios where RAN sharing with common carrier is not enabled. If the sum of the average uplink RB percentages configured for all operators is less than 100%, a higher average uplink RB percentage for an operator results in a higher percentage of common resources that the operator can preempt, and a lower average uplink RB percentage for an operator results in the opposite effect. The impacts on the downlink network performance are the same as described above.

After the function of operator-specific limitation on the maximum number of available RBs is enabled, the peak throughput and scheduling delay of UEs decrease. Specifically, lower maximum percentages of downlink and uplink RBs result in a larger decrease in the peak UE throughput and, in the case of insufficient RBs, a longer UE transmission delay.

After the function of operator-specific guarantee on the minimum number of available RBs is enabled, higher minimum percentages of downlink and uplink RBs result in fewer resources that can be shared by operators and lower peak UE throughput and cell capacity. If the minimum percentages of downlink and uplink RBs for an operator are set to high values (for example, higher than 90%), the UE transmission delay for an operator with a lower RB percentage will increase in the case of insufficient RBs.

After the function of operator-specific guarantee on the minimum number of available RBs is enabled, the limitation on the percentage of scheduled RBs reduces the probability of scheduling UEs, which may affect the service drop rate and access success rate.

NOTE

If multiple operators are configured as a resource group (that is, they form an alliance with each other), the description about operators is also applicable to the resource group.

4.3.3 Requirements

4.3.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021211	Multi-Operator Sharing	NR0S00MULS00	per Cell

Feature ID	Feature Name	Model	Sales Unit
For the license control item NR0S00MULS00: - If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. - If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.3.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.3.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with common carrier	gNBSharingMode.gNBMultiOpSharingMode	<i>Multi-Operator Sharing</i>	Operator-specific limitation on the maximum number of available RBs, operator-specific guarantee on the minimum number of available RBs, and percentage-based dynamic RB sharing among operators require RAN sharing with common carrier to be deployed on the live network.

4.3.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Operator-specific limitation on the maximum number of available RBs and PDSCH MU-MIMO cannot both be enabled.
Low-frequency TDD	PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	Operator-specific limitation on the maximum number of available RBs and PUSCH MU-MIMO cannot both be enabled.
High-frequency TDD	Uplink MU-MIMO for mmWave large-packet UEs	MMW_LRG_PKT_UL_MU_MIMO_SW option of the NRDUCellFeatureSwitch.MmWaveFeatureSwitch parameter	None	Operator-specific limitation on the maximum number of available RBs and uplink MU-MIMO for mmWave large-packet UEs cannot both be enabled.
Low-frequency TDD	Network slice resource management	RB_DYNAMIC_CONTROL_SW option of the NRDUCellAlgoSwitch.NetworkSliceAlgoSwitch parameter	<i>Network Slicing</i>	Network slice resource management can be enabled only when the RB_DYNAMIC_SHARING_SW , MAX_RB_LIMIT_SW , and MIN_RB_GUARANTEE_SW options of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter are all deselected for operator-specific RB management.

RAT	Function Name	Function Switch	Reference	Description
High-frequency TDD	Network slice resource management	RB_DYNAMIC_SLICING_SWITCH option of the NRDUCellAlgorithmSwitch.NetworkSliceAlgorithmSwitch parameter	<i>Network Slicing</i>	Network slice resource management can be enabled only when the RB_DYNAMIC_SHARING_SWITCH option of the NRDUCellAlgorithmSwitch.RANSharingAlgorithmSwitch parameter is deselected for operator-specific RB management.
Low-frequency TDD	QCI-based resource control	RB_DYNAMIC_QCI_CONTROL_SWITCH option of the NRDUCellFeatureSwitch.QciBasedDiffServExpSwitch parameter	<i>QCI-based Resource Control</i>	QCI-based resource control can be enabled only when the RB_DYNAMIC_SHARING_SWITCH , MAX_RB_LIMIT_SWITCH , and MIN_RB_GUARANTEE_SWITCH options of the NRDUCellAlgorithmSwitch.RANSharingAlgorithmSwitch parameter are all deselected for operator-specific RB management.
Low-frequency TDD	Dynamic RB control	RB_DYNAMIC_HIGH_RELIABILITY_SWITCH option of the NRDUCellAlgorithmSwitch.HighReliabilitySwitch parameter	<i>URLLC</i>	Dynamic RB control can be enabled only when the RB_DYNAMIC_SHARING_SWITCH , MAX_RB_LIMIT_SWITCH , and MIN_RB_GUARANTEE_SWITCH options of the NRDUCellAlgorithmSwitch.RANSharingAlgorithmSwitch parameter are all deselected for operator-specific RB management.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Operator-specific guarantee on the minimum number of available RBs	MIN_RB_GUARANTEE_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter	None	Operator-specific limitation on the maximum number of available RBs (controlled by the MAX_RB_LIMIT_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter) and operator-specific guarantee on the minimum number of available RBs (controlled by the MIN_RB_GUARANTEE_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter) are mutually exclusive. That is, they cannot both be enabled.
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	Currently, operator-specific guarantee on the minimum number of available RBs cannot be enabled in a combined cell (with NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	NR DU Cell Networking Mode	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE or HYPER_CELL	<i>Cell Combination Hyper Cell</i>	If the BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter is selected and the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL_COMBINE_MODE or HYPER_CELL , the RB_DYNAMIC_SHARING_SW and MAX_RB_LIMIT_SW options of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter must be deselected.
Low-frequency TDD	Power saving BWP	BWP2_SWITCH option of the NRDUCellUePwrSaving.BwpPwrSavingSw parameter	<i>UE Power Saving</i>	
Low-frequency TDD	FSD transmission mode	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL , and NRDUCellMultiTrp.DataTransMode set to FSD_MODE	<i>Hyper Cell</i>	Operator-specific guarantee on the minimum number of available RBs cannot be enabled in FSD transmission mode.
Low-frequency TDD	RFSP-based RB resource control	RB_DYNAMIC_CONTROL_SW option of the NRDUCellCommonSw.RfspAlgoSwitch parameter	<i>Resource Control in NSA Networking</i>	RFSP-based RB resource control can be enabled only when the RB_DYNAMIC_SHARING_SW , MAX_RB_LIMIT_SW , and MIN_RB_GUARANTEE_SW options of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter are all deselected for operator-specific RB management.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Dynamic user PRB control for FWA	DYN_PRB_CTRL_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	Dynamic user PRB control for FWA can be enabled only when the RB_DYNAMIC_SHARING_SW , MAX_RB_LIMIT_SW , and MIN_RB_GUARANTEE_SW options of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter are all deselected for operator-specific RB management.
High-frequency TDD	Dynamic user PRB control for FWA	DYN_PRB_CTRL_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	Dynamic user PRB control for FWA can be enabled only when the RB_DYNAMIC_SHARING_SW option of the NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter is deselected for operator-specific RB management.

4.3.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Cell Combination	NRDUCell.NrDuCellNetworkingMode set to HYPER_CELL_COMBINE_MODE	<i>Cell Combination</i>	In Cell Combination networking (that is, the NRDUCell.NrDuCellNetworkingMode parameter is set to HYPER_CELL_COMBINE_MODE), when the NRDUCellMultiTrp.DataTransMode parameter is set to BASIC_MODE , the operator-specific percentages of RBs that are actually used will fluctuate around the percentages of RBs that are operator-specifically configured by any of the following functions: percentage-based dynamic RB sharing among operators, operator-specific limitation on the maximum number of available RBs, and operator-specific guarantee on the minimum number of available RBs.
Low-frequency TDD	User PRB control for FWA	PRB_CTRL_SW option of the NRDUCellAlgoSwitch.FwaAlgoSwitch parameter	<i>FWA</i>	For the member operators (configured with the same value of NRDUCellOp.CellResId) in the same operator group, the NRDUCellOpFwaQos MO cannot be used to specify the uplink or downlink RB percentages for different types of users.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Downlink resource isolation	DL_EMBB_PREFE RENTIAL_SCH_S W option of the NRDUCellAlgoS witch.FwaAlgoS witch parameter	<i>FWA</i>	After downlink resource isolation is enabled: If the minimum percentage of downlink RBs in operator-specific RB management has taken effect for FWA users, downlink resource isolation will not take effect for these FWA users; if the minimum percentage of downlink RBs in operator-specific RB management has taken effect for non-FWA users, the average downlink throughput of FWA users will decrease more severely.
Low-frequency TDD	NSA/SA selection based on downlink traffic volume	NSA_SA_DL_SEL_ OPT_SW option of the gNodeBParam.N etworkingOptio nOptSw parameter	<i>NSA-SA Selection Based on User Experience</i>	In RAN sharing with common carrier scenarios, it is recommended that NSA/SA selection based on downlink traffic volume be disabled if operator-specific RB management is enabled and resources in the shared cell are not fully shared. Otherwise, user experience evaluation may be inaccurate, causing user experience to deteriorate.
Low-frequency TDD	CA Performance Boost Based on Multi-dimensional Optimization	CA_PRFM_BOOS T_BSD_ON_MD_ OPT_SW option of the NRDUCellCarrM gmt.CaEnhanced AlgoExtSwitch parameter	<i>Multi-Frequency Smart Aggregation</i>	If percentage-based dynamic RB sharing among operators is enabled, the scheduling priorities of UEs are preferentially modified based on the policies among operators. In this case, CA Performance Boost Based on Multi-dimensional Optimization provides lower gains.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Experience-based smart carrier selection	MULTI_FREQ_SMART_SEL_SW option of the NRCellAlgoSwitch.MultiFreqAlgoSwitch parameter	<i>Multi-Frequency Smart Selection</i>	In RAN sharing with common carrier scenarios, it is recommended that experience-based smart carrier selection be disabled if operator-specific RB management is enabled. Otherwise, user experience evaluation may be inaccurate, causing user experience to deteriorate.
Low-frequency TDD High-frequency TDD	Inter-FR CA (TDD)	INTER_FR_FR1T_OFR2_CA_SW option of the NRDUCellAlgoSwitch.CaAlgoSwitch parameter	<i>Carrier Aggregation</i>	In inter-FR CA scenarios, when operator-specific limitation on the maximum number of available RBs takes effect for the PCell, RB resource estimation is inaccurate during uplink data split. As a result, the average uplink throughput of inter-FR CA UEs fluctuates.
Low-frequency TDD	Intra-base-station DL CoMP	INTRA_GNB_DL_JT_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	- When intra-base-station DL CoMP is enabled together with operator-specific limitation on the maximum number of available RBs, intra-base-station DL CoMP UEs are subject to the limitation on the maximum number of RBs. - When intra-base-station DL CoMP is enabled together with operator-specific limitation on the maximum number of available RBs, if the NRDUCellRes.DIRbMaxRatio parameter is set to a small value (for example, less than 50%) for a secondary operator, UEs of this operator may not be identified as intra-base-station DL CoMP UEs.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Co-MM	INTRA_GNB_SU_COMM_SW option of the NRDUCellCollabServ.MuCellMimoSwitch parameter	<i>Co-MM (Low-Frequency TDD)</i>	When operator-specific limitation on the maximum number of available RBs is enabled together with Co-MM, Co-MM UEs are subject to the limitation on the maximum number of RBs.
Low-frequency TDD	PDSCH MU-MIMO	DL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	When percentage-based dynamic RB sharing among operators and PDSCH MU-MIMO both take effect: The percentage of RBs occupied by each operator can be close to the configured RB percentage when the cell is fully loaded. In addition, when paired UEs that meet certain conditions are preferentially scheduled to guarantee pairing gains, the percentage of RBs occupied by an operator may exceed the configured RB percentage. In this case, the actual percentage of RBs used by an operator fluctuates around the configured value (NRDUCellRes.DIRbAverageRatio or NRDUCellRes.UIRbAverageRatio).

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	When percentage-based dynamic RB sharing among operators and PUSCH MU-MIMO both take effect: The percentage of RBs occupied by each operator can be close to the configured RB percentage when the cell is fully loaded. In addition, when paired UEs that meet certain conditions are preferentially scheduled to guarantee pairing gains, the percentage of RBs occupied by an operator may exceed the configured RB percentage. In this case, the actual percentage of RBs used by an operator fluctuates around the configured value (NRDUCellRes.DIRbAverageRatio or NRDUCellRes.UIRbAverageRatio).
High-frequency TDD	mmWave PUSCH MU-MIMO	UL_MU_MIMO_SW option of the NRDUCellAlgoSwitch.MuMimoSwitch parameter	<i>MIMO (TDD)</i>	In RAN sharing with common carrier scenarios, mmWave PUSCH MU-MIMO does not take effect if operator-specific RB management has been enabled.

4.3.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.3.3.4 Cells

"Operator-specific limitation on the maximum number of available RBs" and "operator-specific guarantee on the minimum number of available RBs" take effect only for low-frequency cells (indicated by **NRDUCell.FrequencyBand** set to values other than **N257**, **N258**, **N260**, and **N261**).

4.3.3.5 Others

None

4.3.4 Operation and Maintenance

4.3.4.1 Data Configuration

4.3.4.1.1 Data Preparation

Table 4-25 describes the parameters used for function activation. No parameters are required for function optimization.

Table 4-25 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
RAN Sharing Algorithm Switch	NRDUCellAlgoSwitch.RanSharingAlgoSwitch	Select the RB_DYNAMIC_SHARING_SW option based on negotiation among operators.
Resource Strategy Type	NRDUCellRes.ResStrategyType	Set this parameter based on the network plan and negotiation among operators.
Resource Usage Object	NRDUCellRes.ResUsageObject	Set this parameter based on the network plan and negotiation among operators.

Parameter Name	Parameter ID	Setting Notes
Downlink RB Average Ratio	NRDUCellRes.DIRbAverageRatio	Set this parameter based on negotiation among operators. It is recommended that the summed average percentages of downlink RBs configured for all operators reach the maximum allowed value of 100%.
Uplink RB Average Ratio	NRDUCellRes.UlRbAverageRatio	Set this parameter based on negotiation among operators. It is recommended that the summed average percentages of uplink RBs configured for all operators reach the maximum allowed value of 100%.
RAN Sharing Algorithm Switch	NRDUCellAlgoSwitch.RanSharingAlgoSwitch	Select the MAX_RB_LIMIT_SW option after negotiation among operators.
Downlink RB Max Ratio	NRDUCellRes.DIRbMaxRatio	It is recommended that the parameter value be greater than 10%. The parameter value must be greater than or equal to the value of NRDUCellRes.DIRbAverageRatio . The number of RBs corresponding to the parameter value must be greater than or equal to the number of RBs in one RBG.
Uplink RB Max Ratio	NRDUCellRes.UlRbMaxRatio	It is recommended that the parameter value be greater than 10%. The parameter value must be greater than or equal to the value of NRDUCellRes.UlRbAverageRatio . The number of RBs corresponding to the parameter value must be greater than or equal to the number of RBs in one RBG.
RAN Sharing Algorithm Switch	NRDUCellAlgoSwitch.RanSharingAlgoSwitch	Select the MIN_RB_GUARANTEE_SW option after negotiation among operators.

Parameter Name	Parameter ID	Setting Notes
Downlink RB Min Ratio	NRDUCellRes.DIRbMinRatio	It is recommended that this parameter be set to a value less than 90%. This parameter value must be less than or equal to the value of NRDUCellRes.DIRbAverageRatio . The sum of NRDUCellRes.DIRbMinRatio configured for all operators must not exceed the maximum value of 100%.
Uplink RB Min Ratio	NRDUCellRes.UlRbMinRatio	It is recommended that this parameter be set to a value less than 90%. This parameter value must be less than or equal to the value of NRDUCellRes.UlRbAverageRatio . The sum of NRDUCellRes.UlRbMinRatio configured for all operators must not exceed the maximum value of 100%.

4.3.4.1.2 Using MML Commands

Before using MML commands, refer to [4.3.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Operator-specific limitation on the maximum number of available RBs in low-frequency scenarios

```
//Adding a resource group and configuring the average percentages of both uplink and downlink RBs and maximum percentages of both uplink and downlink RBs for a cell
ADD NRDUCELLRES: NrDuCellId=1, CellResId=0, ResUsageObject=OPERATOR,
ResStrategyType=MIN_MAX_FOR_TTI_AVG_FOR_MEAN, DIRbAverageRatio=50, UlRbAverageRatio=50,
DIRbMaxRatio=50, UlRbMaxRatio=50;
ADD NRDUCELLRES: NrDuCellId=1, CellResId=1, ResUsageObject=OPERATOR,
ResStrategyType=MIN_MAX_FOR_TTI_AVG_FOR_MEAN, DIRbAverageRatio=50, UlRbAverageRatio=50,
DIRbMaxRatio=50, UlRbMaxRatio=50;
//Associating operators with resource groups
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=0, CellResId=0;
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=1, CellResId=1;
```



```
//Associating a different operator with the same resource group to form an operator group
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=2, CellResId=1;
//Enabling percentage-based dynamic RB sharing among operators
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, RanSharingAlgoSwitch=RB_DYNAMIC_SHARING_SW-1;
//Enabling operator-specific limitation on the maximum number of available RBs, which is mutually
exclusive with operator-specific guarantee on the minimum number of available RBs
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, RanSharingAlgoSwitch=MAX_RB_LIMIT_SW-1;
```

Operator-specific guarantee on the minimum number of available RBs in low-frequency scenarios

```
//Adding a resource group and configuring the average percentages of both uplink and downlink RBs and
minimum percentages of both uplink and downlink RBs for a cell
ADD NRDUCELLRES: NrDuCellId=1, CellResId=0, ResUsageObject=OPERATOR,
ResStrategyType=MEAN_ONLY, DIRbAverageRatio=50, URbAverageRatio=50, DIRbMinRatio=20,
URbMinRatio=20;
ADD NRDUCELLRES: NrDuCellId=1, CellResId=1, ResUsageObject=OPERATOR,
ResStrategyType=MEAN_ONLY, DIRbAverageRatio=50, URbAverageRatio=50, DIRbMinRatio=20,
URbMinRatio=20;
//Associating operators with resource groups
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=0, CellResId=0;
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=1, CellResId=1;
//Associating a different operator with the same resource group to form an operator group
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=2, CellResId=1;
//Enabling percentage-based dynamic RB sharing among operators
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, RanSharingAlgoSwitch=RB_DYNAMIC_SHARING_SW-1;
//Enabling operator-specific guarantee on the minimum number of available RBs, which is mutually
exclusive with operator-specific limitation on the maximum number of available RBs
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, RanSharingAlgoSwitch=MIN_RB_GUARANTEE_SW-1;
```

Percentage-based dynamic RB sharing among operators in high-frequency scenarios

```
//Adding resource groups and configuring the average percentages of both uplink and downlink RBs for a
cell
ADD NRDUCELLRES: NrDuCellId=1, CellResId=0, ResUsageObject=OPERATOR, DIRbAverageRatio=50,
URbAverageRatio=50;
ADD NRDUCELLRES: NrDuCellId=1, CellResId=1, ResUsageObject=OPERATOR, DIRbAverageRatio=50,
URbAverageRatio=50;
//Associating operators with resource groups
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=0, CellResId=0;
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=1, CellResId=1;
//Enabling percentage-based dynamic RB sharing among operators
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, RanSharingAlgoSwitch=RB_DYNAMIC_SHARING_SW-1;
```

Optimization Command Examples

N/A

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling percentage-based dynamic RB sharing among operators
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, RanSharingAlgoSwitch=RB_DYNAMIC_SHARING_SW-0;
//Disabling operator-specific limitation on the maximum number of available RBs
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, RanSharingAlgoSwitch=MAX_RB_LIMIT_SW-0;
//Disabling operator-specific guarantee on the minimum number of available RBs
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, RanSharingAlgoSwitch=MIN_RB_GUARANTEE_SW-0;
```

4.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.3.4.2 Activation Verification

If multiple operators are configured as a resource group (that is, they form an alliance with each other), the description about operators is also applicable to the resource group.

Percentage-based Dynamic RB Sharing Among Operators

Prerequisites:

- The gNodeB is shared between two operators (operators A and B) in RAN sharing with common carrier. The basic networking for RAN sharing with common carrier has been configured.
- **NRDUCellRes.DIRbAverageRatio** and **NRDUCellRes.UlRbAverageRatio** are set to **50** for both operators A and B.

Expected result:

If there are sufficient UEs for each operator and the sum of PRB usages of all operators is equal to 100%, the percentage of RBs used by each operator will approximate to the configured percentage (50% for both operators A and B), which can be verified by checking the values of counters in [Table 4-26](#).

If there are insufficient UEs for any operator or the sum of PRB usages of all operators is not equal to 100%, the percentage of RBs used by each operator will deviate from the configured percentage of RBs to some extent. Further, this deviation will be significant when the configured percentages of RBs differ greatly between operators.

 NOTE

- With dynamic RB sharing, the gNodeB adjusts the scheduling priorities of different operators in each transmission time interval (TTI) to ensure that the resource consumption of each operator approximates to the configured percentage. As such, there may be differences between the counter values and the configured values.
- Percentage of uplink RBs used by an operator = $(N.PRBUUsedOwn.UL.PLMN + N.PRBUUsedOther.UL.PLMN) \text{ for the operator} / (N.PRBUUsedOwn.UL.PLMN + N.PRBUUsedOther.UL.PLMN) \text{ for all operators}$
- Percentage of downlink RBs used by an operator = $(N.PRBUUsedOther.DL.PLMN + N.PRBUUsedOwn.DL.PLMN) \text{ for the operator} / (N.PRBUUsedOther.DL.PLMN + N.PRBUUsedOwn.DL.PLMN) \text{ for all operators}$
- Percentage of RBs used by a resource group = Sum of percentages of RBs used by all operators in the resource group

Table 4-26 Counters

Counter ID	Counter Name
1911822866	N.PRBUUsedOther.UL.PLMN

Counter ID	Counter Name
1911822867	N.PRBUUsedOwn.UL.PLMN
1911822871	N.PRBUUsedOther.DL.PLMN
1911822872	N.PRBUUsedOwn.DL.PLMN
The preceding counters apply only in network sharing scenarios. Msg5 carries the serving PLMN ID of a UE. Before the UE sends Msg5 to the gNodeB, the gNodeB does not know the serving PLMN ID of the UE. Therefore, uplink RBs for transmitting Msg5 and Msg3 are not counted into N.PRBUUsedOther.UL.PLMN and N.PRBUUsedOwn.UL.PLMN. Overheads of SSB transmissions on the downlink common channel cannot be distinguished between operators. Therefore, overheads of SSB transmissions on the downlink common channel and the downlink RBs for transmitting Msg2 and Msg4 to UEs are not counted into N.PRBUUsedOther.DL.PLMN and N.PRBUUsedOwn.DL.PLMN.	

Prerequisites:

- The gNodeB is shared between two operators (operators A and B) in RAN sharing with common carrier. The basic networking for RAN sharing with common carrier has been configured.
- **NRDUCellRes.DIRbAverageRatio** is set to **20** and **40** for operators A and B, respectively. **NRDUCellRes.UIRbAverageRatio** is set to **20** and **40** for operators A and B, respectively.

Expected result:

If the total RBs required by operators A and B do not exceed 100% of the RBs configured for them, RBs are allocated to meet the actual needs of both operators.

If the total RBs required by operators A and B exceed 100% of the RBs configured for them, the average RB percentages (specified by **NRDUCellRes.DIRbAverageRatio** and **NRDUCellRes.UIRbAverageRatio**) configured for both operators are preferentially assured. After that, the remaining RBs are allocated to or preempted by the operators, depending on their weights of average RB percentages configured for them.

Operator-Specific Limitation on the Maximum Number of Available RBs

Prerequisites:

- The gNodeB is shared between two operators (operators A and B) in RAN sharing with common carrier. The basic networking for RAN sharing with common carrier has been configured.
- **NRDUCellRes.DIRbMaxRatio** is set to **60** for operator A and **50** for operator B. **NRDUCellRes.UIRbMaxRatio** is set to **60** for operator A and **50** for operator B.

Expected result:

The percentage of RBs used by each operator does not exceed the configured percentage (60% for operator A and 50% for operator B), which can be verified by checking the values of counters in [Table 4-26](#).

Operator-Specific Guarantee on the Minimum Number of Available RBs

Prerequisites:

- The gNodeB is shared between two operators (operators A and B) in RAN sharing with common carrier. The basic networking for RAN sharing with common carrier has been configured.
- **NRDUCellRes.DIRbMinRatio** and **NRDUCellRes.UlRbMinRatio** are set to 20 for both operators A and B.

Expected result:

If operator A has sufficient UEs running services but operator B does not have UEs running services, operator A does not occupy the RBs configured for operator B, which means that the percentage of RBs used by operator A does not exceed 80%. This can be verified by checking the values of counters in [Table 4-26](#).

NOTE

The function of operator-specific guarantee on the minimum number of available RBs uses the mean-type control policy, which ensures that the minimum number of available RBs for an operator is greater than or equal to the configured minimum number of RBs within a period of time. As such, there may be differences between the counter values and the configured values.

4.3.4.3 Network Monitoring

Check the counters in [Table 4-26](#) to determine whether RBs are shared as expected.

In addition, the total number of used uplink and downlink RBs can be monitored in real time.

- Total number of used uplink RBs = **N.PRBUseOther.UL.PLMN** + **N.PRBUseOwn.UL.PLMN**
- Total number of used downlink RBs = **N.PRBUseOther.DL.PLMN** + **N.PRBUseOwn.DL.PLMN**

4.4 Operator-Specific Roaming Fairness

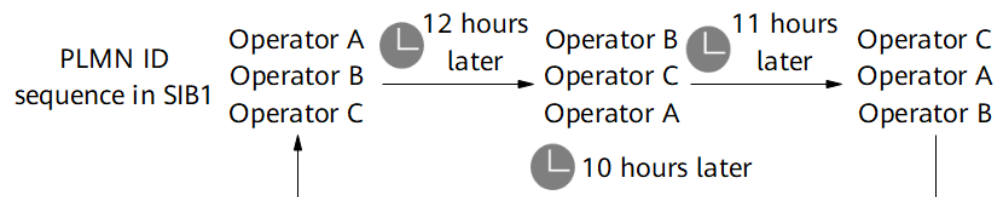
4.4.1 Principles

According to section 4.4 "PLMN selection process" in 3GPP TS 23.122 V15.6.0, a roaming UE randomly selects a PLMN ID in the absence of a prominent PLMN (for example, in terms of signal quality). In such cases, a third-party UE is more likely to select the top entry in the PLMN ID list during roaming to a common-carrier sharing cell, particularly when it is at an airport or harbor. To achieve service provisioning fairness among operators to which a common-carrier sharing cell belongs, PLMN ID round robin can be enabled by selecting the **PLMN_ROUND_ROBIN_SW** option of the

NRDUCellAlgoSwitch.RanSharingAlgoSwitch parameter. The following descriptions assume that operators A, B, and C share a cell.

- If the **PLMN_ROUND_ROBIN_SW** option is deselected, the cell broadcasts a list of PLMN IDs in the following fixed sequence:
 - Operator A
 - Operator B
 - Operator C
- If the **PLMN_ROUND_ROBIN_SW** option is selected, the cell broadcasts a list of operators' PLMN IDs in round robin mode, which rotates the top PLMN ID in the list. This PLMN ID round robin period, also known as the duration for which a PLMN ID stays at the top, is specified by the **NRDUCellOp.PlmnRoundRobinPeriod** parameter. [Figure 4-13](#) shows the sequence of PLMN IDs broadcast in SIB1 when the **NRDUCellOp.PlmnRoundRobinPeriod** parameter is set to 12, 11, and 10 for operators A, B, and C, respectively.

Figure 4-13 PLMN ID round robin



NOTE

- If intra-RAT ANR is enabled on the gNodeB and PLMN ID round robin takes effect, the **NR_NCELL_PLMN_RR_MODE_SW** option of the **NRCellAnr.AnrModeSwitch** parameter must be selected so that the gNodeB can obtain a list of PLMN IDs of the primary and secondary operators to which an external NR cell belongs from the MAE-Access. This is because the primary operator is not always placed at the top of the PLMN ID list after PLMN ID round robin takes effect, in which case the serving cell cannot distinguish between the primary and secondary operators based on the PLMN ID list reported from UEs. As a result, there is a possibility that errors occur when neighbor relationships are added based on cell global identifiers (CGIs). For details, see *ANR*.
- If inter-RAT ANR with NG-RAN is enabled on the eNodeB and PLMN ID round robin takes effect, the **PLMN_RR_INDICATION** option of the **NrNFreq.AggregationAttribute** parameter must be selected so that the eNodeB can obtain a list of PLMN IDs of the primary and secondary operators to which an external NR cell belongs from the MAE-Access. This is because the primary operator is not always placed at the top of the PLMN ID list after PLMN ID round robin takes effect, in which case the serving cell cannot distinguish between the primary and secondary operators based on the PLMN ID list reported from UEs. As a result, there is a possibility that errors occur when neighbor relationships are added based on cell global identifiers (CGIs). For details, see *ANR Management in eRAN Feature Documentation*.

4.4.2 Network Analysis

4.4.2.1 Benefits

RAN sharing with common carrier ensures the fairness of service provisioning among operators to which a common-carrier sharing cell belongs.

4.4.2.2 Impacts

The impacts between this function and other functions are described in [4.4.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.4.3 Requirements

4.4.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021211	Multi-Operator Sharing	NR0S00MULS00	per Cell
For the license control item NR0S00MULS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.4.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.4.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with common carrier	gNBSharingMode.gNBMultiOpSharingMode	<i>Multi-Operator Sharing</i>	Operator-specific roaming fairness requires RAN sharing with common carrier to be deployed on the live network.

4.4.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Operator-specific NCI configuration	gNBSharingMode.MultiplNCISwitch	<i>Multi-Operator Sharing</i>	Operator-specific roaming fairness and operator-specific NCI configuration are mutually exclusive. As such, they cannot both be enabled.

4.4.3.2.3 Function Impacts

None

4.4.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.4.3.4 Others

None

4.4.4 Operation and Maintenance

4.4.4.1 Data Configuration

4.4.4.1.1 Data Preparation

[Table 4-27](#) describes the parameters used for function activation. No parameters are required for function optimization.

Table 4-27 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
RAN Sharing Algorithm Switch	NRDUCellAlgoSwitch.RanSharingAlgoSwitch	To enable PLMN ID round robin, select the PLMN_ROUND_ROBIN_SW option.
PLMN Round Robin Period	NRDUCellOp.PlmnRoundRobinPeriod	Set this parameter based on negotiation among operators.

4.4.4.1.2 Using MML Commands

Before using MML commands, refer to [4.4.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Setting the PLMN ID round robin period for operators
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=0, PlmnRoundRobinPeriod=12;
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=1, PlmnRoundRobinPeriod=11;
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=2, PlmnRoundRobinPeriod=10;
//Enabling PLMN ID round robin
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, RanSharingAlgoSwitch=PLMN_ROUND_ROBIN_SW-1;
```


Optimization Command Examples

N/A

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling PLMN ID round robin  
MOD NRDUCELLALGOSWITCH: NrDuCellId=1, RanSharingAlgoSwitch=PLMN_ROUND_ROBIN_SW-0;
```

4.4.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.4.4.2 Activation Verification

Prerequisites

- The basic networking for multi-operator sharing has been configured.
- PLMN ID round robin in multi-operator sharing has been enabled.
- The PLMN ID round robin period in multi-operator sharing has been correctly configured.

Expected Result

If roaming UEs correctly access the networks of the corresponding PLMN IDs within each PLMN ID round robin period, this function has been activated.

4.4.4.3 Network Monitoring

None

4.5 Operator-Specific Dedicated Cell Reselection Priority Management

4.5.1 Principles

In RAN sharing with common carrier, dedicated cell reselection priorities of frequencies can be configured for operators based on the PLMN IDs of the serving cell using the **NRCeCellOpPolicy** and **gNBFreqPriorityGroup** MOs (however, the frequencies configured in the **gNBFreqPriorityGroup** MO must have been configured in the **NRCeCellFreqRelation** or **NRDUCell** MO). In this way, this configuration enables camping policies for RRC_IDLE UEs to be operator-specific.

- The **NRCellOpPolicy** MO defines parameters related to the operator-specific policies of an NR cell.
The **NRCellOpPolicy.NrCellId**, **NRCellOpPolicy.OperatorId**, and **NRCellOpPolicy.gNBFreqPriorityGroupId** parameters can be used to associate an NR cell of an operator with the ID of a frequency priority group of the gNodeB. The RedCap frequency priority policy used in RedCap scenarios is configured by the **NRCellOpPolicy.RedCapGnbFreqPriGroupId** parameter. For details, see *RedCap*.
- The **gNBFreqPriorityGroup** MO is used to configure dedicated cell reselection priorities for UEs.

Dedicated cell reselection priorities of the same frequency can be different among operators. [Table 4-28](#) provides an example of cell reselection priority configurations for operators A and B in the case of RAN sharing between two operators. It is recommended that the dedicated cell reselection priority be configured for all neighboring frequencies corresponding to the serving PLMN IDs of UEs. The neighboring frequencies for which dedicated cell reselection priorities are not configured will not be delivered to UEs, and reselection to cells working on these frequencies will not be performed for UEs.

Table 4-28 Example of dedicated cell reselection priority configurations based on the PLMN IDs of the serving cell

Operator	Frequency	Priority (in Descending Order)	Dedicated Cell Reselection Priority Setting
Operator A	Serving frequency (NR frequency)	5	gNBFreqPriorityGroup.RatType =NR gNBFreqPriorityGroup.CellReselPri = 5
	Neighboring NR frequency	4	gNBFreqPriorityGroup.RatType =NR gNBFreqPriorityGroup.CellReselPri = 4
	E-UTRAN frequency	3	gNBFreqPriorityGroup.RatType =EUTRAN gNBFreqPriorityGroup.CellReselPri = 3
Operator B	E-UTRAN frequency	6	gNBFreqPriorityGroup.RatType =EUTRAN gNBFreqPriorityGroup.CellReselPri = 6
	Neighboring NR frequency	5	gNBFreqPriorityGroup.RatType =NR gNBFreqPriorityGroup.CellReselPri = 5

Operator	Frequency	Priority (in Descending Order)	Dedicated Cell Reselection Priority Setting
	Serving frequency (NR frequency)	4	gNBFreqPriorityGroup.RatType=NR gNBFreqPriorityGroup.CellReselPri= 4

In low-frequency scenarios, if the **CA_UE_FREQ_STEER_SW** option of the **NRCellAlgoSwitch.MultiFreqAlgoSwitch** parameter is selected in the serving cell for a CA-capable UE, the gNodeB will determine whether to filter out the neighboring frequencies based on CA capabilities of the UE, the frequency of the serving cell on which the UE camps, and network continuity. The frequencies that are not filtered out are delivered to the UE, and the UE can reselect to the corresponding frequency based on the dedicated reselection priorities. For details about frequency filtering scenarios and recommended scenarios for CA UE frequency steering, see *SA Mobility Management in Connected Mode*.

NOTE

This document describes only the content exclusive to operator-specific dedicated cell reselection priority management compared with basic cell reselection functions. For details about basic cell reselection functions, see *SA Mobility Management in Idle Mode* and *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*.

4.5.2 Network Analysis

4.5.2.1 Benefits

In RAN sharing with common carrier, different operators can apply different camping policies to RRC_IDLE UEs.

4.5.2.2 Impacts

The impacts between this function and other functions are described in [4.5.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.5.3 Requirements

4.5.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021211	Multi-Operator Sharing	NR0S00MULS00	per Cell

Feature ID	Feature Name	Model	Sales Unit
For the license control item NR0S00MULS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.5.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.5.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Basic networking for RAN sharing with common carrier	gNBSharingMode.gNBMultiOpSharingMode	<i>Multi-Operator Sharing</i>	Operator-specific dedicated cell reselection priority management requires RAN sharing with common carrier to be deployed on the live network.

4.5.3.2.2 Mutually Exclusive Functions

None

4.5.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Differentiated carrier selection for FWA users	NRCellQciBearer.gNBFreqPriorityGroupId	<i>FWA</i>	In SA networking, operator-specific dedicated cell reselection priority management does not take effect on users with "differentiated carrier selection for FWA users" in effect.
Low-frequency TDD	Inter-frequency handover policy for voice-service-capable UEs	VOICE_INTER_FREQ_HO_POLICY_SW option of the NRCellFreqRelation.AggregationAttribute parameter	<i>SA Mobility Management in Connected Mode</i>	If the inter-frequency handover policy for voice service-capable UEs is enabled for a frequency, the gNodeB will not send an RRCRelease message containing this frequency to a UE that does not support VoNR on its home network in roaming scenarios, and the UE will not reselect this frequency based on operator-specific dedicated cell reselection priorities.
Low-frequency TDD	Voice-capability-based inter-frequency handover back to the HPLMN	VOICE_CAPB_MOB_TO_HPLMN_SW option of the gNBOperator.HplmnAlgoSwitch parameter	<i>SA Mobility Management in Connected Mode</i>	If the voice-capability-based handover back to the HPLMN function is enabled for the home network on the visited network's gNodeB, the gNodeB will send an RRCRelease message to a UE that does not support VoNR on its home network in roaming scenarios. The message contains the frequencies for which the gNBFreqPriorityGroup.RoamingUeHplmnFreqFlag parameter is set to TRUE in the frequency list group bound to the UE on the gNodeB side. In this case, the UE can reselect the corresponding frequency based on operator-specific dedicated cell reselection priorities.

4.5.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.5.3.4 Others

None

4.5.4 Operation and Maintenance

4.5.4.1 Data Configuration

4.5.4.1.1 Data Preparation

Table 4-29 describes the parameters used for function activation. No parameters are required for function optimization.

Table 4-29 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellOpPolicy.NrCellId	Set this parameter based on the network plan.
Operator ID	NRCellOpPolicy.OperatorId	Set this parameter based on the network plan.
gNodeB Frequency Priority Group ID	NRCellOpPolicy.gNBFrequencyPriorityGroupId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
gNodeB Frequency Priority Group ID	gNBFreqPriorityGroup.gNBFreqPriorityGroupId	Set this parameter based on the network plan.
Frequency Index	gNBFreqPriorityGroup.FreqIndex	Set this parameter based on the network plan.
RAT Type	gNBFreqPriorityGroup.RatType	- To configure the dedicated cell reselection priority of an E-UTRAN frequency, set this parameter to EUTRAN. - To configure the dedicated cell reselection priority of an NR frequency, set this parameter to NR.
Downlink EARFCN	gNBFreqPriorityGroup.DlEarfcn	Set this parameter based on the network plan.
SSB Frequency Position	gNBFreqPriorityGroup.SsbFreqPos	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	gNBFreqPriorityGroup.SsbDescMethod	Set this parameter to SSB_DESC_TYPE_GSCN .
Cell Reselection Priority	gNBFreqPriorityGroup.CellReselPri	Set this parameter based on the network plan.
Cell Reselection Sub-Priority	gNBFreqPriorityGroup.CellReselSubPri	Set this parameter based on the network plan.

4.5.4.1.2 Using MML Commands

Before using MML commands, refer to [4.5.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring the dedicated cell reselection priorities of E-UTRAN frequencies for UEs
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=0, RatType=EUTRAN,
DlEarfcn=1444, CellReselPri=1, CellReselSubPri=0DOT2;
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=1, RatType=EUTRAN,
```

```

DIEarfcn=1888, CellReselPri=1, CellReselSubPri=0DOT2;
//Configuring the dedicated cell reselection priorities of NR frequencies for UEs
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=2, RatType=NR, SsbFreqPos=26639,
CellReselPri=5, CellReselSubPri=0DOT2;
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=3, RatType=NR, SsbFreqPos=26638,
CellReselPri=4, CellReselSubPri=0DOT2;
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=4, RatType=NR, SsbFreqPos=26639,
CellReselPri=5, CellReselSubPri=0DOT2;
ADD GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=5, RatType=NR, SsbFreqPos=26637,
CellReselPri=3, CellReselSubPri=0DOT2;
//Configuring parameters related to operator-specific policies of NR cells
ADD NRCELLOPPOLICY: NrCellId=1, OperatorId=0, gNBFreqPriorityGroupId=0;
ADD NRCELLOPPOLICY: NrCellId=1, OperatorId=1, gNBFreqPriorityGroupId=1;
ADD NRCELLOPPOLICY: NrCellId=2, OperatorId=0, gNBFreqPriorityGroupId=0;
ADD NRCELLOPPOLICY: NrCellId=2, OperatorId=1, gNBFreqPriorityGroupId=1;

```

Optimization Command Examples

N/A

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```

//Configuring the invalid value for the parameters related to operator-specific cell reselection policies of NR
cells
MOD NRCELLOPPOLICY: NrCellId=1, OperatorId=0, gNBFreqPriorityGroupId=255;
MOD NRCELLOPPOLICY: NrCellId=1, OperatorId=1, gNBFreqPriorityGroupId=255;
MOD NRCELLOPPOLICY: NrCellId=2, OperatorId=0, gNBFreqPriorityGroupId=255;
MOD NRCELLOPPOLICY: NrCellId=2, OperatorId=1, gNBFreqPriorityGroupId=255;
//Removing the redundant configurations of the frequency priority groups of the gNodeB
RMV GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=0;
RMV GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=1;
RMV GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=2;
RMV GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=0, FreqIndex=3;
RMV GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=4;
RMV GNBREQPRIORITYGROUP: gNBFreqPriorityGroupId=1, FreqIndex=5;

```

4.5.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.5.4.2 Activation Verification

Prerequisites

- The basic networking for multi-operator sharing has been configured.
- The frequency priorities have been configured for the serving PLMNs of UEs.
- The frequencies supported by UEs are present in the frequency priority group configured for the corresponding operators.

Expected Result

This function has been activated if the RRCRelease message sent from the gNodeB to a UE contains the *CellReselectionPriorities* IE, and the frequencies and

frequency priorities supported by the UE are delivered based on the configurations.

4.5.4.3 Network Monitoring

Check the value of the **N.User.RRCCConn.Avg.Operator** counter to obtain the operator-specific number of UEs in cells working on different frequencies. Check whether UE distribution matches frequency priority configurations. Specifically, the higher the dedicated cell reselection priority of a frequency, the larger the expected number of UEs in the cell working on that frequency.

4.6 Operator-Specific NCI Configuration

4.6.1 Principles

An NR Cell Identity (NCI) is 36 bits long, and is composed of a gNodeB ID (22 to 32 bits long) and a cell ID (4 to 14 bits long). An operator uses the same gNodeB ID length across an NR system. In multi-operator sharing scenarios, cellIdentity values (indicating NCI information) in the PLMN-IdentityList IE broadcast in SIB1 can be operator-specific. In these scenarios, cellIdentity and NCI have the same settings of **gNBOperator.gNBId** and **NRDUCellOp.CellId**.

Operator-specific NCI Configuration

On the gNodeB side: In multi-operator sharing, the operator-specific NCI configuration function can be enabled by setting the gNodeB-level parameter **gNBSharingMode.MultipleNCISwitch** to **ON**. This function takes effect after the corresponding application is manually restarted by running the **RST APP** command. [Table 4-30](#) describes the parameters related to this function.

Table 4-30 Parameters on the gNodeB side

Parameter Type	Parameter Name	Configuration Information
Local cell parameter	gNBOperator.gNBId	gNodeB ID of the local cell
Local cell parameter	gNBOperator.gNBIdLength	Length of the gNodeB ID of the local cell
Local cell parameter	gNBOperator.UserLabel	User label
Local cell parameter	NRDUCellOp.CellId	Cell ID of the local cell
Local cell parameter	NRDUCellOp.CellPrimaryOpFlag	A flag that uniquely identifies the primary operator of a cell ^a

Parameter Type	Parameter Name	Configuration Information
External neighboring cell parameter	NRExternalNCell.gNBId	gNodeB ID of an external neighboring cell
External neighboring cell parameter	NRExternalNCell.CellId	Cell ID of an external neighboring cell
External neighboring cell parameter	NRExternalNCellPlmn.SharedPlmnGnbId	gNodeB ID of the secondary operator in multi-operator sharing
External neighboring cell parameter	NRExternalNCellPlmn.SharedPlmnCellId	Cell ID of the secondary operator in multi-operator sharing
External neighboring cell parameter	gNBExternalPlmn.gNBIdLength	(Preferred) Length of the gNodeB ID specific to the PLMN if an external cell's operator-specific PLMN information is not present in the gNBOperator MO configurations of the local gNodeB
External neighboring cell parameter	gNodeBFunction.gNBIdLength	(Alternative) Length of the gNodeB ID specific to the PLMN if an external cell's operator-specific PLMN information is not present in the gNBOperator MO configurations of the local gNodeB
External neighboring cell parameter	gNBOperator.gNBIdLength	Length of the gNodeB ID specific to the PLMN if an external cell's operator-specific PLMN information is already present in the gNBOperator MO configurations of the local gNodeB
a: The NRDUCellOp.CellId parameter for the primary operator (specified by the NRDUCellOp.CellPrimaryOpFlag parameter) of a cell must be set to the same value as the NRDUCell.CellId parameter. The primary operator of a cell (specified by the NRDUCellOp.CellPrimaryOpFlag parameter) can be different from the primary operator of the gNodeB (specified by the gNBOperator.OperatorType parameter). However, it is recommended that these primary operators be the same.		

On the eNodeB side: When the operator-specific NCI configuration function takes effect on the gNodeB, gNodeB information (including the gNodeB ID, cell ID, and PLMN ID) needs to be specified on the eNodeB in LTE-to-NR procedures (such as LTE-to-NR handover and EN-DC X2 self-setup in NSA networking). To ensure

normal operation of LTE-to-NR services, the following parameters listed in [Table 4-31](#) must be reconfigured on the eNodeB:

Table 4-31 Parameters on the eNodeB side

Parameter Name	Configuration Information
NrExternalCell.GnodebId	gNodeB ID on the gNodeB side
NrExternalCell.CellId	Cell ID on the gNodeB side
NrExternalCellPlmn.SharedPlmnGnodebId	gNodeB ID of the secondary operator in multi-operator sharing
NrExternalCellPlmn.SharedPlmnCellId	Cell ID of the secondary operator in multi-operator sharing
NCellPlmnList.gNBIdLength	Length of the gNodeB ID
NbrGnodebPolicy.LteToNrPlmnSelectionMode	PLMN ID filling method in LTE-to-NR procedures: • If this parameter is set to NOT_CFG, the filling method takes the value of the EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode parameter. • If this parameter is set to a value other than NOT_CFG, the filling method takes the value of the NbrGnodebPolicy.LteToNrPlmnSelectionMode parameter.
EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode	PLMN ID filling method in LTE-to-NR procedures: • If this parameter is set to PRIMARY_PLMN_ID, the eNodeB fills in the primary-operator PLMN ID of the target NR cell. • If this parameter is set to SELECTED_PLMN_ID, the eNodeB fills in the PLMN ID selected by the UE that intends to be served by the target NR cell. • If this parameter is set to HYBRID_PLMN_ID, the eNodeB fills in the primary-operator PLMN ID of the target NR cell when different PLMNs of the same NR cell use the same NCI, but the PLMN ID selected by the UE that intends to be served by the target NR cell in other scenarios.

 NOTE

If the neighboring gNodeB fills in the PLMN ID using a method different from that specified by the **EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode** parameter, the **NbrGnodebPolicy.LteToNrPlmnSelectionMode** parameter can be configured for this gNodeB to specify how to fill in the PLMN ID in LTE-to-NR procedures. The **NbrGnodebPolicy.LteToNrPlmnSelectionMode** parameter has a higher priority than the **EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode** parameter. Specifically,

With NCIs being operator-specific, each operator's own gNodeB ID, cell ID, and PLMN ID are used over each interface. The details are as follows:

- The *Global gNB ID* IE over the NG interface carries the PLMN ID and gNodeB ID of each operator. The messages involved are as follows:
NG SETUP REQUEST, RAN CONFIGURATION UPDATE, UPLINK RAN CONFIGURATION TRANSFER, DOWNLINK RAN CONFIGURATION TRANSFER, PWS RESTART INDICATION, PWS FAILURE INDICATION, SON Configuration Transfer, and HANDOVER REQUIRED
- The *Global gNB ID* IE over the Xn interface carries the PLMN ID and gNodeB ID of each operator. The messages involved are as follows:
XN SETUP REQUEST, XN SETUP RESPONSE, XN REMOVAL REQUEST, XN REMOVAL RESPONSE, and HANDOVER REQUEST
- The *Global en-gNB ID* IE over the X2 interface carries the PLMN ID and gNodeB ID of each operator. The messages involved are as follows:
EN-DC X2 SETUP REQUEST, EN-DC X2 SETUP RESPONSE, EN-DC X2 REMOVAL REQUEST, and EN-DC X2 REMOVAL REQUEST
- Operators' PLMN IDs and cell IDs in the NCGIs exchanged over each standard interface are set to their respective PLMN IDs and cell IDs.

NOTE

After the operator-specific NCI configuration function is enabled, the RETRIEVE UE CONTEXT REQUEST message over the Xn interface carries target cell information (including the PLMN ID, gNodeB ID, cell ID, PCI, and DL ARFCN) due to the introduction of a Huawei proprietary protocol. This enables the source base station to quickly detect the corresponding neighboring cell.

If NCIs are operator-specific in the peer non-Huawei gNodeB interconnected with the local Huawei gNodeB, the latter may fill in an incorrect target gNodeB ID for an outgoing handover intended for a UE belonging to a secondary operator. In this case, the **NRExternalgNB** MO can be configured to solve the compatibility issue between the peer non-Huawei gNodeB and the local Huawei gNodeB.

NRExternalgNB.Mcc, **NRExternalgNB.Mnc**, and **NRExternalgNB.gNBId** indicate the primary-operator MCC, MNC, and gNodeB ID of the peer non-Huawei gNodeB, respectively. **NRExternalgNB.PlmnPolicy** indicates the PLMN selection policy applied to the primary and secondary operators for NG-C link setup between the peer non-Huawei gNodeB and the 5GC.

- If **NRExternalgNB.PlmnPolicy** is set to **PRIMARY_PLMN_PREFER**, the primary-operator PLMN ID and gNodeB ID are used for NG-C link setup with the 5GC for the primary and secondary operators.
- If **NRExternalgNB.PlmnPolicy** is set to **SELECTED_PLMN_PREFER**, the primary and secondary operators' own PLMN IDs and gNodeB IDs are used for NG-C link setup with the 5GC for the primary and secondary operators.

Operator-specific NCI Configuration Enhancement

The operator-specific NCI configuration enhancement function can be enabled by setting the gNodeB-level parameter **gNBSharingMode.MultipleNciEnhanceSw** to **ON**. This function takes effect after the corresponding application is manually restarted by running the **RST APP** command. Before this function is enabled, operator-specific NCI configuration needs to be enabled (by setting **gNBSharingMode.MultipleNCISwitch** to **ON**).

When the operator-specific NCI configuration function is enabled and the **gNBSharingMode.MultipleNciEnhanceSw** parameter is set to **OFF**:

- The 3GPP-defined NCI (corresponding to **gNBOperator.gNBId** and **NRDUCellOp.CellId**) must be different for operators sharing a cell.
- The primary operator's information must be specified in the **NR CGI, TAC, RANAC, Connectivity Support**, or **Configured TAC Indication** field in the *Served Cell Information NR* IE included in the following messages: XN SETUP REQUEST, XN SETUP RESPONSE, NG-RAN NODE CONFIGURATION UPDATE, and NG-RAN NODE CONFIGURATION UPDATE ACKNOWLEDGE.

When the operator-specific NCI configuration function is enabled and the **gNBSharingMode.MultipleNciEnhanceSw** parameter is set to **ON**:

- The protocol-defined NCI (corresponding to **gNBOperator.gNBId** and **NRDUCellOp.CellId**) can be identical for operators sharing a cell.
- Operators' respective information is specified in the **NR CGI, TAC, RANAC, Connectivity Support**, or **Configured TAC Indication** field in the *Served Cell Information NR* IE included in the following messages: XN SETUP REQUEST, XN SETUP RESPONSE, NG-RAN NODE CONFIGURATION UPDATE, and NG-RAN NODE CONFIGURATION UPDATE ACKNOWLEDGE.

The **gNBSharingMode.MultipleNCISwitch** and **gNBSharingMode.MultipleNciEnhanceSw** parameters can be configured on the gNodeB in one of the following ways:

- Configuration 1: **gNBSharingMode.MultipleNCISwitch** is set to **OFF**.
- Configuration 2: **gNBSharingMode.MultipleNCISwitch** and **gNBSharingMode.MultipleNciEnhanceSw** are set to **ON**.
- Configuration 3: **gNBSharingMode.MultipleNCISwitch** is set to **ON** and **gNBSharingMode.MultipleNciEnhanceSw** is set to **OFF**.

If the **gNBSharingMode.MultipleNCISwitch** and **gNBSharingMode.MultipleNciEnhanceSw** parameters are configured differently between gNodeBs and neighbor relationships are configured between them, the **NRExternalgNB.PlmnPolicy** parameter must be configured as follows:

- When neighbor relationships are configured between the gNodeB using configuration 1 and the gNodeB using configuration 2, set the **NRExternalgNB.PlmnPolicy** parameter on both gNodeBs as follows: On the gNodeB using configuration 1, set this parameter to **SELECTED_PLMN_PREFER** for the gNodeB using configuration 2; on the gNodeB using configuration 2, set this parameter to **PRIMARY_PLMN_PREFER** for the gNodeB using configuration 1.
- When neighbor relationships are configured between the gNodeB using configuration 2 and the gNodeB using configuration 3 and operators sharing the former gNodeB have the same NCI, set the **NRExternalgNB.PlmnPolicy** parameter on the gNodeB using configuration 3 to **SELECTED_PLMN_PREFER** for the gNodeB using configuration 2.
- When neighbor relationships are configured between the gNodeB using configuration 1 and the gNodeB using configuration 3, the **NRExternalgNB.PlmnPolicy** parameter does not need to be configured.

During Xn setup and Xn update procedures, gNodeBs can exchange the configurations of **gNBSharingMode.MultipleNCISwitch** and **gNBSharingMode.MultipleNciEnhanceSw**.

If there is an Xn interface between neighboring gNodeBs of specific configurations (for example, between configuration-1 gNodeB and configuration-2 gNodeB or between configuration-2 gNodeB and configuration-3 gNodeB) and the **EXTERNAL_GNB_SELF_CONFIG_SW** option of the **gNodeBAlgo.SelfConfigSwitch** parameter is selected, the local gNodeB can adaptively generate the **NRExternalgNB** MO specific to the neighboring gNodeB based on their **gNBSharingMode.MultipleNCISwitch** and **gNBSharingMode.MultipleNciEnhanceSw** configurations, reducing the workload on neighboring gNodeB configuration.

 NOTE

- When the operator-specific NCI configuration function is enabled, configurations of the following objects must be different among operators during Xn, X2, and NG interface configuration: Xn (**gNBCUXnOp.gNBCuXnId**), X2 (**gNBCUX2Op.gNBCuX2Id**), and NG (**gNBCUNgOp.gNBCuNgId**). After this function is enabled, transmission reconstruction may lead to redundant SCTP links. As such, it is recommended that Xn self-removal and X2 self-removal be enabled. For details about X2 self-removal and Xn self-removal, see *X2 and S1 Self-Management in NSA Networking* and *NG and Xn Self-Management*, respectively.
- When the operator-specific NCI configuration function is enabled, the eNodeB, gNodeB, and MAE-Access must be upgraded to a version that supports this function. If not, the eNodeB and gNodeB of an earlier version cannot interconnect with the gNodeB enabled with this function, and neighboring base stations cannot be identified based on neighboring cell configurations on the MAE-Access.

4.6.2 Network Analysis

4.6.2.1 Benefits

gNodeB IDs and cell IDs can be operator-specific.

4.6.2.2 Impacts

The impacts between this function and other functions are described in [4.6.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.6.3 Requirements

4.6.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021211	Multi-Operator Sharing	NR0S00MULS00	per Cell

Feature ID	Feature Name	Model	Sales Unit
For the license control item NR0S00MULS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.6.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.6.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with common carrier	gNBSharingMode .gNBMultiOpSharingMode set to SHARED_FREQ	<i>Multi-Operator Sharing</i>	Operator-specific NCI configuration requires RAN sharing with common carrier or RAN sharing with dedicated carrier to be deployed on the live network.
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with dedicated carrier	gNBSharingMode .gNBMultiOpSharingMode set to SEPARATED_FREQ	<i>Multi-Operator Sharing</i>	

4.6.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Operator-specific roaming fairness	NRDUCellAlgoSwitch.RanSharingAlgoSwitch	<i>Multi-Operator Sharing</i>	Operator-specific roaming fairness and operator-specific NCI configuration are mutually exclusive. They cannot both be enabled.
Low-frequency TDD	Operator-specific eXn interface configuration	gNBDUExnOp.OperatorId , gNBDUExnOp.gNBduExnId , and gNBDUExnOp.UpEpGroupId	<i>Multi-Operator Sharing</i>	Operator-specific eXn interface configuration and operator-specific NCI configuration are mutually exclusive. They cannot both be enabled.
Low-frequency TDD High-frequency TDD	SA X2 setup	X2_SETUP_FOR_SA_SW option of the gNBInterfaceParam.X2ConfigSw parameter	<i>X2 Self-Management in SA Networking</i>	SA X2 setup and operator-specific NCI configuration are mutually exclusive. They cannot both be enabled.
Low-frequency TDD High-frequency TDD	X2-U IP address sending control	X2U_IPADDR_SEND_SW option of the gNodeBParam.NsaDcOptExtSwitch parameter	None	X2-U IP address sending control and operator-specific NCI configuration are mutually exclusive. They cannot both be enabled.

4.6.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD	Xn self-management	gNBXnBlackWhitelist.XnListType	<i>NG and Xn Self-Management</i>	An operator-specific gNodeB ID must be configured if operator-specific NCI configuration is enabled on the peer gNodeB that is added to the Xn blacklist or whitelist in Xn self-management.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	NCL update triggered by messages transmitted over the Xn interface between gNodeBs	NR_NR_ANR_SW option of the NRCellAlgoSwitch h.AnrSwitch parameter	<i>ANR</i>	When an Xn link is configured between a shared gNodeB of a version earlier than V100R017C10 and a gNodeB of V100R017C10 with gNBSharingMode.MultipleNciEnhanceSw set to ON, the "NCL update triggered by messages transmitted over the Xn interface between gNodeBs" function of the former gNodeB requires that the Xn interface of the primary operator work properly. When (1) an Xn link is configured between a non-shared gNodeB of a version earlier than V100R017C10 and a gNodeB of V100R017C10 with gNBSharingMode.MultipleNciEnhanceSw set to ON, and (2) the operator to which the former gNodeB belongs is a secondary operator for the latter gNodeB, then the "NCL update triggered by messages transmitted over the Xn interface between gNodeBs" function will not take effect on the former gNodeB. NOTE It is recommended that the gNBSharingMode.MultipleNciEnhanceSw parameter be set to ON for the shared gNodeBs after all of them have been upgraded to V100R017C10. This prevents the operator-specific NCI configuration function from being affected.
Low-frequency TDD	X2/Xn-based network-wide synchronization deviation detection	gNBInterfaceParam.X2XnSyncDiffDetectSwitch	<i>Synchronization</i>	In NSA networking, when multiple NR operators work in RAN sharing mode and operator-specific NCI configuration is enabled, X2/Xn-based network-wide synchronization deviation detection does not take effect.

4.6.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.6.3.4 Others

When operator-specific NCI configuration is enabled on the gNodeB and RAN sharing with common carrier is enabled on the eNodeB, continuity must be ensured between the X2 IP address of each operator configured on the gNodeB and the X2 IP address configured on the eNodeB.

4.6.4 Operation and Maintenance

4.6.4.1 Data Configuration

4.6.4.1.1 Data Preparation

- The parameters used for function activation are as follows:
 - gNodeB: [Table 4-32](#)
 - eNodeB: [Table 4-40](#)
- The parameters used for neighboring cell configuration related to this function are as follows:
 - gNodeB: [Table 4-33](#) to [Table 4-38](#)
 - eNodeB: [Table 4-41](#) to [Table 4-43](#)
- The parameters used for function optimization are as follows:
 - gNodeB: [Table 4-39](#)
 - eNodeB: N/A

Configuring Parameters on the gNodeB

Table 4-32 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Multiple NR Cell Identity Switch	<i>gNBSharingMode.MultipleNCISwitch</i>	Set this parameter to ON when different operators need to use different NCIs.
Multiple NR Cell Identity Enhance Switch	<i>gNBSharingMode.MultipleNciEnhanceSw</i>	If operators need to use their respective NCGIs over the Xn and NG interfaces, set this parameter to ON .
gNodeB ID	<i>gNBOperator.gNBId</i>	Set these parameters based on the network plan. If the settings of the three parameters need to be identical across different operators, set the <i>gNBSharingMode.MultipleNciEnhanceSw</i> parameter to ON .
gNodeB ID Length	<i>gNBOperator.gNBIdLength</i>	
Cell ID	<i>NRDUCellOp.CellId</i>	
User Label	<i>gNBOperator.UserLabel</i>	Set this parameter based on the network plan.
Cell Primary Operator Flag	<i>NRDUCellOp.CellPrimaryOpFlag</i>	Set this parameter based on the network plan. Deactivate the cell before reconfiguring this parameter.
gNodeB CU Xn ID	<i>gNBCUXnOp.gNBCuXnId</i>	Set this parameter based on the network plan. The Xn objects must be different for each operator.
gNodeB CU X2 ID	<i>gNBCUX2Op.gNBCuX2Id</i>	Set this parameter based on the network plan. The X2 objects must be different for each operator.
gNodeB CU NG ID	<i>gNBCUNgOp.gNBCuNgId</i>	Set this parameter based on the network plan. The NG objects must be different for each operator.

Table 4-33 Parameters for configuring an NR external neighboring cell of the gNodeB

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	NRExternalNCell.Mcc	- If operator-specific NCI configuration is enabled on the peer gNodeB, set these parameters separately to the primary operator's MCC, MNC, gNodeB ID, and cell ID of an external neighboring NR cell. (An operator is considered the primary operator if its NRDUCellOp.CellPrimaryOpFlag is set to TRUE.) - If operator-specific NCI configuration is disabled on the peer gNodeB, set these parameters separately to the primary operator's MCC, MNC, gNodeB ID, and cell ID of an external neighboring NR cell.
Mobile Network Code	NRExternalNCell.Mnc	
gNodeB ID	NRExternalNCell.gNBId	
Cell ID	NRExternalNCell.CellId	
Physical Cell ID	NRExternalNCell.PhysicalCellId	Set this parameter based on the network plan.
SSB Frequency Position	NRExternalNCell.SsbFrequencyPos	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Tracking Area Code	NRExternalNCell.Tac	Set this parameter to the invalid TAC (4294967295) for an external neighboring cell for which the NRExternalNCell.NrNetworkingOption parameter is set to NSA . Set this parameter to a valid TAC (ranging from 0 to 16777215) for an external neighboring cell for which the NRExternalNCell.NrNetworkingOption parameter is set to SA or SA_NSA .
NR Networking Option	NRExternalNCell.NrNetworkingOption	Set this parameter based on the network plan.
PLMN Reserved Flag	NRExternalNCell.PlmnReservedFlag	Set this parameter to TRUE if the gNodeB serving an NR external neighboring cell works in RAN sharing with dedicated carrier, and the external neighboring cell belongs to a secondary operator.

Table 4-34 Parameters for configuring the NR external neighboring cell PLMN list of the gNodeB

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NRExternalNCellPlmn.gNBId	- If operator-specific NCI configuration is enabled on the peer gNodeB, set these parameters separately to the primary operator's MCC, MNC, gNodeB ID, and cell ID of an external neighboring NR cell. (An operator is considered the primary operator if its NRDUCellOp.CellPrimaryOpFlag is set to TRUE.) - If operator-specific NCI configuration is disabled on the peer gNodeB, set these parameters separately to the primary operator's MCC, MNC, gNodeB ID, and cell ID of an external neighboring NR cell.
Cell ID	NRExternalNCellPlmn.CellId	
Mobile Country Code	NRExternalNCellPlmn.Mcc	
Mobile Network Code	NRExternalNCellPlmn.Mnc	
Shared Mobile Country Code	NRExternalNCellPlmn.SharedMcc	Separately set these parameters to the secondary-operator MCC, MNC, gNodeB ID, and cell ID of an NR external neighboring cell.
Shared Mobile Network Code	NRExternalNCellPlmn.SharedMnc	
Shared PLMN gNodeB ID	NRExternalNCellPlmn.SharedPlmnNBId	
Shared PLMN Cell ID	NRExternalNCellPlmn.SharedPlmnCellId	

Parameter Name	Parameter ID	Setting Notes
Tracking Area Code	NRExternalNCellPlmn.Tac	Set this parameter to the secondary-operator TAC of an NR external neighboring cell. Set this parameter to the invalid TAC (4294967295) for an external neighboring cell for which the NRExternalNCellPlmn.NrNetworkingOption parameter is set to NSA . Set this parameter to a valid TAC (ranging from 0 to 16777215) for an external neighboring cell for which the NRExternalNCellPlmn.NrNetworkingOption parameter is set to SA or SA_NSA .
NR Networking Option	NRExternalNCellPlmn.NrNetworkingOption	Set this parameter to the secondary-operator NR networking option of an NR external neighboring cell.
Note: When operator-specific NCI configuration is enabled on the gNodeB, the preceding parameters do not need to be configured for primary-operator-exclusive or secondary-operator-exclusive NR cells.		

Table 4-35 Parameters for configuring a relationship between NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCeIlRelatIon.NrCellId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	NRCellRelation.Mcc	• If operator-specific NCI configuration is enabled on the gNodeB serving a neighboring NR cell, separately set these parameters to the primary-operator MCC, MNC, gNodeB ID, and cell ID of the neighboring NR cell. (If an operator's NRDUCellOp.CellPrimaryOpFlag is set to TRUE, it is the primary operator.) • If operator-specific NCI configuration is disabled on the gNodeB serving a neighboring NR cell, separately set these parameters to the primary-operator MCC, MNC, gNodeB ID, and cell ID of a neighboring NR cell.
Mobile Network Code	NRCellRelation.Mnc	
gNodeB ID	NRCellRelation.gNBId	
Cell ID	NRCellRelation.CellId	

Table 4-36 Parameters for configuring an external E-UTRAN cell of the gNodeB

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	gNBEutraExternal-Cell.Mcc	- If the eNodeB serving an external E-UTRAN cell works in RAN sharing with common carrier, RAN sharing with special dedicated carrier, or hybrid RAN sharing, separately set these parameters to the MCC and MNC of the primary operator to which the external E-UTRAN cell belongs. - If the eNodeB serving an external E-UTRAN cell works in RAN sharing with dedicated carrier or does not work in RAN sharing, separately set these parameters to the MCC and MNC of the operator to which the external E-UTRAN cell belongs.
Mobile Network Code	gNBEutraExternal-Cell.Mnc	
eNodeB ID	gNBEutraExternal-Cell.EnodebId	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	gNBEutraExternal-Cell.CelId	Set this parameter based on the network plan.
Downlink EARFCN	gNBEutraExternal-Cell.DlEarfcn	Set this parameter based on the network plan.
Physical Cell ID	gNBEutraExternal-Cell.PhysicalCellId	Set this parameter based on the network plan.
Tracking Area Code	gNBEutraExternal-Cell.Tac	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	gNBEutraExternal-Cell.AggregationAttribute	Select the MASTER_PLMN_RESERVED_FLAG option of the gNBEutraExternal-Cell.AggregationAttribute parameter if the eNodeB serving an external E-UTRAN cell works in RAN sharing with special dedicated carrier, and the external cell belongs to a secondary operator.

Table 4-37 Parameters for configuring the E-UTRAN external-cell PLMN list of the gNodeB

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	gNBEutraExtCellPlmn.Mcc	Separately set these parameters to the MCC and MNC of the primary operator to which the external E-UTRAN cell belongs.
Mobile Network Code	gNBEutraExtCellPlmn.Mnc	
eNodeB ID	gNBEutraExtCellPlmn.EnodebId	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	gNBEutraExtCellPlmn.CellId	Set this parameter based on the network plan.
Shared Mobile Country Code	gNBEutraExtCellPlmn.SharedMcc	Separately set these parameters to the MCC and MNC of the secondary operator to which the external E-UTRAN cell belongs.
Shared Mobile Network Code	gNBEutraExtCellPlmn.SharedMnc	
Note: The preceding parameters need to be configured only when the eNodeB serving an external E-UTRAN cell works in RAN sharing with common carrier or RAN sharing with special dedicated carrier.		

Table 4-38 Parameters for configuring the relationship of an NR cell with an E-UTRAN cell

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellEutraNRelation.NrCellId	Set this parameter based on the network plan.
Mobile Country Code	NRCellEutraNRelation.Mcc	• If the eNodeB serving a neighboring E-UTRAN cell works in RAN sharing with common carrier, RAN sharing with special dedicated carrier, or hybrid RAN sharing, separately set these parameters to the MCC and MNC of the primary operator to which the neighboring E-UTRAN cell belongs. • If the eNodeB serving a neighboring E-UTRAN cell works in RAN sharing with dedicated carrier or does not work in RAN sharing, separately set these parameters to the MCC and MNC of the operator to which the neighboring E-UTRAN cell belongs.
Mobile Network Code	NRCellEutraNRelation.Mnc	
eNodeB ID	NRCellEutraNRelation.EnodebId	Set this parameter based on the network plan.
Cell ID	NRCellEutraNRelation.CellId	Set this parameter based on the network plan.

Table 4-39 Parameters used for optimization

Parameter Name	Parameter ID	Setting Notes
PLMN Policy	NRExternalgNB.PlmnPolicy	If there are issues related to compatibility with peer non-Huawei gNodeBs, the following mechanism applies: - If NRExternalgNB.PlmnPolicy is set to PRIMARY_PLMN_PREFER, the primary operator's PLMN ID and gNodeB ID are used for NG-C link setup with the 5GC for the primary and secondary operators. - If NRExternalgNB.PlmnPolicy is set to SELECTED_PLMN_PREFER, the primary and secondary operators' own PLMN IDs and gNodeB IDs are used for NG-C link setup with the 5GC for the primary and secondary operators.
Mobile Country Code	NRExternalgNB.Mcc	If there are issues related to compatibility with peer non-Huawei gNodeBs, separately set these parameters to the primary-operator MCC, MNC, and gNodeB ID of an external NR cell.
Mobile Network Code	NRExternalgNB.Mnc	
gNodeB ID	NRExternalgNB.gNBId	
gNodeB ID Length	gNBExternalPlmn.gNBIdLength	If an external cell's operator-specific PLMN information is not present in the gNBOperator MO configurations of the local gNodeB, configure this parameter to specify the gNodeB ID length corresponding to the PLMN.

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	gNBExternalPlmn.Mcc	If an external cell's operator-specific PLMN information is not present in the gNBOperator MO configurations of the local gNodeB, configure this parameter to specify the MCC corresponding to the PLMN.
Mobile Network Code	gNBExternalPlmn.Mnc	If an external cell's operator-specific PLMN information is not present in the gNBOperator MO configurations of the local gNodeB, configure this parameter to specify the MNC corresponding to the PLMN.
Note: On the MAE-Access, check for to-be-confirmed neighboring NR cells in the ANR controlled mode before running the ADD NREXTERNALGNB, MOD NREXTERNALGNB, or RMV NREXTERNALGNB command. If to-be-confirmed neighboring NR cells are found, confirm them on the MAE-Access, and run the intended command after the neighboring NR cells are added. Otherwise, an incorrect gNodeB ID is carried in the signaling messages related to handovers to the to-be-confirmed neighboring NR cells, leading to handover failures.		

Configuring Parameters on the eNodeB

Table 4-40 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
LTE to NR PLMN Selection Mode	EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode	Set these parameters when operator-specific NCI configuration is enabled on the gNodeB.
LTE to NR PLMN Selection Mode	NbrGnodebPolicy.LteToNrPlmnSelectionMode	Set EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode to SELECTED_PLMN_ID or HYBRID_PLMN_ID . If the PLMN ID of a neighboring gNodeB is filled in using a method different from that specified by EnodebAlgoExtSwitch.LteToNrPlmnSelectionMode , set NbrGnodebPolicy.LteToNrPlmnSelectionMode to PRIMARY_PLMN_ID or HYBRID_PLMN_ID .
gNodeB ID Length	NCellPlmnList.gNBIdLength	This parameter must be configured for the primary operator and can be optionally configured for a secondary operator. If this parameter is not configured, the value configured for the primary operator takes effect. The values of this parameter configured for the primary and secondary operators must be the same as those of gNBOperator.gNBIdLength configured for the primary and secondary operators on the gNodeB.

Table 4-41 Parameters for configuring an external NR cell of the eNodeB

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NrExternalCell.<i>GnodebId</i>	- If operator-specific NCI configuration is enabled on the gNodeB, separately set these parameters to the primary-operator MCC, MNC, gNodeB ID, and cell ID of an external NR cell. (If an operator's NRDUCellOp.<i>CellPrimaryOpFlag</i> is set to TRUE, it is the primary operator.) - If operator-specific NCI configuration is disabled on the gNodeB, separately set these parameters to the primary-operator MCC, MNC, gNodeB ID, and cell ID of an external NR cell.
Cell ID	NrExternalCell.<i>CellId</i>	
Mobile Country Code	NrExternalCell.<i>Mcc</i>	
Mobile Network Code	NrExternalCell.<i>Mnc</i>	
Physical Cell ID	NrExternalCell.<i>PhyCellId</i>	Set this parameter based on the network plan.
Tracking Area Code	NrExternalCell.<i>Tac</i>	Set this parameter based on the network plan.
Downlink ARFCN	NrExternalCell.<i>DLArfcn</i>	Set this parameter based on the network plan.
NR Networking Option	NrExternalCell.<i>NrNetworkingOption</i>	Set this parameter based on the network plan.

Table 4-42 Parameters for configuring the NR external-cell PLMN list of the eNodeB

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NrExternalCellPlmn.GnodebId	- If operator-specific NCI configuration is enabled on the gNodeB, separately set these parameters to the primary-operator MCC, MNC, gNodeB ID, and cell ID of an external NR cell. (If an operator's NRDUCellOp.CellPrimaryOpFlag is set to TRUE, it is the primary operator.) - If operator-specific NCI configuration is disabled on the gNodeB, separately set these parameters to the primary-operator MCC, MNC, gNodeB ID, and cell ID of an external NR cell.
Cell ID	NrExternalCellPlmn.CellId	
Mobile Country Code	NrExternalCellPlmn.Mcc	
Mobile Network Code	NrExternalCellPlmn.Mnc	
Shared Mobile Country Code	NrExternalCellPlmn.SharedMcc	Separately set these parameters to the secondary-operator MCC, MNC, gNodeB ID, and cell ID of an external NR cell.
Shared Mobile Network Code	NrExternalCellPlmn.SharedMnc	
Shared PLMN gNodeB ID	NrExternalCellPlmn.SharedPlmnGnodebId	
Shared PLMN Cell ID	NrExternalCellPlmn.SharedPlmnCellId	
Tracking Area Code	NrExternalCellPlmn.Tac	This parameter indicates the ID of base station serving the external NR cell for a RAN sharing operator. The ID uniquely identifies the base station for the operator.
Note: When operator-specific NCI configuration is enabled on the gNodeB, the preceding parameters do not need to be configured for primary-operator-exclusive or secondary-operator-exclusive NR cells.		

Table 4-43 Parameters for configuring a relationship of an E-UTRAN cell with an NR cell

Parameter Name	Parameter ID	Setting Notes
Local cell ID	NrNRRelationship.LocalCellId	Set this parameter based on the network plan.
Mobile Country Code	NrNRRelationship.Mcc	- If operator-specific NCI configuration is enabled on the gNodeB serving a neighboring NR cell, separately set these parameters to the primary-operator MCC, MNC, gNodeB ID, and cell ID of the neighboring NR cell. (If an operator's NRDUCellOp.CellPrimaryOpFlag is set to TRUE, it is the primary operator.) - If operator-specific NCI configuration is disabled on the gNodeB serving a neighboring NR cell, separately set these parameters to the primary-operator MCC, MNC, gNodeB ID, and cell ID of a neighboring NR cell.
Mobile Network Code	NrNRRelationship.Mnc	
gNodeB ID	NrNRRelationship.GnodebId	
Cell ID	NrExternalCellPlmn.CellId	

4.6.4.1.2 Using MML Commands

Before using MML commands, refer to [4.6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

This section uses NSA and SA hybrid networking as an example. The configurations in this networking are the same as those in NSA networking and SA networking. For details about the basic configurations of RAN sharing with common carrier, see [4.1.4.1.2 Using MML Commands](#). For this example, assume that both the local and neighboring base stations work in RAN sharing with common carrier.

Activation Command Examples

Example 1: In new deployments, RAN sharing with common carrier is used, and each operator has its own gNodeB ID and cell ID.

- Primary operator: For the local base station, the gNodeB ID is 111 and consists of 24 bits, and the IDs of its two served cells are 11 and 21, separately. For the neighboring base station, the gNodeB ID is 211, and the IDs of its two served cells are 31 and 41, separately.
- Secondary operator: For the local base station, the gNodeB ID is 112 and consists of 24 bits, and the IDs of its two served cells are 12 and 22, separately. For the neighboring base station, the gNodeB ID is 212, and the IDs of its two served cells are 32 and 42, separately.

S1, X2, NG, and Xn interface configuration is the same as that in RAN sharing with common carrier. For configuration examples of S1 and X2 interfaces, see [4.1.4.1.2 Using MML Commands](#). For configuration examples of NG and Xn interfaces, see [4.1.4.1.2 Using MML Commands](#).

gNodeB configurations:

```
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Configuring the sharing mode, gNodeB-level operator information, and cell-level operator
information
//Configuring the sharing mode of the gNodeB
MOD GNBSHARINGMODE: gNBMultiOpSharingMode=SHARED_FREQ;
//Manually restarting the application referenced by the gNodeB function
RST APP: AID=1;
//Configuring the gNodeB ID and gNodeB ID length for the primary operator of the gNodeB
ADD GNBOPERATOR: OperatorId=0, OperatorName="unicom", OperatorType=PRIMARY_OPERATOR,
Mcc="460", Mnc="01", NrNetworkingOption=SA_NSA, gNBIdLength=24, gNBId=111,
UserLabel="AirportSite";
//Configuring the gNodeB ID and gNodeB ID length for the secondary operator of the gNodeB
ADD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,
Mcc="460", Mnc="03", NrNetworkingOption=SA_NSA, gNBIdLength=24, gNBId=112,
UserLabel="AirportSiteTelecom";
//Configuring operator information for cell 1 with cell IDs being operator-specific
ADD NRDUCELL: NrDuCellId=1, OperatorId=0, CellId=11, CellPrimaryOpFlag=TRUE;
ADD NRDUCELL: NrDuCellId=1, OperatorId=1, CellId=12, CellPrimaryOpFlag=FALSE;
//Configuring operator information for cell 2 with cell IDs being operator-specific
ADD NRDUCELL: NrDuCellId=2, OperatorId=0, CellId=21, CellPrimaryOpFlag=TRUE;
ADD NRDUCELL: NrDuCellId=2, OperatorId=1, CellId=22, CellPrimaryOpFlag=FALSE;
//Configuring interfaces (all interfaces are required to work in independent mode)
//S1 and X2 interface configuration is the same as that in RAN sharing with common carrier. For details, see
the configuration examples for basic networking for RAN sharing with common carrier.
//NG and Xn interface configuration is the same as that in RAN sharing with common carrier. For details,
see the configuration examples for basic networking for RAN sharing with common carrier.
//Configuring neighboring cells
//Assuming that the local and neighboring base stations work in RAN sharing with common carrier and
NCIs are operator-specific in the two cells served by the neighboring base station
//Adding NR external neighboring cells, with Mcc and Mnc respectively set to the MCC and MNC of the
primary operator
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=211, CellId=31, PhysicalCellId=3, Tac=1,
SsbFreqPos=26639, NrNetworkingOption=SA_NSA;
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=211, CellId=41, PhysicalCellId=4, Tac=1,
SsbFreqPos=26638, NrNetworkingOption=SA_NSA;
//Adding the secondary-operator information to the external neighboring cell PLMN list, with Mcc and Mnc
respectively set to the MCC and MNC of the primary operator
ADD NREXTERNALNCELLPLMN: Mcc="460", Mnc="01", gNBId=211, CellId=31, SharedMcc="460",
SharedMnc="03", Tac=1, NrNetworkingOption=SA_NSA, SharedPlmnNbId=212, SharedPlmnCellId=32;
ADD NREXTERNALNCELLPLMN: Mcc="460", Mnc="01", gNBId=211, CellId=41, SharedMcc="460",
SharedMnc="03", Tac=1, NrNetworkingOption=SA_NSA, SharedPlmnNbId=212, SharedPlmnCellId=42;
//Adding the neighbor relationships with NR cells served by the neighboring base station (assuming that
NrCellId of the two cells of the local base station is 1 and 2, separately), with Mcc and Mnc respectively set
```

```
to the MCC and MNC of the primary operator
ADD NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=211, CellId=31;
ADD NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=211, CellId=41;
ADD NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=211, CellId=31;
ADD NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=211, CellId=41;
//Enabling operator-specific NCI configuration
MOD GNBSHARINGMODE: MultipleNCISwitch=ON, MultipleNciEnhanceSw=ON;
RST APP: AID=1;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
```

eNodeB configurations:

```
//Configuring the LTE-to-NR PLMN selection mode
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode= SELECTED_PLMN_ID;
//(Optional) Configuring the LTE-to-NR PLMN selection mode for a specific gNodeB
ADD NBRGNODEBPOLICY: Mcc="460", Mnc="01", GnodebId=111,
LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//Configuring gNBIdLength for each involved operator
ADD NCELLPLMNLIST: Mcc="460", Mnc="01", RatType=NR, gNBIdLength=24;
ADD NCELLPLMNLIST: Mcc="460", Mnc="03", RatType=NR, gNBIdLength=24;
//Adding external NR cells with primary-operator NR cell information specified
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=111, CellId=11, DlArfcn=636666, PhyCellId=1,
Tac=1;
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=111, CellId=21, DlArfcn=636666, PhyCellId=2,
Tac=1;
//Configuring the external-cell PLMN list with SharedPlmnGnodebId and SharedPlmnCellId set to the
secondary-operator gNodeB ID and cell ID, respectively
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="01", GnodebId=111, CellId=11, SharedMcc="460",
SharedMnc="03", SharedPlmnGnodebId=112, SharedPlmnCellId=12;
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="01", GnodebId=111, CellId=21, SharedMcc="460",
SharedMnc="03", SharedPlmnGnodebId=112, SharedPlmnCellId=22;
//Configuring neighbor relationships with primary-operator NR cell information specified
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=111, CellId=11;
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=111, CellId=21;
```

Example 2: Deploy multi-NCGI configuration in reconstruction with the planned gNodeB ID and cell ID of the primary and secondary operators unchanged.

- Before reconstruction, the PLMN ID in the NCGI is set to the primary operator's PLMN ID.
For the local base station, the gNodeB ID is 111, and the IDs of its two served cells are 1 and 2, separately. For the neighboring base station, the gNodeB ID is 211, and the IDs of its two served cells are 3 and 4, separately.
- After reconstruction, operators' own PLMN IDs are used, and different operators use different NCGIs.
Both the gNodeB IDs and cell IDs planned for the primary and secondary operators are the same. For the local base station, the gNodeB ID is 111, and the IDs of its two served cells are 1 and 2, separately. For the neighboring base station, the gNodeB ID is 211, and the IDs of its two served cells are 3 and 4, separately.

gNodeB configurations:

```
//Configuring interfaces
//(Optional) If Xn is shared before reconstruction and independent Xn configurations are required after
reconstruction, see the configuration examples for basic networking for RAN sharing with common carrier.
//(Optional) If X2 is shared before reconstruction and independent X2 configurations are required after
reconstruction, see the configuration examples for basic networking for RAN sharing with common carrier.
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Configuring the gNodeB ID and gNodeB ID length for each operator
//Configuring the gNodeB ID and gNodeB ID length for the primary operator of the gNodeB
```

```
MOD GNBOperator: OperatorId=0, OperatorName="unicom", OperatorType=PRIMARY_OPERATOR,
Mcc="460", Mnc="01", NrNetworkingOption=SA_NSA, gNBid=111, gNBidLength=24;
//Configuring the gNodeB ID and gNodeB ID length for the secondary operator of the gNodeB
MOD GNBOperator: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,
Mcc="460", Mnc="03", NrNetworkingOption=SA_NSA, gNBid=111, gNBidLength=24;
//Configuring the cell ID for each operator and specifying the primary operator for the cell
MOD NRDUCELL: NrDuCellId=1, OperatorId=0, CellId=1, CellPrimaryOpFlag=TRUE;
MOD NRDUCELL: NrDuCellId=1, OperatorId=1, CellId=1, CellPrimaryOpFlag=FALSE;
MOD NRDUCELL: NrDuCellId=2, OperatorId=0, CellId=2, CellPrimaryOpFlag=TRUE;
MOD NRDUCELL: NrDuCellId=2, OperatorId=1, CellId=2, CellPrimaryOpFlag=FALSE;
//Configuring neighboring cells (Note that neighboring relationships do not need to be adjusted because
the planned data for gNBid and CellId remains unchanged for the primary operator.)
MOD NREXTERNALCELLPLMN: CellId=3, gNBid=211, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03", SharedPlmnGnbid=211, SharedPlmnCellId=3, Tac=1;
MOD NREXTERNALCELLPLMN: CellId=4, gNBid=211, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03", SharedPlmnGnbid=211, SharedPlmnCellId=4, Tac=1;
//Enabling operator-specific NCI configuration
MOD GNBSHARINGMODE: MultipleNCISwitch=ON, MultipleNciEnhanceSw=ON;
RST APP: AID=1;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
```

The **gNBSharingMode.MultipleNCISwitch** and **gNBSharingMode.MultipleNciEnhanceSw** parameters can be configured on the gNodeB in one of the scenarios detailed in [Operator-specific NCI Configuration Enhancement](#).

Assume that the gNodeB IDs of the gNodeB using configuration 1, gNodeB using configuration 2, and gNodeB using configuration 3 are 211, 111, and 311, respectively, and neighbor relationships exist among the gNodeBs.

//Configuring the gNodeB using configuration 1 with the gNodeB using configuration 2 specified

```
ADD NREXTERNALGNB: Mcc="460", Mnc="01", gNBid=111, PlmnPolicy=SELECTED_PLMN_PREFER;
```

//Configuring the gNodeB using configuration 2 with the gNodeB using configuration 1 specified

```
ADD NREXTERNALGNB: Mcc="460", Mnc="01", gNBid=211, PlmnPolicy=PRIMARY_PLMN_PREFER;
```

//Configuring the gNodeB using configuration 3 with the gNodeB using configuration 2 specified

```
ADD NREXTERNALGNB: Mcc="460", Mnc="01", gNBid=111, PlmnPolicy=SELECTED_PLMN_PREFER;
```

eNodeB configurations:

```
//Configuring the eNodeB
//Configuring the LTE-to-NR PLMN selection mode
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode= SELECTED_PLMN_ID;
//Optional) Configuring the LTE-to-NR PLMN selection mode for a specific gNodeB
ADD NBRGNODEBPOLICY: Mcc="460", Mnc="01", GnodebId=111,
LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//Configuring gNBidLength for each involved operator
ADD NCELLPLMNLIST: Mcc="460", Mnc="01", RatType=NR, gNBidLength=24;
ADD NCELLPLMNLIST: Mcc="460", Mnc="03", RatType=NR, gNBidLength=24;
//Configuring neighboring cells
//Configuring the external-cell PLMN list with SharedPlmnGnodebId and SharedPlmnCellId set to the
secondary-operator gNodeB ID and cell ID, respectively
MOD NREXTERNALCELLPLMN: Mcc="460", Mnc="01", GnodebId=111, CellId=1, SharedMcc="460",
SharedMnc="03", SharedPlmnGnodebId=111, SharedPlmnCellId=1;
MOD NREXTERNALCELLPLMN: Mcc="460", Mnc="01", GnodebId=111, CellId=2, SharedMcc="460",
SharedMnc="03", SharedPlmnGnodebId=111, SharedPlmnCellId=2;
```

Example 3: Reconfiguration for the primary and secondary operators in reconstruction

- Before reconstruction:
For the local base station, the gNodeB ID is 110, and the IDs of its two served cells are 1 and 2, separately. For the neighboring base station, the gNodeB ID is 210, and the IDs of its two served cells are 3 and 4, separately.
- After reconstruction:
 - Primary operator: A different gNodeB ID is added and the cell IDs remain unchanged, and the **gNBOperator.gNBIdLength** and **gNodeBFunction.gNBIdLength** parameters are set to the same value (24).
For the local base station, the gNodeB ID is 111, and the IDs of its two served cells are 1 and 2, separately. For the neighboring base station, the gNodeB ID is 211, and the IDs of its two served cells are 3 and 4, separately.

 NOTE

The **NRDUCellOp.CellId** parameter for the primary operator (specified by **NRDUCellOp.CellPrimaryOpFlag**) of a cell must be set to the same value as the **NRDUCell.CellId** parameter. As such, the cell ID of the primary operator remains unchanged after reconstruction.

- Secondary operator: A different gNodeB ID and cell ID are added, and the **gNBOperator.gNBIdLength** and **gNodeBFunction.gNBIdLength** parameters are set to the same value (24).
For the local base station, the gNodeB ID is 112, and the IDs of its two served cells are 12 and 22, separately. For the neighboring base station, the gNodeB ID is 212, and the IDs of its two served cells are 32 and 42, separately.

gNodeB configurations:

```
//Configuring interfaces
//(Optional) If Xn is shared before reconstruction and independent Xn configurations are required after reconstruction, see the configuration examples for basic networking for RAN sharing with common carrier.
//(Optional) If X2 is shared before reconstruction and independent X2 configurations are required after reconstruction, see the configuration examples for basic networking for RAN sharing with common carrier.
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Configuring different gNodeB IDs and gNodeB ID lengths for different operators
//Configuring the gNodeB ID and gNodeB ID length for the primary operator of the gNodeB
MOD GNBOPERATOR: OperatorId=0, OperatorName="unicom", OperatorType=PRIMARY_OPERATOR,
Mcc="460", Mnc="01", NrNetworkingOption=SA_NSA, gNBId=111, gNBIdLength=24;
//Configuring the gNodeB ID and gNodeB ID length for the secondary operator of the gNodeB
MOD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,
Mcc="460", Mnc="03", NrNetworkingOption=SA_NSA, gNBId=112, gNBIdLength=24;
//Configuring operator information for cell 1 with cell IDs being operator-specific
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=0, CellId=1, CellPrimaryOpFlag=TRUE;
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=1, CellId=12, CellPrimaryOpFlag=FALSE;
//Configuring operator information for cell 2 with cell IDs being operator-specific
MOD NRDUCELLOP: NrDuCellId=2, OperatorId=0, CellId=2, CellPrimaryOpFlag=TRUE;
MOD NRDUCELLOP: NrDuCellId=2, OperatorId=1, CellId=22, CellPrimaryOpFlag=FALSE;
//Configuring neighboring cells
//(Optional) Intra-base-station neighboring cell: Removing and then reconfiguring a neighbor relationship using the primary-operator gNodeB ID and cell ID of a neighboring cell (required when the primary operator of a neighboring cell is different from that configured in the gNBOperator.MO, or the gNBOperator.gNBId parameter for the primary operator of a neighboring cell is set to a different value from the gNodeBFunction.gNBId parameter)
RMV NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=110, CellId=2;
ADD NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=111, CellId=2;
RMV NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=110, CellId=1;
ADD NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=111, CellId=1;
```

```
//Optional Inter-base-station neighboring cell: Removing a neighbor relationship and an external cell and
then reconfiguring an external cell and a neighbor relationship using the primary-operator gNodeB ID and
cell ID of a neighboring cell (required when the primary operator of a neighboring cell is different from that
configured in the gNBOperator MO, or the gNBOperator.gNBId parameter for the primary operator of a
neighboring cell is set to a different value from the gNodeBFunction.gNBId parameter)
//Cells 1 and 3, and cells 2 and 4 are used as examples in the following descriptions. If other neighbor
relationships (those between cells 1 and 4 and between cells 2 and 3) are existent, modify the related
parameter values accordingly.
RMV NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=210, CellId=3;
RMV NREXTERNALNCELLPLMN: CellId=3, gNBId=210, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03";
RMV NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=210, CellId=3;
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=211, CellId=3, PhysicalCellId=3, Tac=1,
SsbFreqPos=26639;
ADD NREXTERNALNCELLPLMN: CellId=3, gNBId=211, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03", SharedPlmnGnbId=212, SharedPlmnCellId=32, Tac=1;
ADD NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=211, CellId=3;
RMV NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=210, CellId=4;
RMV NREXTERNALNCELLPLMN: CellId=4, gNBId=210, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03";
RMV NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=210, CellId=4;
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=211, CellId=4, PhysicalCellId=4, Tac=1,
SsbFreqPos=26639;
ADD NREXTERNALNCELLPLMN: CellId=4, gNBId=211, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03", SharedPlmnGnbId=212, SharedPlmnCellId=42, Tac=1;
ADD NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=211, CellId=4;
//Enabling operator-specific NCI configuration
MOD GNBSHARINGMODE: MultipleNCISwitch=ON, MultipleNciEnhanceSw=ON;
RST APP: AID=1;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
```

eNodeB configurations:

```
//Configuring the eNodeB
//Configuring the LTE-to-NR PLMN selection mode
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode= SELECTED_PLMN_ID;
//Optional Configuring the LTE-to-NR PLMN selection mode for a specific gNodeB
ADD NBRGNODEBPOLICY: Mcc="460", Mnc="01", GnodebId=111,
LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//Configuring gNBIdLength for each involved operator
ADD NCELLPLMNLIST: Mcc="460", Mnc="01", RatType=NR, gNBIdLength=24;
ADD NCELLPLMNLIST: Mcc="460", Mnc="03", RatType=NR, gNBIdLength=24;
//Configuring neighboring cells (ensure that pre-reconstruction neighboring cell configurations have been
removed from the eNodeB and gNodeB)
//Configuring external NR cells
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=111, CellId=1, DIARfcn=636666, PhyCellId=1,
Tac=1;
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=111, CellId=2, DIARfcn=636666, PhyCellId=2,
Tac=1;
//Configuring the external-cell PLMN list with SharedPlmnGnodebId and SharedPlmnCellId set to the
secondary-operator gNodeB ID and cell ID, respectively
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="01", GnodebId=111, CellId=1, SharedMcc="460",
SharedMnc="03", SharedPlmnGnodebId=112, SharedPlmnCellId=12;
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="01", GnodebId=111, CellId=2, SharedMcc="460",
SharedMnc="03", SharedPlmnGnodebId=112, SharedPlmnCellId=22;
//Configuring neighbor relationships
ADD NRNRRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=111, CellId=1;
ADD NRNRRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=111, CellId=2;
```

Example 4: Configurations of the primary operator remain unchanged and those of the secondary operator are adjusted in reconstruction.

- Before reconstruction:

For the local base station, the gNodeB ID is 111, and the IDs of its two served cells are 1 and 2, separately. For the neighboring base station, the gNodeB ID is 211, and the IDs of its two served cells are 3 and 4, separately.

- After reconstruction:
 - Primary operator: Configurations remain unchanged.
For the local base station, the gNodeB ID is 111, and the IDs of its two served cells are 1 and 2, separately. For the neighboring base station, the gNodeB ID is 211, and the IDs of its two served cells are 3 and 4, separately.
 - Secondary operator: A different gNodeB ID and cell ID are added, and the **gNBOperator.gNBIdLength** and **gNodeBFunction.gNBIdLength** parameters are set to the same value (24).
For the local base station, the gNodeB ID is 112, and the IDs of its two served cells are 12 and 22, separately. For the neighboring base station, the gNodeB ID is 212, and the IDs of its two served cells are 32 and 42, separately.

gNodeB configurations:

```
//Configuring interfaces
//(Optional) If Xn is shared before reconstruction and independent Xn configurations are required after
reconstruction, see the configuration examples for basic networking for RAN sharing with common carrier.
//(Optional) If X2 is shared before reconstruction and independent X2 configurations are required after
reconstruction, see the configuration examples for basic networking for RAN sharing with common carrier.
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Configuring different gNodeB IDs and gNodeB ID lengths for different operators
//Configuring the gNodeB ID and gNodeB ID length for the primary operator of the gNodeB
MOD GNBOPERATOR: OperatorId=0, OperatorName="unicom", OperatorType=PRIMARY_OPERATOR,
Mcc="460", Mnc="01", NrNetworkingOption=SA_NSA, gNBId=111, gNBIdLength=24;
//Configuring the gNodeB ID and gNodeB ID length for the secondary operator of the gNodeB
MOD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,
Mcc="460", Mnc="03", NrNetworkingOption=SA_NSA, gNBId=112, gNBIdLength=24;
//Enabling cell IDs to be operator-specific and specifying the primary operator of a cell
MOD NRDUCELL: NrDuCellId=1, OperatorId=0, CellId=1, CellPrimaryOpFlag=TRUE;
MOD NRDUCELL: NrDuCellId=1, OperatorId=1, CellId=12, CellPrimaryOpFlag=FALSE;
//Configuring neighboring cells
//(Optional) Intra-base-station neighboring cell: Removing and then reconfiguring a neighbor relationship
using the primary-operator gNodeB ID and cell ID of a neighboring cell (required when the primary
operator of a neighboring cell is different from that configured in the gNBOperator MO, or the
gNBOperator.gNBId parameter for the primary operator of a neighboring cell is set to a different value
from the gNodeBFunction.gNBId parameter; this operation is not required in the given scenario)
//(Optional) Inter-base-station neighboring cell: Removing a neighbor relationship and an external cell and
then reconfiguring an external cell and a neighbor relationship using the primary-operator gNodeB ID and
cell ID of a neighboring cell (required when the primary operator of a neighboring cell is different from that
configured in the gNBOperator MO, or the gNBOperator.gNBId parameter for the primary operator of a
neighboring cell is set to a different value from the gNodeBFunction.gNBId parameter; neighbor
relationship modification is not needed in the given scenario and only the secondary-operator gNodeB ID
and cell ID need to be added)
MOD NREXTERNALNCELLPLMN: CellId=3, gNBId=211, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03", SharedPlmnNbId=212, SharedPlmnCellId=32, Tac=1;
MOD NREXTERNALNCELLPLMN: CellId=4, gNBId=211, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03", SharedPlmnNbId=212, SharedPlmnCellId=42, Tac=1;
//Enabling operator-specific NCI configuration
MOD GNBSHARINGMODE: MultipleNCISwitch=ON, MultipleNciEnhanceSw=ON;
RST APP: AID=1;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
```

eNodeB configurations:

```
//Configuring the LTE-to-NR PLMN selection mode
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode= SELECTED_PLMN_ID;
//(Optional) Configuring the LTE-to-NR PLMN selection mode for a specific gNodeB
ADD NBRGNODEBPOLICY: Mcc="460", Mnc="01", GnodebId=111,
```

```
LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//Configuring gNBIdLength for each involved operator
ADD NCELLPLMNLIST: Mcc="460", Mnc="01", RatType=NR, gNBIdLength=24;
ADD NCELLPLMNLIST: Mcc="460", Mnc="03", RatType=NR, gNBIdLength=24;
//Configuring neighboring cells (ensure that pre-reconstruction neighboring cell configurations have been
removed from the eNodeB and gNodeB)
//Configuring external NR cells
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=111, CellId=1, DIARfcn=636666, PhyCellId=1,
Tac=1;
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=111, CellId=2, DIARfcn=636666, PhyCellId=2,
Tac=1;
//Configuring the external-cell PLMN list with SharedPlmnGnodebId and SharedPlmnCellId set to the
secondary-operator gNodeB ID and cell ID, respectively
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="01", GnodebId=111, CellId=1, SharedMcc="460",
SharedMnc="03", SharedPlmnGnodebId=212, SharedPlmnCellId=12;
ADD NREXTERNALCELLPLMN: Mcc="460", Mnc="01", GnodebId=111, CellId=2, SharedMcc="460",
SharedMnc="03", SharedPlmnGnodebId=212, SharedPlmnCellId=22;
//Configuring neighbor relationships
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=111, CellId=1;
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=111, CellId=2;
```

Optimization Command Examples

gNodeB configurations (eNodeB configuration is not required):

```
//Applying the primary-operator PLMN ID and gNodeB ID for NG-C link setup between the 5GC and the
peer non-Huawei gNodeB with NCIs being operator-specific
ADD NREXTERNALGNB: Mcc="460", Mnc="01", gNBId=211, PlmnPolicy=PRIMARY_PLMN_PREFER;
//Running the following command if the PLMN of an external cell is not configured on the local base station
ADD GNBEXTERNALPLMN: Mcc="460", Mnc="10", gNBIdLength=24;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

gNodeB configurations:

```
//Deactivating cells
DEA NRCELL: NrCellId=0;
DEA NRCELL: NrCellId=1;
//Modifying the primary operator flag of cells
MOD NRDUCELLOP: NrDuCellId=1, OperatorId=0, CellPrimaryOpFlag=FALSE;
MOD NRDUCELLOP: NrDuCellId=2, OperatorId=0, CellPrimaryOpFlag=FALSE;
//Disabling operator-specific NCI configuration
MOD GNBSHARINGMODE: MultipleNCISwitch=OFF;
RST APP: AID=1;
//Activating cells
ACT NRCELL: NrCellId=0;
ACT NRCELL: NrCellId=1;
//Changing the X2 interface mode from the independent mode to the shared mode
//Removing the X2 interface binding relationship and the corresponding X2 object for the primary operator
RMV GNBCUX2OP: gNBCuX2Id=0, OperatorId=0;
RMV GNBCUX2: gNBCuX2Id=0;
//Binding the primary operator to the configured X2 interface of operator 1
ADD GNBCUX2OP: gNBCuX2Id=1, OperatorId=0;
//((Optional, depending on the operator's plan) Changing the Xn interface mode from the
independent mode to the shared mode
//Removing the Xn interface binding relationship and the corresponding Xn object for the primary operator
RMV GNBCUXNOP: gNBCuXnId=0, OperatorId=0;
RMV GNBCUXN: gNBCuXnId=0;
//Binding the primary operator to the configured Xn interface of operator 1
ADD GNBCUXNOP: gNBCuXnId=1, OperatorId=0;
```

eNodeB configurations:

MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode = PRIMARY_PLMN_ID;

4.6.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.6.4.2 Activation Verification

Prerequisites

According to the network plan, the gNodeB is shared by multiple operators, and the settings of the **gNBOperator.gNBId** and **NRDUCellOp.CellId** parameters vary with operators.

Expected Result

Check SIB1. Specifically, log in to the MAE-Access, and choose **Monitor > Signaling Trace > Signaling Trace Management > Trace Type > NR > Application Layer > Uu Interface Trace**. The value of **cellIdentity** in the **PLMN-Identity** field in SIB1 varies with operators.

4.6.4.3 Network Monitoring

None

4.7 Operator-Specific Function Configuration

In this version, configurations of specific functions can be operator-specific.

- Operator-specific TAC configuration. For details, see [Operator-Specific TAC Configuration](#).
- Operator-specific RANAC configuration. For details, see [Operator-Specific RANAC Configuration](#).
- Operator-specific VoNR configuration. For details, see [Operator-Specific VoNR Configuration](#).
- Operator-specific NSA data split policy configuration. For details, see *EPC-based NSA Basics*, *EPC-based NSA Performance Enhancement*, and *EPC-based NSA Experience Improvement*.

4.7.1 Principles

In RAN sharing with common carrier, operator-specific function configurations are as follows.

Operator-Specific TAC Configuration

[Table 4-44](#) describes TAC-related parameters.

Table 4-44 TAC-related parameters

Parameter	Description	Remarks
gNBTrackingArea.Tac gNBTrackingArea.TrackingAreaId	These parameters collectively specify the TAC supported by the gNodeB.	gNodeB-level parameter
NRExternalNCell.Tac	This parameter specifies the TAC of the primary operator to which an NR external neighboring cell belongs.	Cell-level parameter
NRExternalNCellPlmn.Tac	This parameter specifies the TAC of the secondary operator to which an NR external neighboring cell belongs.	Cell-level parameter
NRDUCell.TrackingAreaId	This parameter uniquely identifies the TAC of a local cell.	Cell-level parameter
NRDUCellOp.TrackingAreaId	This parameter uniquely identifies the TAC of an operator in an NR DU cell.	Cell-level parameter

In RAN sharing with common carrier, the **NRDUCellOp.TrackingAreaId** parameter specifies an operator-specific TAC of a shared cell.

- If **NRDUCellOp.TrackingAreaId** is set to a value ranging from 0 to 65535 for an operator, system information is broadcast in the tracking area corresponding to the TAC defined in the **gNBTrackingArea** MO associated with the **NRDUCellOp.TrackingAreaId** parameter.
- If **NRDUCellOp.TrackingAreaId** is not configured for an operator (that is, the default value **4294967295** indicating an invalid configuration, is retained), system information is broadcast in the tracking area corresponding to the TAC defined in the **gNBTrackingArea** MO associated with the **NRDUCell.TrackingAreaId** parameter.

 NOTE

If TACs are required to be operator-specific on the NR side, the LTE side must be upgraded to eRAN16.1 to adapt to such configurations. Operator-specific TACs on the LTE side must be configured based on the operator-specific TACs configured on the NR side. Otherwise, there is a possibility that inter-RAT handovers from E-UTRAN cells to NG-RAN cells fail.

Operator-Specific RANAC Configuration

Table 4-45 describes RANAC-related parameters.

Table 4-45 RANAC-related parameters

Parameter	Description	Remarks
NRDUCell.RanNotificationAreald	This parameter uniquely identifies the RANAC of a local cell.	Cell-level parameter
NRDUCellOp.RanNotificationAreald	This parameter uniquely identifies the RANAC of an operator in an NR DU cell.	Cell-level parameter

In RAN sharing with common carrier, the **NRDUCellOp.RanNotificationAreald** parameter specifies an operator-specific RANAC of a shared cell.

- If **NRDUCellOp.RanNotificationAreald** is set to a value ranging from 0 to 255 for an operator, the value of **NRDUCellOp.RanNotificationAreald** is broadcast in system information.
- If **NRDUCellOp.RanNotificationAreald** is not configured (that is, the default value **65535** indicating an invalid configuration, is retained), the value of **NRDUCell.RanNotificationAreald** is broadcast in system information.

 NOTE

Assume that a UE enters the inactive state in the coverage area of a base station of V100R017C10SPC130 or earlier. After the UE moves to the coverage area of a base station on which operator-specific RANAC configuration is applied, RAN paging over the Xn link fails for the UE if the UE is served by a secondary operator. In this case, the source gNodeB triggers the UE to transit to the idle state after 10s and waits for the core network to initiate 5GC paging.

Operator-Specific VoNR Configuration

In RAN sharing with common carrier, the **OPERATOR_VONR_SW** option of the **NRCellOpPolicy.VoicePolicySwitch** parameter can be selected to enable VoNR for an operator in a shared cell.

- When the **VONR_SW** option (cell-level VoNR switch) of the **NRCellAlgoSwitch.VonrSwitch** parameter is selected, whether the VoNR function takes effect for an operator is determined by the setting of the **OPERATOR_VONR_SW** option (operator-level VoNR switch) of the **NRCellOpPolicy.VoicePolicySwitch** parameter.
- When the **VONR_SW** option of the **NRCellAlgoSwitch.VonrSwitch** parameter is deselected, the VoNR function does not take effect for an operator regardless of the setting of the **OPERATOR_VONR_SW** option of the **NRCellOpPolicy.VoicePolicySwitch** parameter.

For details about VoNR, see *VoNR*.

4.7.2 Network Analysis

4.7.2.1 Benefits

Function configurations can be operator-specific.

4.7.2.2 Impacts

The impacts between this function and other functions are described in [4.7.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.7.3 Requirements

4.7.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021211	Multi-Operator Sharing	NR0S00MULS00	per Cell
For the license control item NR0S00MULS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.7.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.7.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with common carrier	gNBSharingMode.gNBMultiOpSharingMode	<i>Multi-Operator Sharing</i>	Operator-specific function configuration requires RAN sharing with common carrier to be deployed on the live network.
Low-frequency TDD	VoNR	VONR_SW option of the NRCellAlgoSwitch.VonrSwitch parameter	<i>VoNR</i>	VoNR can be enabled for an operator only when the cell-level VoNR switch is turned on.

4.7.3.2.2 Mutually Exclusive Functions

None

4.7.3.2.3 Function Impacts

None

4.7.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.7.3.4 Others

None

4.7.4 Operation and Maintenance

4.7.4.1 Data Configuration

4.7.4.1.1 Data Preparation

Table 4-46 describes the parameters used for function activation. No parameters are required for function optimization.

Table 4-46 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Tracking Area Code	gNBTrackingArea.Tac	Set this parameter based on the network plan.
Tracking Area ID	gNBTrackingArea.TrackingAreaId	Set this parameter based on the network plan.
Tracking Area ID	NRDUCellOp.TrackingAreaId	Set this parameter based on the network plan if different operators need to be configured with different TACs.
Voice Policy Switch	NRCellOpPolicy.VoicePolicySwitch	Select the OPERATOR_VONR_SW option of the NRCellOpPolicy.VoicePolicySwitch parameter.
RAN Notification Area ID	NRDUCellOp.RANNotificationAreaId	Set this parameter based on the network plan.

4.7.4.1.2 Using MML Commands

Before using MML commands, refer to **4.6.3 Requirements** and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Configuring different TACs and RANACs for operators
ADD GNBTRACKINGAREA: TrackingAreald=1, Tac=1;
ADD GNBTRACKINGAREA: TrackingAreald=0, Tac=0;
MOD NRDUCELL: NrDuCellId=1, OperatorId=0, TrackingAreald=1, RanNotificationAreald=1;
MOD NRDUCELL: NrDuCellId=1, OperatorId=1, TrackingAreald=0, RanNotificationAreald=0;
//Enabling VoNR
ADD NRCELLPOLICY: NrCellId=1, OperatorId=0, VoicePolicySwitch=OPERATOR_VONR_SW-1;
```

Optimization Command Examples

N/A

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Configuring the same TAC and RANAC for operators
MOD NRDUCELL: NrDuCellId=1, OperatorId=1, TrackingAreald=4294967295,
RanNotificationAreald=65535;
MOD NRDUCELL: NrDuCellId=1, OperatorId=0, TrackingAreald=4294967295,
RanNotificationAreald=65535;
//Disabling VoNR
MOD NRCELLPOLICY: NrCellId=1, OperatorId=0, VoicePolicySwitch=OPERATOR_VONR_SW-0;
```

4.7.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.7.4.2 Activation Verification

Operator-specific TAC Configuration

Prerequisites:

The gNodeB is shared by multiple operators based on the network plan, and different operators are configured with different TACs.

Expected result:

Check SIB1. Specifically, log in to the MAE-Access, and choose **Monitor > Signaling Trace > Signaling Trace Management > Trace Type > NR > Application Layer > Uu Interface Trace**. The TACs corresponding to **PLMN-Identity** values in SIB1 vary with operators.

```
systemInformationBlockType1
  cellSelectionInfo
    q-RxLevMin: ---- 0xfffffc0 (-64)
  cellAccessRelatedInfo
    plmn-IdentityList
      PLMN-IdentityInfo
        plmn-IdentityList
          PLMN-Identity
```

```
mcc
MCC-MNC-Digit:0x4 (4)
MCC-MNC-Digit:0x6 (6)
MCC-MNC-Digit:0x0 (0)
mnc
MCC-MNC-Digit:0x6 (0)
MCC-MNC-Digit:0x1 (1)
trackingAreaCode: ---- 00000000000000000000000000000001 (00 00 01)
cellIdentity: ---- 000000000000000000000000000000010001 (00 00 00 01 01)
cellReservedForOperatorUse: ---- notReserved (1)
PLMN-IdentityInfo
plmn-IdentityList
PLMN-Identity
mcc
MCC-MNC-Digit:0x4 (4)
MCC-MNC-Digit:0x6 (6)
MCC-MNC-Digit:0x0 (0)
mnc
MCC-MNC-Digit:0x0 (0)
MCC-MNC-Digit:0x3 (3)
trackingAreaCode: ---- 00000000000000000000000000000000 (00 00 00)
cellIdentity: ---- 000000000000000000000000000000010001 (00 00 00 01 01)
cellReservedForOperatorUse: ---- notReserved (1)
```

Operator-Specific VoNR Configuration

- Prerequisites:
- The cell-level VoNR switch has been turned on. For details about how to turn on the cell-level VoNR switch, see *VoNR*.
 - The operator-level VoNR switch has been turned on.
- Expected result:
1. After the cell-level VoNR switch is turned on for a period of time, turn off the operator-level VoNR switch for one operator. The values of counters listed in [Table 4-47](#) decrease.
 2. Disable VoNR for all operators. The values of the counters listed in [Table 4-47](#) are 0.

Table 4-47 Counters

Counter ID	Counter Name
1911820520	N.QosFlow.Max.Cell5QI
1911820521	N.QosFlow.Avg.Cell5QI
1911820488	N.User.RRCCConn.Active.DuCell5QI.Max
1911820489	N.User.RRCCConn.Active.DuCell5QI.Avg
1911820510	N.QosFlow.Est.Att.Cell5QI
1911820511	N.QosFlow.Est.Succ.Cell5QI

4.7.4.3 Network Monitoring

Operator-Specific TAC Configuration

Operator-specific TAC configuration has no impact on network performance. The performance counters listed in [4.1.4.3 Network Monitoring](#) can be used to check whether this function has taken effect.

Operator-Specific VoNR Configuration

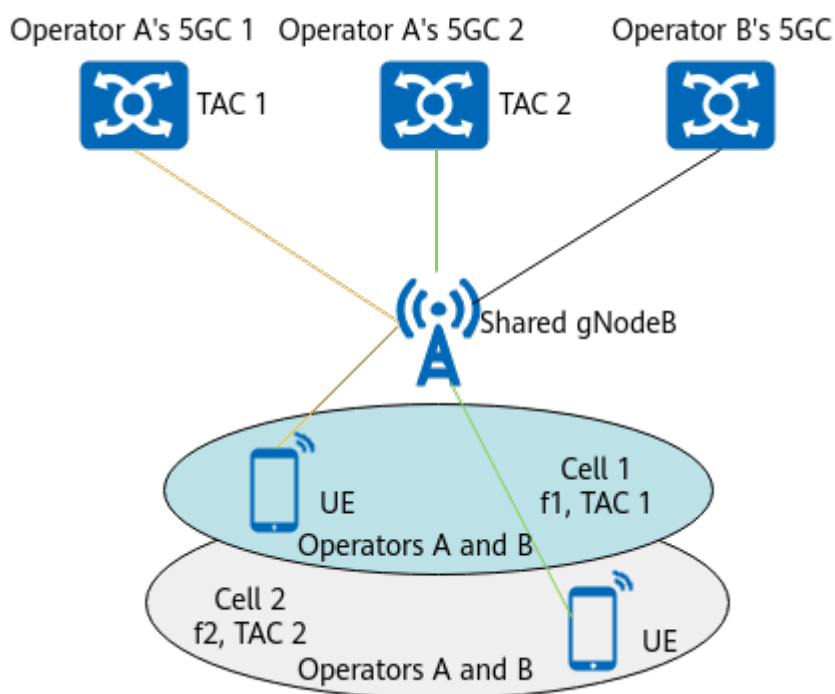
Operator-specific VoNR configuration does not affect voice quality and voice KPIs. Refer to the network monitoring methods presented in *VoNR* to check whether this function has taken effect.

4.8 Frequency-based 5GC Selection

4.8.1 Principles

In multi-operator sharing with a gNodeB connected to different 5GCs of a single operator, the gNodeB selects 5GCs for UEs when they access the network based on their serving frequencies. [Figure 4-14](#) shows an example, in which the gNodeB selects 5GC 1 and 5GC 2 for a UE served by frequency f1 and a UE served by frequency f2, respectively.

Figure 4-14 Different 5GCs selected for UE access in multi-operator sharing



As shown in [Figure 4-14](#), 5GC 1 and 5GC 2 manage UEs separately depending on their serving frequencies. Given this, different TACs must be configured. After the

function takes effect, a UE selects an AMF in the 5GC based on the configured TAC. For details about the AMF selection procedure, see *NG-flex*. The **gNBCUNgTrackingArea.TrackingAreaId** parameter is specified on the gNodeB to bind a TAC to an NG interface (specified by the **gNBCUNgTrackingArea.gNBCuNgId** parameter). When frequency-based 5GC selection takes effect, scenario-specific 5GC selection for UEs is performed as follows:

- For an RRC_IDLE UE: Different tracking areas (specified by the **NRDUCellOp.TrackingAreaId** parameter) are configured for an operator per frequency to enable the 5GC to promptly obtain the UE's location after the UE reselects from F1 to F2. Upon detecting that the TAC changes after the reselection, the UE initiates a location update. After the location update, the gNodeB selects a 5GC based on the TAC corresponding to the UE's serving frequency.
- For an RRC_CONNECTED UE: The gNodeB routes the UE's services to a 5GC based on the TAC corresponding to the UE's serving frequency.
- For an RRC_INACTIVE UE: When an RRC connection resume procedure is triggered, the gNodeB checks whether specific conditions are met. If all the conditions are met, the gNodeB responds that the RRC connection resume has failed, in which case RRC connection establishment is performed to enable the UE to access the cell from RRC_IDLE. If not all the conditions are met, the RRC connection resume has succeeded. For details about the RRC_INACTIVE state, see *SA Mobility Management in Inactive Mode*. The conditions are as follows:

Condition 1: A UE's serving PLMN is equal to the counterpart collectively represented by **gNBNgHoFreqRelation.Mcc** and **gNBNgHoFreqRelation.Mnc**.

Condition 2: The frequency of the source cell is equal to the value of **gNBNgHoFreqRelation.SourceCellSsbFreqPos**.

Condition 3: The frequency of the target cell is equal to the value of **gNBNgHoFreqRelation.TargetCellSsbFreqPos**.

NOTE

- When **gNBNgHoFreqRelation.SsbDescMethod** is set to **SSB_DESC_TYPE_GSCN**, **gNBNgHoFreqRelation.SourceCellSsbFreqPos** and **gNBNgHoFreqRelation.TargetCellSsbFreqPos** indicate the corresponding GSCNs.
- When **gNBNgHoFreqRelation.SsbDescMethod** is set to **SSB_DESC_TYPE_NARFCN**, **gNBNgHoFreqRelation.SourceCellSsbFreqPos** and **gNBNgHoFreqRelation.TargetCellSsbFreqPos** indicate the corresponding absolute frequency numbers.
- For inter-cell RRC connection reestablishment: The gNodeB checks whether the preceding conditions are met. If all the conditions are met, the reestablishment is not allowed. In this case, an RRC connection establishment procedure is triggered to enable the UE to access the cell from RRC_IDLE. If not all the conditions are met, the reestablishment is allowed.
- For inter-frequency handovers: The gNodeB determines whether to trigger an NG-based handover to change the serving 5GC of the UE. In this case, the gNodeB checks whether the preceding conditions are met. If all the conditions are met, the gNodeB triggers an NG-based handover. If not all the conditions are met, an NG-based handover is not triggered.

4.8.2 Network Analysis

4.8.2.1 Benefits

None

4.8.2.2 Impacts

The impacts between this function and other functions are described in [4.8.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.8.3 Requirements

4.8.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021211	Multi-Operator Sharing	NR0S00MULS00	per Cell
For the license control item NR0S00MULS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.8.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.8.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with common carrier	gNBSharingMode.gNBMultiOpSharingMode set to SHARED_FREQ	<i>Multi-Operator Sharing</i>	Frequency-based 5GC selection requires RAN sharing with common carrier or RAN sharing with dedicated carrier to be deployed on the live network.
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with dedicated carrier	gNBSharingMode.gNBMultiOpSharingMode set to SEPARATED_FREQ	<i>Multi-Operator Sharing</i>	

4.8.3.2.2 Mutually Exclusive Functions

None

4.8.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Network slice deployment	Multiple parameters are required for network slicing deployment. For details, see <i>Network Slicing</i> .	<i>Network Slicing</i>	When both network slicing introduction and frequency-based 5GC selection are enabled, there must be an NG interface for which gNBCUNg.CuNgType is set to DEFAULT_CP or DEFAULT_CP_UP among an operator's NG interfaces corresponding to the same gNBCUNgTrackingArea.TrackingAreaId value. Otherwise, the gNodeB cannot select a default AMF for a network slice UE when neither the S-NSSAIs supported by the gNodeB nor the Temp ID is reported from the UE, resulting in access failures. In live network planning, if a gNBCUNg MO of the default type (gNBCUNg.CuNgType set to DEFAULT_CP or DEFAULT_CP_UP) is configured, there must be an available NG link under the gNBCUNg MO of the default type. Otherwise, services will be affected.

4.8.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.8.3.4 Others

None

4.8.4 Operation and Maintenance

4.8.4.1 Data Configuration

4.8.4.1.1 Data Preparation

Table 4-48 Parameters used to specify tracking area information for a gNodeB NG interface

Parameter Name	Parameter ID	Setting Notes
gNodeB CU NG ID	gNBCUNgTrackingArea.gNBCuNgId	Set this parameter based on the network plan.
Tracking Area ID	gNBCUNgTrackingArea.TrackingAreald	Set this parameter based on the network plan. Note that different 5GCs correspond to different tracking area IDs.

NOTICE

A cell's **NRDUCell.TrackingAreald** or **NRDUCellOp.TrackingAreald** parameter value must be the same as the **gNBCUNgTrackingArea.TrackingAreald** parameter value for any of the NG interfaces of the involved operator so that the UEs' services in the cell can be routed to the corresponding 5GCs based on TAC information. Otherwise, the UEs fail to access the cell, or the cell is deactivated.

Table 4-49 Parameters used to specify frequency relationships for NG-based handovers

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	gNBNGHoFreqRelation.Mcc	Set this parameter based on the network plan.
Mobile Network Code	gNBNGHoFreqRelation.Mnc	Set this parameter based on the network plan.
SSB Frequency Position Describe Method	gNBNGHoFreqRelation.SsbDescMethod	Set this parameter based on the network plan.
Source Cell SSB Frequency Position	gNBNGHoFreqRelation.SourceCellSsbFreq-Pos	When gNBNGHoFreqRelation.SsbDescMethod is set to SSB_DESC_TYPE_GSCN, this parameter indicates the GSCN corresponding to the source cell's SSB. When gNBNGHoFreqRelation.SsbDescMethod is set to SSB_DESC_TYPE_NARFCN, this parameter indicates the absolute frequency number corresponding to the source cell's SSB.
Target Cell SSB Frequency Position	gNBNGHoFreqRelation.TargetCellSsbFreq-Pos	When gNBNGHoFreqRelation.SsbDescMethod is set to SSB_DESC_TYPE_GSCN, this parameter indicates the GSCN corresponding to the target cell's SSB. When gNBNGHoFreqRelation.SsbDescMethod is set to SSB_DESC_TYPE_NARFCN, this parameter indicates the absolute frequency number corresponding to the target cell's SSB.

4.8.4.1.2 Using MML Commands

Before using MML commands, refer to [4.8.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After

Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

The following examples assume that frequency-based 5GC selection is to be deployed with RAN sharing with common carrier. Given this assumption, operator A has two 5GCs, NG object IDs are 1 (corresponding to TAC 1) and 2 (corresponding to TAC 2), and cells 1 and 2 working on different frequencies correspond to TAC 1 and TAC 2, respectively.

```
//Configuring relationships between TACs and tracking area IDs on the gNodeB
ADD GNBTRACKINGAREA: TrackingAreaId=1, Tac=1;
ADD GNBTRACKINGAREA: TrackingAreaId=2, Tac=2;
//Configuring different TACs for cells
MOD NRDUCELL: NrDuCellId=1, NrDuCellName="1", DuplexMode=CELL_TDD, CellId=1, PhysicalCellId=1,
FrequencyBand=N78, Dlnarfcn=650000, ULBandwidth=CELL_BW_100M, DLBandwidth=CELL_BW_100M,
SlotAssignment=4_1_DDSU, SlotStructure=SS2, TrackingAreaId=1, SsbFreqPos=8000,
LogicalRootSequenceIndex=0, CustomizedBwConfigInd=NOT_CONFIG;
MOD NRDUCELL: NrDuCellId=2, NrDuCellName="2", DuplexMode=CELL_TDD, CellId=2, PhysicalCellId=2,
FrequencyBand=N78, Dlnarfcn=650000, ULBandwidth=CELL_BW_100M, DLBandwidth=CELL_BW_100M,
SlotAssignment=4_1_DDSU, SlotStructure=SS2, TrackingAreaId=2, SsbFreqPos=8010,
LogicalRootSequenceIndex=0, CustomizedBwConfigInd=NOT_CONFIG;
//Configuring TACs for NG interfaces on the gNodeB
ADD GNBCUNTRACKINGAREA: gNBCuNgId=1, TrackingAreaId=1;
ADD GNBCUNTRACKINGAREA: gNBCuNgId=2, TrackingAreaId=2;
//Configuring frequency relationships for NG-based handovers
ADD GNBNGHOFREQRELATION: Mcc="460", Mnc="01", SsbDescMethod=SSB_DESC_TYPE_GSCN,
SourceCellSsbFreqPos=8000, TargetCellSsbFreqPos=8010;
```

Optimization Command Examples

N/A

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Removing relationships between NG interfaces and TACs
RMV GNBCUNTRACKINGAREA: gNBCuNgId=1, TrackingAreaId=1;
RMV GNBCUNTRACKINGAREA: gNBCuNgId=2, TrackingAreaId=2;
//Removing frequency relationships for NG-based handovers
RMV GNBNGHOFREQRELATION: Mcc="460", Mnc="01", SourceCellSsbFreqPos=8000,
TargetCellSsbFreqPos=8010;
```

4.8.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.8.4.2 Activation Verification

Prerequisites

Frequency-based 5GC selection is deployed with RAN sharing with common carrier. Operator A has two 5GCs, NG object IDs are 1 (corresponding to TAC 1) and 2 (corresponding to TAC 2), and cells 1 and 2 working on different frequencies correspond to TAC 1 and TAC 2, respectively.

Procedure

- Step 1** Start NG signaling tracing on the MAE as follows: Log in to the MAE and choose **Monitor > Signaling Trace > Signaling Trace Management**. On the displayed page, choose **Trace Type > NR > Application Layer > NG Interface Trace**.
- Step 2** Enable a UE of operator A to access cell 1, and check the tracing result of the NG interface toward the 5GC corresponding to NG interface ID 1. The INIT_UE_MSG message of the UE is displayed.
- Step 3** Enable a UE of operator A to access cell 2, and check the tracing result of the NG interface toward the 5GC corresponding to NG interface ID 2. The INIT_UE_MSG message of the UE is displayed.

----End

4.8.4.3 Network Monitoring

NG-based handovers of UEs served by the gNodeB are operator-specifically counted into the counters listed in [Table 4-50](#).

Table 4-50 Counters related to NG-based handovers specific to a PLMN

Counter ID	Counter Name
1911827140	N.HO.Ng.IntergNB.PrepAttIn.PLMN
1911827143	N.HO.Ng.IntergNB.ExecAttIn.PLMN
1911827147	N.HO.Ng.IntergNB.ExecSucIn.PLMN
1911822985	N.HO.InterFreq.Ng.IntergNB.ExecAttOut.PLMN
1911823005	N.HO.InterFreq.Ng.IntergNB.PrepAttOut.PLMN
1911823015	N.HO.InterFreq.Ng.IntergNB.ExecSucOut.PLMN
1911827137	N.HO.Ng.IntergNB.Duration.Avg.PLMN

4.9 Operator-Specific Paging Resource Management

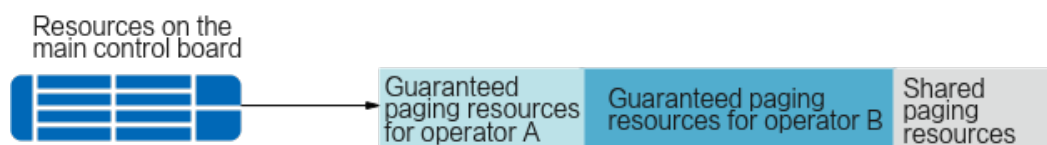
4.9.1 Principles

In multi-operator sharing scenarios, operators share the paging resources on the main control board of a base station, and the percentage of paging resources available for a single operator cannot be guaranteed. Against this background, operator-specific paging resource management is introduced to guarantee the percentage of paging resources available for each operator when paging congestion occurs. This function is controlled by the **gNBSharingMode.OpPagingResAdaptSw** parameter.

- If this parameter is set to **OFF**, operator-specific paging resource management is disabled. In this case, operators share paging resources and there is no operator-specific guarantee on paging resources.
- If this parameter is set to **CONFIG_MODE**, operator-specific paging resource configuration is enabled, and the percentage of paging resources available for an operator is specified by the **gNBOperator.PagingResRatio** parameter. The actual percentage of paging resources of the base station allocated to the operator will be greater than or equal to the setting of this parameter. Specifically, paging messages of the operator are processed preferentially using the paging resources that are guaranteed by the operator-specific percentage. If such paging resources have been used up, paging messages of the operator will be processed using the paging resources that are shared among operators. If the shared paging resources have also been used up, paging messages of the operator will be discarded. Note that the idle paging resources left within the guaranteed percentage for an operator at the end of a period can be shared with other operators in a subsequent period. The sum of percentages of paging resources configured for all operators must be less than or equal to 100%. If the sum of percentages is equal to 100%, the percentage of paging resources configured for any operator cannot be set to 0.
- If this parameter is set to **AUTO_MODE**, operator-specific paging resource adaptation is enabled, and the base station starts a seven-day learning phase about the changes in paging resource consumption of all operators. After the seven-day learning phase is complete, the base station adaptively and periodically adjusts the operator-specific percentages of guaranteed paging resources based on the current paging resource consumption of all operators. Specifically, paging messages of an operator are processed preferentially using the paging resources that are guaranteed by the operator-specific percentage. If such paging resources have been used up, paging messages of the operator will be processed using the paging resources that are shared among operators. Note that the idle paging resources left within the guaranteed percentage for an operator at the end of a period can be shared with other operators in a subsequent period.

Figure 4-15 shows an example of paging resource allocation for two operators.

Figure 4-15 Paging resource allocation



4.9.2 Network Analysis

4.9.2.1 Benefits

In multi-operator sharing scenarios with some operators encountering paging congestion, operator-specific paging resource management prevents these operators from using up all of the paging resources, thereby avoiding a decrease in the paging success rates of other operators.

When operator-specific paging resource configuration is enabled:

- The **gNBOperator.PagingResRatio** parameter indicates the percentage of paging resources available for an operator in multi-operator sharing scenarios. A smaller value of this parameter for an operator indicates fewer paging resources that can be occupied by the operator. This means a higher probability that paging packets are discarded and a lower paging success rate for the operator. For the other operators, by contrast, this means more paging resources that can be occupied, a lower probability that paging packets are discarded, and a higher paging success rate.
- A larger value of the **gNBOperator.PagingResRatio** parameter for an operator indicates more paging resources that can be occupied by the operator. This means a lower probability that paging packets are discarded and a higher paging success rate for the operator. For the other operators, by contrast, this means fewer paging resources that can be occupied, a higher probability that paging packets are discarded, and a lower paging success rate.

4.9.2.2 Impacts

The impacts between this function and other functions are described in [4.9.3.2.3 Function Impacts](#).

This function has no impact on the network.

4.9.3 Requirements

4.9.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021211	Multi-Operator Sharing	NR0S00MULS00	per Cell

Feature ID	Feature Name	Model	Sales Unit
For the license control item NR0S00MULS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

4.9.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.9.3.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with common carrier	gNBSharingMode .gNBMultiOpSharingMode set to SHARED_FREQ	<i>Multi-Operator Sharing</i>	Operator-specific paging resource management requires RAN sharing with common carrier or RAN sharing with dedicated carrier to be deployed on the live network.
Low-frequency TDD High-frequency TDD	Basic networking for RAN sharing with dedicated carrier	gNBSharingMode .gNBMultiOpSharingMode set to SEPARATED_FREQ	<i>Multi-Operator Sharing</i>	

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Operator-specific paging resource configuration	gNBSharingMode .OpPagingResAda ptSw set to CONFIG_MODE	<i>Multi-Operator Sharing</i>	The configured value of the gNBOperator.PagingResRatio parameter (indicating the operator-specific paging resource percentage) takes effect only when operator-specific paging resource configuration is enabled.

4.9.3.2.2 Mutually Exclusive Functions

None

4.9.3.2.3 Function Impacts

None

4.9.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

4.9.4 Operation and Maintenance

4.9.4.1 Data Configuration

4.9.4.1.1 Data Preparation

Table 4-51 describes the parameters used for function activation.

Table 4-51 Parameters used for activation

Parameter Name	Parameter ID	Setting Notes
Operator Paging Resource Adaptation Sw	gNBSharingMode.OpPagingResAdaptSw	Set this parameter based on the network plan.
Paging Resource Ratio	gNBOperator.PagingResRatio	Set this parameter based on the network plan.

4.9.4.1.2 Using MML Commands

Before using MML commands, refer to [4.9.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

```
//Enabling operator-specific paging resource adaptation
MOD GNBSHARINGMODE: OpPagingResAdaptSw=AUTO_MODE;
//Specifying the operator-specific paging resource percentages and enabling operator-specific paging resource configuration
MOD GNBOPERATOR: OperatorId=0, PagingResRatio=60;
MOD GNBOPERATOR: OperatorId=1, PagingResRatio=20;
MOD GNBSHARINGMODE: OpPagingResAdaptSw=CONFIG_MODE;
```

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

```
//Disabling operator-specific paging resource configuration
MOD GNBSHARINGMODE: OpPagingResAdaptSw=OFF;
```

4.9.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.9.4.2 Activation Verification

Operator-Specific Paging Resource Configuration

Prerequisites:

- The gNodeB is shared by two operators (operators A and B) in RAN sharing with common carrier. The basic networking for multi-operator sharing has been configured, and operator-specific paging resource configuration has been enabled.
- **gNBOperator.PagingResRatio** is set to **40** for operator A and **60** for operator B.

Expected result:

The percentage of paging resources used by each operator can be obtained from the values of counters in [Table 4-52](#). These configured percentages guarantee the paging resources that can be used by operators. In multi-operator sharing scenarios with some operators encountering paging congestion, these configured percentages of paging resources prevent other operators from encountering UE access failures due to no paging resources.

Operator-specific Paging Resource Adaptation

Prerequisites:

The gNodeB is shared by two operators (operators A and B) in RAN sharing with common carrier. The basic networking for multi-operator sharing has been configured, and operator-specific paging resource adaptation has been enabled.

Expected result:

The percentage of paging resources used by each operator can be obtained from the values of counters in [Table 4-52](#). These percentages are adaptively configured to guarantee the paging resources that can be used by operators. In multi-operator sharing scenarios with some operators encountering paging congestion, these configured percentages of paging resources prevent other operators from encountering UE access failures due to no paging resources.

In the **LST GNBOPERATOR** command output, if the value of the **gNBOperator.OpPagingResAdaptRatio** parameter (indicating the adaptively configured percentage of paging resources for an operator, with its values automatically recorded by the gNodeB) adaptively changes, operator-specific paging resource adaptation has taken effect.

4.9.4.3 Network Monitoring

Monitor the performance of the feature by observing the performance counters listed in [Table 4-52](#).

Table 4-52 Performance counters

Counter ID	Counter Name
1911823022	N.Paging.NG.Rx.PLMN
1911840481	N.Paging.NG.Rx.Discard.PLMN
1911840480	N.Paging.Ng.Dis.PchCong.PLMN

5 RAN Sharing with Dedicated Carrier

5.1 Basic Networking for RAN Sharing with Dedicated Carrier

5.1.1 Principles

5.1.1.1 Overview

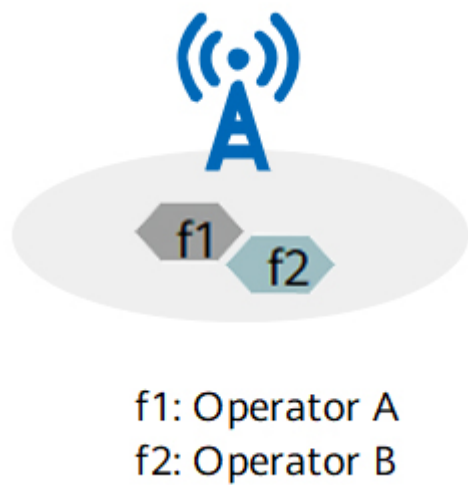
RAN sharing with dedicated carrier allows the gNodeB, but not cells served by the gNodeB, to be shared among operators. In a primary-operator cell, the PLMN ID of the primary operator is broadcast, whereas in a secondary-operator cell, the PLMN IDs of the primary and secondary operators are broadcast but the primary-operator PLMN ID is automatically configured as **reserved**. For details about the **reserved** state, see section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331 V15.5.0. The operator information of a cell is collectively configured using the **NRDUCellOp.NrDuCellId** and **NRDUCellOp.OperatorId** parameters in the **NRDUCellOp** MO. In RAN sharing with dedicated carrier, a maximum of one **NRDUCellOp** instance can be configured for each cell.

NOTE

After operator-specific NCI configuration (controlled by the **gNBSharingMode.MultipleNCISwitch** parameter) takes effect, only the PLMN ID of the operator to which a cell belongs is broadcast in the system information. Specifically, the cell belonging to the primary operator broadcasts the PLMN ID of the primary operator, whereas the cell of a secondary operator broadcasts the PLMN ID of the secondary operator. For details, see [5.3 Operator-Specific NCI Configuration](#).

[Figure 5-1](#) shows spectrum assignment in RAN sharing with dedicated carrier, whereby different operators use different frequencies.

Figure 5-1 Spectrum assignment in RAN sharing with dedicated carrier



RAN sharing with dedicated carrier can be used when:

- Operators that have their own radio spectrum share gNodeB hardware resources.
- An operator shares some of its licensed radio spectrum exclusively with another operator on a per cell basis.

RAN sharing with dedicated carrier is enabled by setting the **gNBSharingMode.gNBMultiOpSharingMode** parameter to **SEPARATED_FREQ**. The **gNBOperator.NrNetworkingOption** parameter specifies the networking option of an operator.

The evolution of networking modes has created a need for multi-operator sharing in the following networking scenarios: NSA networking, NSA and SA hybrid networking, and SA networking. [Table 5-1](#) provides details.

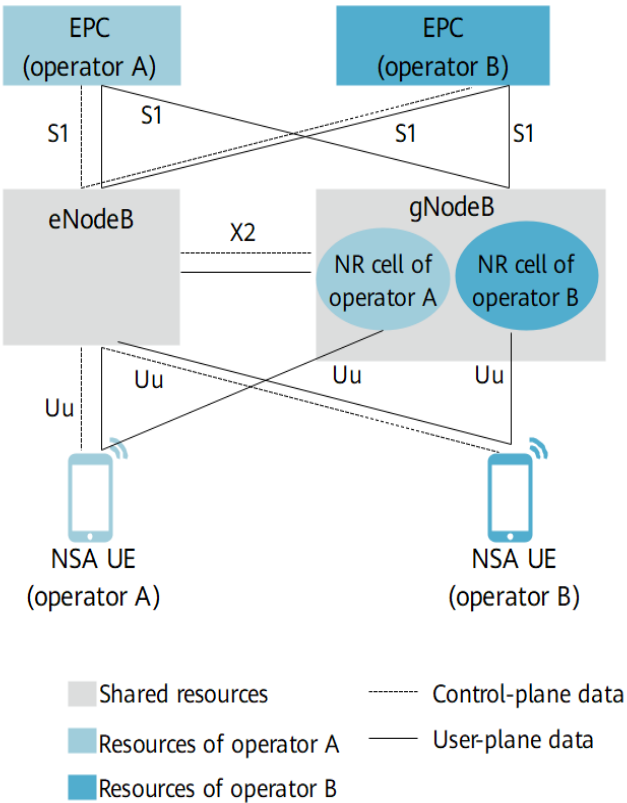
Table 5-1 Networking modes in multi-operator sharing

Networki ng Mode	Networking of Operators	Network Architecture
NSA networkin g	All operators use NSA networking and only NSA UEs can access the gNodeB.	Figure 5-2 shows the network architecture.

Networki ng Mode	Networking of Operators	Network Architecture
NSA and SA hybrid networkin g	Both NSA and SA UEs can access the gNodeB. The following two scenarios are involved: - Scenario 1: At least one operator uses NSA and SA hybrid networking. - Scenario 2: Some operators use SA networking, while others use NSA networking.	- Scenario 1: Figure 5-3 shows the network architecture. - Scenario 2: Figure 5-4 shows the network architecture.
SA networkin g	All operators use SA networking and only SA UEs can access the gNodeB.	Figure 5-5 shows the network architecture.

- RAN sharing with dedicated carrier in NSA networking
The **gNBOperator.NrNetworkingOption** parameter is set to **NSA** for all operators. In this networking mode, each cell served by the gNodeB belongs to only one operator.

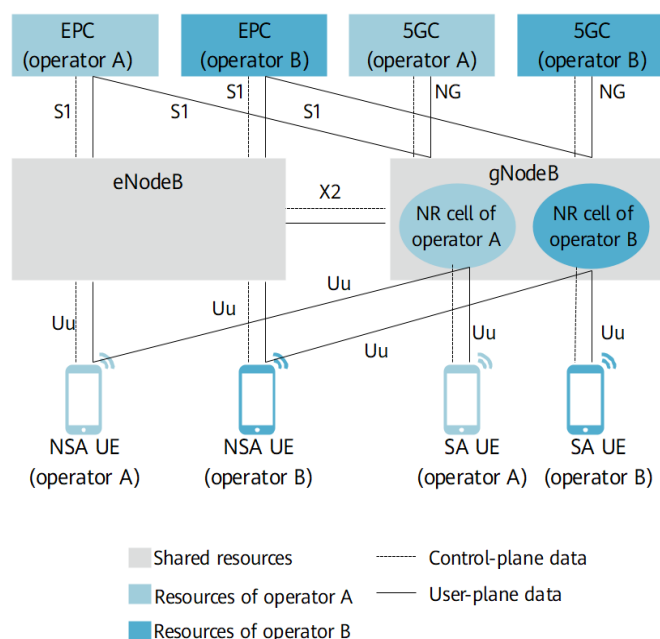
Figure 5-2 RAN sharing with dedicated carrier in NSA networking (using Option 3x as an example)



- RAN sharing with dedicated carrier in NSA and SA hybrid networking

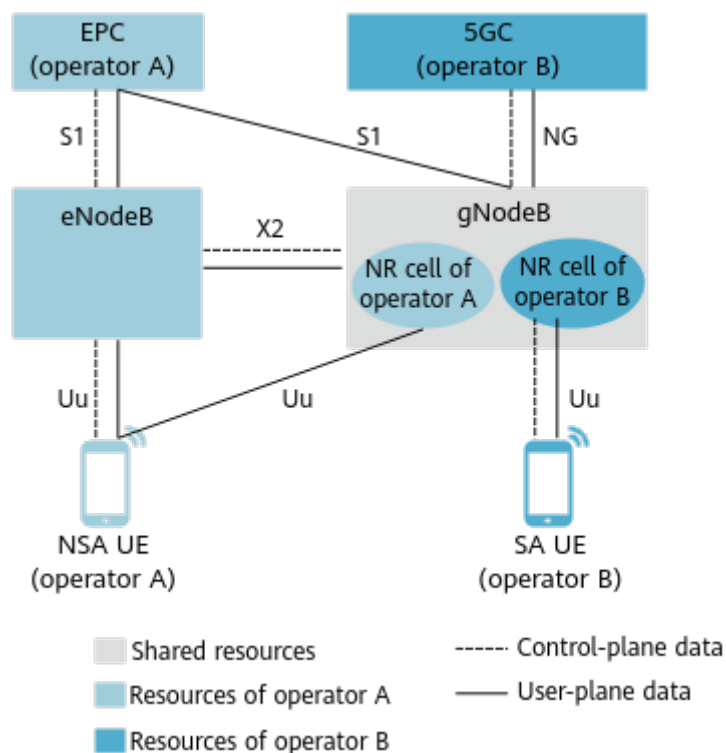
The **gNBOperator.NrNetworkingOption** parameter can be set to **SA_NSA** for all operators. Alternatively, this parameter can be set to **SA_NSA** for some operators and to **SA** or **NSA** for others. The following figure shows an example of the network architecture when the **gNBOperator.NrNetworkingOption** parameter is set to **SA_NSA** for all operators (operators A and B). In this scenario, each cell served by the gNodeB belongs to only one operator.

Figure 5-3 Scenario 1: RAN sharing with dedicated carrier in NSA and SA hybrid networking (using Option 3x and Option 2 as an example)



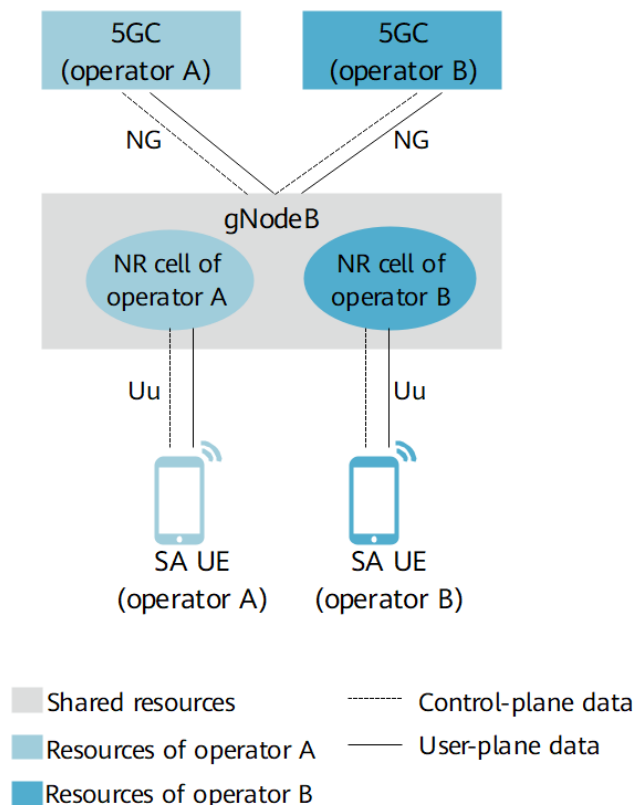
In this networking mode, the **gNBOperator.NrNetworkingOption** parameter can also be set to **SA** for some operators and to **NSA** for others. The following figure shows an example of the network architecture when this parameter is set to **NSA** for operator A and **SA** for operator B. In this scenario, each cell served by the gNodeB belongs to only one operator.

Figure 5-4 Scenario 2: RAN sharing with dedicated carrier in NSA and SA hybrid networking (using Option 3x and Option 2 as an example)



- RAN sharing with dedicated carrier in SA networking
The **gNBOperator.NrNetworkingOption** parameter is set to **SA** for all operators. In this networking mode, each cell served by the gNodeB belongs to only one operator.

Figure 5-5 RAN sharing with dedicated carrier in SA networking (using Option 2 as an example)



NOTE

Sharing on the gNodeB side does not necessarily require sharing on the eNodeB side, meaning that sharing on the eNodeB side can be enabled or not. In the preceding examples of RAN sharing with dedicated carrier in NSA networking and RAN sharing with dedicated carrier in NSA and SA hybrid networking, sharing is enabled on both the eNodeB and gNodeB. For details about multi-operator sharing on the eNodeB side, see *RAN Sharing of eRAN Feature Documentation*.

RAN sharing with dedicated carrier differs from the independent operator mode as follows:

- Mobility management. For details, see [5.1.1.3 Mobility Management in Idle Mode](#) and [5.1.1.4 Mobility Management in Connected Mode](#).
- Interface management. For details, see [5.1.1.2 Network Sharing Among More Operators](#).
- Feature and capacity management. For details, see [5.1.1.6 License Management](#).
- OSS management. For details, see [5.1.1.7 OSS Management](#).

5.1.1.2 Network Sharing Among More Operators

Network sharing among more operators in RAN sharing with dedicated carrier is the same as that in RAN sharing with common carrier. For details, see [4.1.1.2 Network Sharing Among More Operators](#).

5.1.1.3 Mobility Management in Idle Mode

5.1.1.3.1 NSA Mobility Management in Idle Mode

In NSA networking, for an operator with the **gNBOperator.NrNetworkingOption** parameter set to **NSA**, the *PLMN-IdentityInfo* IE contained in the system information does not contain TAC information. For details, see section 6.3.2 "Radio resource control information elements" of 3GPP TS 38.331 V15.5.0.

5.1.1.3.2 SA Mobility Management in Idle Mode

In SA networking, dedicated-carrier sharing cells are independent among operators. As such, SA mobility management in idle mode specific to RAN sharing with dedicated carrier is not required.

The state of a cell is signaled to UEs in SIB1. If the NG link of the primary operator or a secondary operator is faulty or not set up, the gNodeB automatically changes the state to **Barred** for the cell of the corresponding operator.

5.1.1.4 Mobility Management in Connected Mode

5.1.1.4.1 NSA Mobility Management in Connected Mode

In NSA networking, NSA DC UEs access both an eNodeB and a gNodeB to use the radio resources of both base stations for transmission. The eNodeB serves as the MeNB and connects to the EPC, and the gNodeB serves as the SgNB and connects to the eNodeB over an X2 interface. The UEs connect to both the PCell served by the MeNB and the PSCell served by the SgNB. For details, see *EPC-based NSA Basics*, *EPC-based NSA Performance Enhancement*, and *EPC-based NSA Experience Improvement*. In regard to NSA mobility management in connected mode, RAN sharing with dedicated carrier mainly differs from the independent operator mode as follows (refer to section 9.1.4 "Messages for E-UTRAN-NR Dual Connectivity Procedures" of 3GPP TS 36.423 V15.4.0 for details).

Configuration of Shared Neighboring Cells

Table 5-2 Parameters for configuring shared neighboring cells in NSA networking

Scenario	NE	MO	MO Configuration Information
Intra-RAT cases	gNodeB	NRExternalNC ellPlmn	NR external-cell PLMN list
		NRCellRelation	New neighbor relationship
	eNodeB	NrExternalCell Plmn	NR external-cell PLMN list
		NrNRRelationsh ip	New neighbor relationship
Inter-RAT cases	No related configuration is involved, as management of mobility from NR cells to E-UTRAN cells is not required.		

On a gNodeB with a dedicated-carrier sharing cell configured as a neighboring cell, the information about the shared neighboring cell must be configured for the eNodeB to add the PSCell or for the gNodeB to change the PSCell.

Intra-RAT cases

- gNodeB: When a neighboring NR cell belongs to a secondary operator of a neighboring base station, the following configuration is required for the gNodeB to change the PSCell.
 - Use the **NRExternalNCellPlmn** MO to configure the NR external neighboring cell PLMN list.
 - Use the **NRCellRelation** MO to configure a new neighbor relationship.
- eNodeB: When a neighboring NR cell belongs to a secondary operator of a neighboring base station, the following configuration is required for the eNodeB to add the PSCell.
 - Use the **NrExternalCellPlmn** MO to configure the NR external neighboring cell PLMN list.
 - Use the **NrNRRelationship** MO to configure a new neighbor relationship.

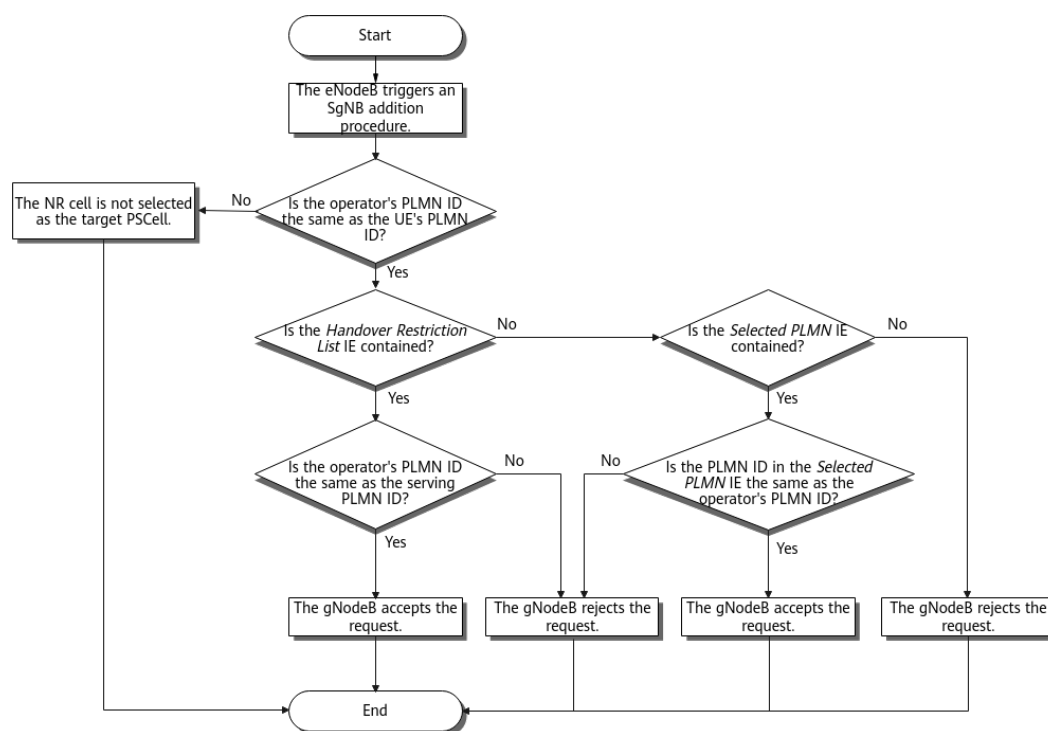
Inter-RAT cases

Management of mobility from NR cells to E-UTRAN cells is not required, and there is no need to perform related configuration.

Adding the PSCell

Figure 5-6 shows the procedure for adding the PSCell.

Figure 5-6 Procedure for adding the PSCell



In an eNodeB-triggered SgNB addition procedure, the eNodeB checks whether an NR cell meets any of the following conditions:

- The primary-operator PLMN ID of the NR cell is the same as the PLMN ID or EPLMN ID of the UE and no secondary-operator PLMN ID is configured for the NR cell.
- The secondary-operator PLMN ID of the NR cell is the same as the PLMN ID or EPLMN ID of the UE.

If any of these conditions is met, the eNodeB selects the NR cell as the target PSCell and sends a signaling message to the gNodeB.

The gNodeB checks whether the *Handover Restriction List* IE is contained in the SGNB ADDITION REQUEST message sent by the eNodeB.

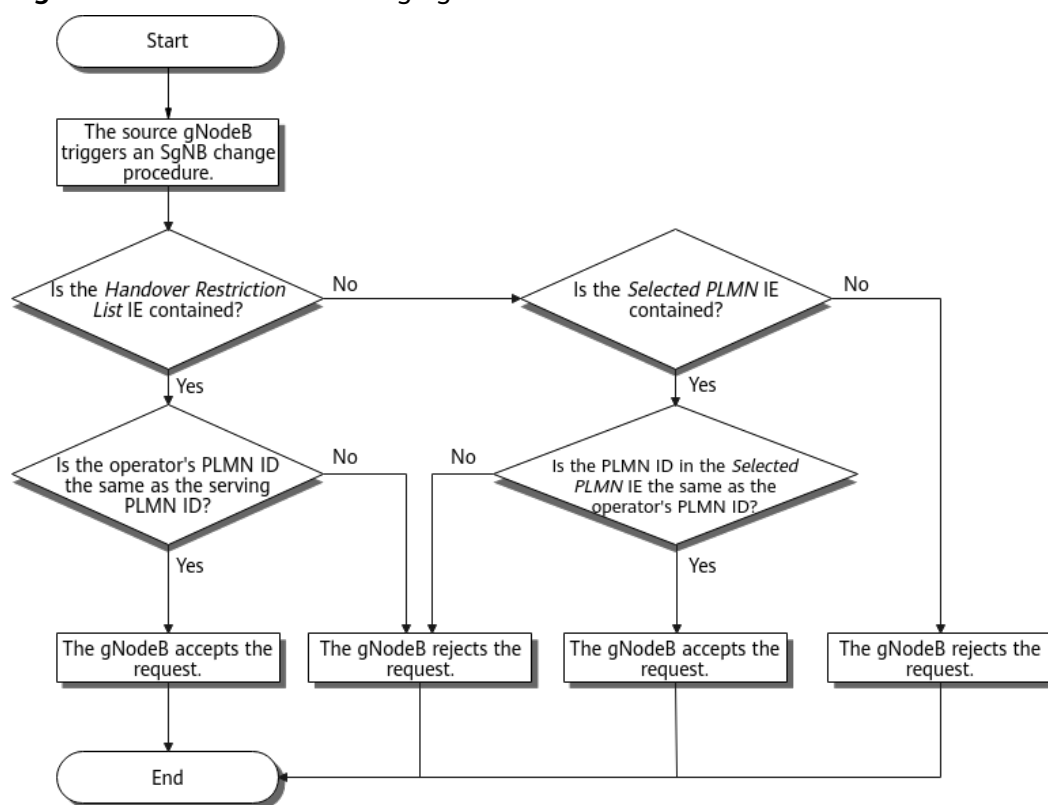
- If it is contained, the serving PLMN ID indicated in the *Handover Restriction List* IE is compared with the primary-operator PLMN ID or secondary-operator PLMN ID of the target NR cell.
 - If the value is the same as the primary-operator PLMN ID and no secondary-operator PLMN ID is configured, or if the value is the same as the secondary-operator PLMN ID, the gNodeB accepts the request.
 - If the value is not the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the gNodeB rejects the request.
- If it is not contained, the gNodeB checks whether the *Selected PLMN* IE is contained in the SGNB MODIFICATION REQUEST or SGNB ADDITION REQUEST message sent by the eNodeB.
 - If it is contained, the PLMN ID indicated in the *Selected PLMN* IE is compared with the primary-operator PLMN ID or secondary-operator PLMN ID of the target NR cell.
 - If the value is the same as the primary-operator PLMN ID and no secondary-operator PLMN ID is configured, or if the value is the same as the secondary-operator PLMN ID, the gNodeB accepts the request.
 - If the value is not the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the gNodeB rejects the request.
 - If it is not contained, the gNodeB rejects the request.

If none of these conditions is met, the eNodeB does not select the NR cell as the target PSCell.

Changing the PSCell

Figure 5-7 shows the procedure for changing the PSCell.

Figure 5-7 Procedure for changing the PSCell



In an SgNB change procedure triggered by the source gNodeB, the source gNodeB selects an NR cell as the target PSCell based on the neighbor relationship, and sends a signaling message to the target gNodeB through the eNodeB. The target gNodeB checks whether the *Handover Restriction List* IE is contained in the SGNB ADDITION REQUEST message sent by the eNodeB.

- If it is contained, the serving PLMN ID indicated in the *Handover Restriction List* IE is compared with the primary-operator PLMN ID or secondary-operator PLMN ID of the target NR cell.
 - If the value is the same as the primary-operator PLMN ID and no secondary-operator PLMN ID is configured, or if the value is the same as the secondary-operator PLMN ID, the target gNodeB accepts the request.
 - If the value is not the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the target gNodeB rejects the request.
- If it is not contained, the target gNodeB checks whether the *Selected PLMN* IE is contained in the SGNB ADDITION REQUEST message sent by the eNodeB.
 - If it is contained, the PLMN ID indicated in the *Selected PLMN* IE is compared with the primary-operator PLMN ID or secondary-operator PLMN ID of the target NR cell.
 - If the value is the same as the primary-operator PLMN ID and no secondary-operator PLMN ID is configured, or the value is the same as the secondary-operator PLMN ID, the target gNodeB accepts the request.
 - If the value is not the same as the primary-operator PLMN ID or secondary-operator PLMN ID, the target gNodeB rejects the request.

- If it is not contained, the target gNodeB rejects the request.

5.1.1.4.2 SA Mobility Management in Connected Mode

On a gNodeB with a dedicated-carrier sharing cell configured as a neighboring cell, the information about the shared neighboring cell must be configured to help determine the target cell for a handover.

Intra-RAT cases

- gNodeB: When a neighboring NR cell belongs to a secondary operator of a neighboring base station, the following configuration is required for the source gNodeB to select the target NR cell during an intra-RAT handover.
 - Use the **NRExternalNCellPlmn** MO to configure the NR external neighboring cell PLMN list.
 - Use the **NRCellRelation** MO to configure a new neighbor relationship.
- eNodeB: No configuration is required.

Inter-RAT cases

- gNodeB: When a neighboring E-UTRAN cell belongs to a secondary operator of a neighboring base station, the following configuration is required for the source gNodeB to select the target E-UTRAN cell during an inter-RAT handover.
 - Use the **gNBEutraExtCellPlmn** MO to configure the E-UTRAN external-cell PLMN list.
 - Use the **NRCellEutraNRelation** MO to configure a new neighbor relationship.
- eNodeB: When a neighboring NR cell belongs to a secondary operator of a neighboring base station, the following configuration is required for the source eNodeB to select the target NR cell during an inter-RAT handover.
 - Use the **NrExternalCellPlmn** MO to configure the NR external-cell PLMN list.
 - Use the **NrNRelationship** MO to configure a new neighbor relationship.

5.1.1.5 Interface Management

Operator-specific eXn interface configuration enables operator-specific data transmission over the eXn interface(s). This function helps avoid mutual impacts between operator-specific services when the eXn interface(s) carry coordination-based services (such as inter-gNodeB CA) under non-ideal backhaul mode in RAN sharing with dedicated carrier. For details about the eXn interface(s), see *eXn Self-Management*. In RAN sharing with dedicated carrier, a gNodeB allows operators to share an eXn interface or use separated eXn interfaces by configuring the **gNBDUExnOP** and **gNBDUExn** MOs. [Table 5-3](#) describes the configuration requirements. If the local and peer base stations use inconsistent eXn modes (either separated eXn or shared eXn), operator-specific eXn transmission path setup in separated mode must be enabled. For details, see [Operator-specific eXn Transmission Path Setup in Separated Mode](#).

Table 5-3 Basic configuration requirements

Scenario	User-Plane IP Address Planning and Configuration	Endpoint Group Configuration	Interface Configuration
Separate eXn	Different host eXn IP addresses are planned, and a unique user-plane host is configured (using the USERPLANEHOST MO) for each operator.	A unique endpoint group is configured (using the gNBDUExnOP MO) for each operator.	If the gNBDUExnOP MO instances are configured for operators, each operator must use unique settings of both gNBDUExnOp.OperatorId and gNBDUExnOp.gNBDuExnId . If an operator does not have a gNBDUExnOP MO instance or experiences an eXn link fault, the operator will have no available eXn link.
Shared eXn	One host eXn IP address is planned, and the same user-plane host is configured (using the USERPLANEHOST MO) for all operators.	Different operators share an endpoint group (configured using the gNBDUExn MO).	The same settings of gNBDUExn.gNBDuExnId and gNBDUExn.UpEpGroupId are specified for all operators.

Note:

- A base station does not support concurrent configuration of separated and shared eXn interfaces. That is, a base station does not support concurrent configuration of the **gNBDUExnOP** and **gNBDUExn** MOs.
- In the configuration of the eXn interface(s), the local and peer base stations must use the same eXn mode—either separated eXn or shared eXn—for all involved operators. If they use different eXn modes, coordination-based services between the local and peer base stations will be interrupted. In separated eXn mode, the **gNBDUExnOP** MO must be configured for both the local and peer base stations.
- The value of **gNBDUExnOp.OperatorId** must be consistent with the value of **gNBOperator.OperatorId**.
- The configuration requirements for interface-related parameters are as follows:
 - The value of **gNBCUNg.UpEpGroupId** configured for an NG interface must differ from that of **gNBDUExnOp.UpEpGroupId** configured for an eXn interface.
 - The value of **gNBCUS1.UpEpGroupId** configured for an S1 interface must differ from that of **gNBDUExnOp.UpEpGroupId** configured for an eXn interface.
 - If the value of **gNBCUXn.UpEpGroupId** configured for an Xn interface differs from that of **gNBDUExnOp.UpEpGroupId** configured for an eXn interface, the referenced user-plane host ID (specified by **USERPLANEHOST.UPHOSTID**) must be different between them.
 - If the value of **gNBCUX2.UpEpGroupId** configured for an X2 interface differs from that of **gNBDUExnOp.UpEpGroupId** configured for an eXn interface, the referenced user-plane host ID (specified by **USERPLANEHOST.UPHOSTID**) must be different between them.
- Changing the value of **gNBDUExnOp.UpEpGroupId** may interrupt inter-base-station coordination-based services.
- **USERPLANEHOST.FLAG** referenced by the **EPGROUP** MO that is associated with **gNBDUExnOp.UpEpGroupId** must be set to **MASTER**.

 NOTE

When two or more operators share a base station, some of these operators may form an alliance. This alliance is established by setting **gNBUExnOp.UpEpGroupId** to the same value for all member operators in the alliance. The member operators in the alliance share the eXn user-plane path. It is recommended that operators have consistent alliance relationships across all of the shared base stations.

Operator-specific eXn Transmission Path Setup in Separated Mode

When the eXn interface(s) are configured in RAN sharing with dedicated carrier, the local and peer base stations must use the same eXn mode (either separated eXn or shared eXn). If they use different eXn modes, coordination-based services between the local and peer base stations will be interrupted. The operator-specific eXn transmission path setup in separated mode function is used to make the coordination-based services between base stations immune from different eXn modes configured for the base stations. This function is controlled by the **FLEX_EXNOP_SW** option of the **gNBInterfaceParam.InterfacePolicySwitch** parameter. If (1) this function is enabled on all base stations that support coordination-based services, and (2) an operator has different eXn modes configured on the local and peer base stations, this function can allow the operator to establish an eXn user-plane transmission path between base stations in separated eXn mode. This ensures normal running of the operator's coordination-based services between base stations.

Other interface management functions in RAN sharing with dedicated carrier are the same as those in RAN sharing with common carrier. For details, see [4.1.1.6 Interface Management](#).

5.1.1.6 License Management

A license is an authorization agreement between the vendor and the customer in terms of the application scope and service lifetime of products.

In RAN sharing with dedicated carrier, a license purchase order is placed by only one party, which can either be the primary operator or a joint venture with multiple operators. Each NE has only one license file, which does not distinguish between operators.

License management includes feature and capacity license management.

5.1.1.6.1 Feature License Management

Feature licenses are shared among operators, and their sales are not operator-specific. Cell-level features can be independently enabled in cells of specific operators.

5.1.1.6.2 Capacity License Management

Capacity licenses are shared among operators, and their sales are not operator-specific. Percentages of the licensed limits defined by some capacity licenses, including the RRC Connected User License and RE(Resource Element) License, can be operator-specific. In this way, this mechanism is the same as that in RAN

sharing with common carrier. For details, see [5.2 Operator-Specific License Management](#).

5.1.1.7 OSS Management

The OSS mainly provides configuration management and performance management for both shared and non-shared gNodeBs.

Configuration Management

In RAN sharing with dedicated carrier, OSS management supports operator-specific independent cell-level parameter configuration so that their parameter values do not need to be negotiated. The values of base-station-level parameters need to be negotiated among operators.

Cell-level common configurations and base-station-level common configurations are reported over the northbound interface (Itf-N) of the OSS. The reported configurations are the same as those in the independent operator mode.

The scope of operator information query on the OSS depends on operator-specific account permissions.

- Only information about a secondary operator can be queried using an account of the secondary operator.
- Information about all operators can be queried using an account of the primary operator.

Performance Management

On the gNodeB, statistics collection on some key performance counters is operator-specific. For details, see [5.1.4.3 Network Monitoring](#).

Performance counter reporting over the Itf-N of the OSS is operator-specific, meaning the OSS reports counters specific to each operator.

5.1.2 Network Analysis

5.1.2.1 Benefits

The gains in RAN sharing with dedicated carrier are the same as those in RAN sharing with common carrier. For details, see [4.1.2.1 Benefits](#).

5.1.2.2 Impacts

The impacts between this function and other functions are described in [5.1.3.2.3 Function Impacts](#).

If the configured percentage of licensed resources for an operator is smaller than the actual percentage required for the operator, UEs of the operator may fail to access the network due to insufficient licensed resources.

Operator-specific eXn interface configuration triggers establishment of new eXn user-plane paths, consuming the specifications of user-plane peers of an endpoint group. Such specifications include the specifications of the entire base station and

a single endpoint group. If data traffic cannot be transmitted on a newly established eXn user-plane path, ALM-25954 User Plane Fault will be reported. After an eXn user-plane path is established, operator-specific data is transmitted over the eXn interface for coordination-based services (such as inter-gNodeB CA) under non-ideal backhaul mode. As a result, the operator-specific transmission bandwidth occupied by coordination-based services may change, reducing the probability of transmission congestion for single operators. Other network impacts are as follows:

- Coordination data needs to be split based on operators before being transmitted. As a result, the average user-plane CPU usage (indicated by **VS.NRBoard.UPlane.CPULoad.Avg**) and the maximum user-plane CPU usage (indicated by **VS.NRBoard.UPlane.CPULoad.Max**) may increase on a baseband processing unit.
- Values of counters in the function subsets related to eXn-interface data transmission may change. Such function subsets include IP Measurement (Network.IP), gNodeB Transmission Interface Measurement (Traffic.TRPIP.gNodeB), and Cell User Quantity Measurement (Traffic.User.Cell).
- Values of counters in the function subsets related to coordination-based services may also change. Taking inter-gNodeB CA as an example, these function subsets include CA Measurement (CA.Cell) and CA Cell MAC Measurement (Traffic.MAC.CA.Cell).
- Base stations exchange more cell configuration information through the eXn interface(s). Consequently, in the **SCTP Link Measurement** function subset, the values of counters related to received and transmitted traffic increase.

For details about the network impacts of operator-specific eXn transmission path setup in separated mode, refer to the aforesaid network impacts of operator-specific eXn interface configuration. In addition, enabling operator-specific eXn transmission path setup in separated mode raises the probability of data traffic not being transmitted properly on newly established eXn user-plane paths. In this case, ALM-25954 User Plane Fault is reported more frequently.

5.1.3 Requirements

5.1.3.1 Licenses

Feature ID	Feature Name	Model	Sales Unit
FOFD-021211	Multi-Operator Sharing	NR0S00MULS00	per Cell

Feature ID	Feature Name	Model	Sales Unit
For the license control item NR0S00MULS00: • If the feature is enabled for an activated cell, the cell will not be automatically deactivated when the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation. However, after the cell is reestablished, the cell cannot be activated again due to license insufficiency. • If the feature is not enabled for an activated cell and the number of units of the license for the feature is insufficient due to license expiration or forcible license file installation, the cell cannot be activated again after the feature is enabled and the parameters that cause automatic cell reset are modified.			

 NOTE

The insufficiency of certain feature licenses affects cell activation. For details, see the **Is Cell Activation Affected by License Insufficiency** column in *License Control Item Lists*.

5.1.3.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

5.1.3.2.1 Prerequisite Functions

Function Name	Function Switch	Reference	Description
Basic networking for RAN sharing with dedicated carrier	gNBSharingMode.gNBMultiOpSharing-Mode set to SEPARATED_FREQ	<i>Multi-Operator Sharing</i>	Operator-specific eXn interface configuration requires RAN sharing with dedicated carrier to be deployed on the live network.

5.1.3.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Operator-specific NCI configuration	gNBSharingMode.MultipleNCISwitch	<i>Multi-Operator Sharing</i>	Operator-specific eXn interface configuration does not work with operator-specific NCI configuration. They cannot be both enabled.

5.1.3.2.3 Function Impacts

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	NSA networking based on EPC	NRCellAlgoSwitch.NsaDcSwitch	<i>EPC-based NSA Basics</i>	When both DC in NSA networking based on EPC and basic networking for RAN sharing with dedicated carrier are enabled, data split is only supported between an E-UTRAN cell and an NR cell that belong to the same operator.
Low-frequency TDD	Intra-base-station UL CoMP	INTRA_GNB_UL_COMP_SW option of the NRDUCellAlgoSwitch.CompSwitch parameter	<i>CoMP</i>	The PLMN list of a cooperating cell for a UE must contain the ID of the PLMN with which the UE has registered.
High-frequency TDD	Inter-frequency MLB for mmWave PCells	MMW_PCELL_INTEFREQ_MLB_SW option of the NRCellMlb.MlbAlgoEnhSwitch parameter	<i>Mobility Load Balancing in Connected Mode</i>	If a UE only supports certain operators in multi-operator sharing scenarios, the UE may fail to be handed over blindly during load redistribution.

RAT	Function Name	Function Switch	Reference	Description
Low-freque ncy TDD High-freque ncy TDD	Direct IPsec self-update	GEPMODELPARA .DIRIPSECUPDAT ESW	<i>eXn Self-Management</i>	When operators in an alliance use separated eXn objects and have inconsistent alliance relationships on neighboring base stations: • After direct IPsec self-update is enabled, user-plane peers may be added to an endpoint group, consuming the specifications of user-plane peers of the endpoint group. Such specifications include the specifications of the entire base station and a single endpoint group. • After direct IPsec self-update is enabled, eXn self-update may fail. If this is the case, ALM-25954 User Plane Fault is reported.

RAT	Function Name	Function Switch	Reference	Description
Low-frequency TDD High-frequency TDD	Direct IPsec self-update	GEPMODELPARA.DIRIPSECUPDATESW	<i>eXn Self-Management</i>	If (1) operator-specific eXn transmission path setup in separated mode is not enabled for all base stations that support coordination-based services, and (2) an operator has different eXn modes configured on the local and peer base stations, there are the following impact relationships: • After direct IPsec self-update is enabled, the local and peer base stations may add user-plane peers to an endpoint group, consuming the specifications of user-plane peers of the endpoint group. Such specifications include the specifications of the entire base station and a single endpoint group. • After direct IPsec self-update is enabled, eXn self-update may fail on the local and peer base stations. If this is the case, ALM-25954 User Plane Fault is reported.

5.1.3.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910.

DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

All NR-capable main control boards and baseband processing units support this function. For details, see the BBU technical specifications in *3900 & 5900 Series Base Station Product Documentation*.

RF Modules

This function does not depend on RF modules.

5.1.3.4 Others

None

5.1.4 Operation and Maintenance

5.1.4.1 Data Configuration

[5.1.1.5 Interface Management](#) presents a brief setting note on transmission configuration.

- For details about S1 and X2 interface configuration on the eNodeB side in endpoint mode, see *S1 and X2 Self-Management of eRAN Feature Documentation*. For details about S1 and X2 interface configuration on the eNodeB side in link mode, see *Transmission Resource Management of eRAN Feature Documentation*.
- For details about NG and Xn interface configuration on the gNodeB side, see *NG and Xn Self-Management*.
- For details about eXn interface configuration on the gNodeB side, see *eXn Self-Management*.

In RAN sharing with dedicated carrier, reconfiguration is required in any of the five scenarios described below.

NOTICE

A change to the sharing mode of the gNodeB does not take effect until the application referenced by the gNodeB function is manually restarted by running the **RST APP** command. The ID of the application referenced by the gNodeB function can be queried using the **LST GNODEBFUNCTION** command.

You are not advised to add or remove a carrier or reset a cell on a multi-carrier RF module during peak hours, as adding or removing a carrier interrupts signals of other carriers on the RF module.

You are not advised to add or remove a carrier on a multi-carrier high-frequency RF module during peak hours, as adding or removing a carrier causes other carriers on the RF module to be reestablished, which results in service drops on these carriers.

Scenario 1: Changing the Sharing Mode of the gNodeB from the Independent Operator Mode to RAN Sharing with Dedicated Carrier

In this scenario, the sharing mode of the gNodeB needs to be changed and cell-level operator information (about one primary operator and one secondary operator) needs to be added. The required MO reconfiguration is the same as that in RAN sharing with common carrier. For details, see [Scenario 1: Changing the Sharing Mode of the gNodeB from the Independent Operator Mode to RAN Sharing with Common Carrier](#).

Scenario 2: Changing the Sharing Mode of the gNodeB from RAN Sharing with Dedicated Carrier to the Independent Operator Mode

In this scenario, cell-level operator information needs to be removed and the sharing mode of the gNodeB needs to be changed. The required MO reconfiguration is the same as that in RAN sharing with common carrier. For details, see [Scenario 2: Changing the Sharing Mode of the gNodeB from RAN Sharing with Common Carrier to the Independent Operator Mode](#).

Scenario 3: Changing the Sharing Mode of the gNodeB from RAN Sharing with Dedicated Carrier to RAN Sharing with Common Carrier

In this scenario, cell-level operator information and the sharing mode of the gNodeB need to be changed. The required MO reconfiguration is as follows:

- NSA networking
 - a. Deactivate cells (using the **NRCCell** MO).
 - b. Configure the gNodeB to work in RAN sharing with common carrier (using the **gNBSharingMode** MO).
 - c. Add operators to the shared cells (using the **NRDUCellOp** MO).
 - d. Activate cells (using the **NRCCell** MO).
- NSA and SA hybrid networking

The configuration is the same as that in NSA networking.
- SA networking

The configuration is the same as that in NSA networking.

Scenario 4: Adding a Secondary Operator

In this scenario, a secondary operator needs to be added. The required MO reconfiguration is the same as that in RAN sharing with common carrier. For details, see [Scenario 4: Adding a Secondary Operator](#).

Scenario 5: Removing a Secondary Operator

In this scenario, a secondary operator needs to be deleted. The required MO reconfiguration is the same as that in RAN sharing with common carrier. For details, see [Scenario 5: Removing a Secondary Operator](#).

5.1.4.1.1 Data Preparation

The parameters for initial configuration related to different networking options are described as follows:

- NSA networking
 - [Table 5-4](#), [Table 5-5](#), and [Table 5-6](#) describe the parameters used for function activation.
 - The parameters used for neighboring cell configuration related to this function are as follows:
 - gNodeB: [Table 5-7](#) to [Table 5-9](#)
 - eNodeB: [Table 5-13](#) to [Table 5-15](#)
 - No parameters are required for function optimization.
- NSA and SA hybrid networking
 - [Table 5-4](#), [Table 5-5](#), and [Table 5-6](#) describe the parameters used for function activation.
 - The parameters used for neighboring cell configuration related to this function are as follows:
 - gNodeB: [Table 5-7](#) to [Table 5-12](#)
 - eNodeB: [Table 5-13](#) to [Table 5-15](#)
 - No parameters are required for function optimization.
- SA networking
 - [Table 5-4](#), [Table 5-5](#), and [Table 5-6](#) describe the parameters used for function activation.
 - [Table 5-7](#) to [Table 5-12](#) describe the parameters used for neighboring cell configuration related to this function.
 - No parameters are required for function optimization.

Configuring Parameters on the gNodeB

Table 5-4 Parameters for configuring the sharing mode of the gNodeB and operator information

Parameter Name	Parameter ID	Setting Notes
gNodeB Multi Operator Sharing Mode	gNBSharingMode.gNBMultiOpSharingMode	In RAN sharing with dedicated carrier, set this parameter to SEPARATED_FREQ .
Operator ID	gNBOperator.OperatorId	Set this parameter based on the network plan.
Mobile Country Code	gNBOperator.Mcc	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Mobile Network Code	gNBOperator.Mnc	Set this parameter based on the network plan.
Operator Name	gNBOperator.Operator Name	Set this parameter based on the network plan.
Operator Type	gNBOperator.Operator Type	Set this parameter based on the network plan.
NR Networking Option	gNBOperator.NrNetworkingOption	• In NSA networking, set this parameter to NSA for all operators. • In NSA and SA hybrid networking, the following settings are supported: – Set this parameter to SA_NSA for all operators. – Set this parameter to SA_NSA for some operators and to SA or NSA for other operators. – Set this parameter to SA for some operators and to NSA for other operators. • In SA networking, set this parameter to SA for all operators.
NR DU Cell ID ^a	NRDUCellOp.NrDuCellId	Set this parameter based on the network plan.
Operator ID	NRDUCellOp.OperatorId	Set this parameter based on the network plan.
a: If the NRDUCell.DuplexMode parameter is set to CELL_SUL , the NRDUCellOp MO does not need to be configured.		

Table 5-5 Parameters for configuring a tracking area

Parameter Name	Parameter ID	Setting Notes
Tracking Area Code	gNBTrackingArea.Tac	Set this parameter to the invalid TAC (4294967295) for an operator for which the gNBOperator.NrNetworkingOption parameter is set to NSA . Set this parameter to a valid TAC (ranging from 0 to 16777215) for an operator for which the gNBOperator.NrNetworkingOption parameter is set to SA or SA_NSA .
Tracking Area ID	gNBTrackingArea.TrackingAreaId	Set this parameter based on the network plan.
NR DU Cell ID	NRDUCell.NrDuCellId	Set this parameter based on the network plan.
Tracking Area ID	NRDUCell.TrackingAreaId	Set this parameter to the tracking area ID that corresponds to the invalid TAC for the cell of an operator for which the gNBOperator.NrNetworkingOption parameter is set to NSA . Set this parameter to the tracking area ID that corresponds to a valid TAC for the cell of an operator for which the gNBOperator.NrNetworkingOption parameter is set to SA or SA_NSA .

Table 5-6 Parameters for configuring interfaces

Parameter Name	Parameter ID	Setting Notes
gNodeB CU S1 ID	gNBCUS1Op.gNBCuS1Id	Configure these parameters only when NSA Option 3x is used on live networks.
Operator ID	gNBCUS1Op.OperatorId	

Parameter Name	Parameter ID	Setting Notes
gNodeB CU NG ID	gNBCUNgOp.gNBCuNgId	Configure these parameters only when the gNBOperator.NrNetworkingOption parameter is set to SA or SA_NSA for an operator.
Operator ID	gNBCUNgOp.OperatorId	
gNodeB CU Xn ID	gNBCUXnOp.gNBCuXnId	Set these parameters based on the network plan. It is recommended that one operator be configured with only one Xn object.
Operator ID	gNBCUXnOp.OperatorId	
gNodeB CU X2 ID	gNBCUX2Op.gNBCuX2Id	Set these parameters based on the network plan.
Operator ID	gNBCUX2Op.OperatorId	
gNodeB DU eXn ID	gNBDUExn.gNBDuExnId	Set these parameters based on the network plan.
User Plane Endpoint Group ID	gNBDUExn.UpEpGroupId	
Operator ID	gNBDUExnOp.OperatorId	Set these parameters based on the network plan. You are advised to replan the IP addresses.
gNodeB DU eXn ID	gNBDUExnOp.gNBDuExnId	
User Plane Endpoint Group ID	gNBDUExnOp.UpEpGroupId	
Interface Policy Switch	gNBInterfaceParam.InterfacePolicySwitch	Select the FLEX_EXNOP_SW option when operator-specific eXn transmission path setup in separated mode needs to be enabled.
Note: This table lists only the interface configuration parameters that must be additionally prepared for multi-operator sharing. For details about other interface configuration parameters, see <i>X2 and S1 Self-Management in NSA Networking</i> , <i>NG and Xn Self-Management</i> , <i>eXn Self-Management</i> , and <i>S1 and X2 Self-Management in eRAN Feature Documentation</i> .		

Table 5-7 Parameters for configuring an NR external neighboring cell of the gNodeB

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	NRExternalNCell.Mcc	- If the gNodeB serving an NR external neighboring cell works in multi-operator sharing, set these parameters to the MCC and MNC of the primary operator to which the external NR cell belongs, separately. - If the gNodeB serving an NR external neighboring cell does not work in multi-operator sharing, set these parameters to the MCC and MNC of the operator to which the external NR cell belongs, separately.
Mobile Network Code	NRExternalNCell.Mnc	
gNodeB ID	NRExternalNCell.gNBId	Set this parameter based on the network plan. gNodeB IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NRExternalNCell.CellId	Set this parameter based on the network plan.
Physical Cell ID	NRExternalNCell.PhysicalCellId	Set this parameter based on the network plan.
SSB Frequency Position	NRExternalNCell.SsbFrequencyPos	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Tracking Area Code	NRExternalNCell.Tac	Set this parameter to the invalid TAC (4294967295) for an external neighboring cell for which the NRExternalNCell.NrNetworkingOption parameter is set to NSA . Set this parameter to a valid TAC (ranging from 0 to 16777215) for an external neighboring cell for which the NRExternalNCell.NrNetworkingOption parameter is set to SA or SA_NSA .
NR Networking Option	NRExternalNCell.NrNetworkingOption	Set this parameter based on the network plan.
PLMN Reserved Flag	NRExternalNCell.PlmnReservedFlag	Set this parameter to TRUE if the gNodeB serving an NR external neighboring cell works in RAN sharing with dedicated carrier, and the external neighboring cell belongs to a secondary operator.

Table 5-8 Parameters for configuring the NR external neighboring cell PLMN list of the gNodeB

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NRExternalNCellPlmn.gNBId	Set this parameter based on the network plan. gNodeB IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NRExternalNCellPlmn.CellId	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	NRExternalNCellPlmn.<i>Mcc</i>	Set these parameters to the MCC and MNC of the primary operator to which the NR external neighboring cell belongs, separately.
Mobile Network Code	NRExternalNCellPlmn.<i>Mnc</i>	
Shared Mobile Country Code	NRExternalNCellPlmn.<i>SharedMcc</i>	Set these parameters to the MCC and MNC of the secondary operator to which the NR external neighboring cell belongs, separately.
Shared Mobile Network Code	NRExternalNCellPlmn.<i>SharedMnc</i>	
Tracking Area Code	NRExternalNCellPlmn.<i>Tac</i>	Set this parameter to the secondary-operator TAC of an NR external neighboring cell. Set this parameter to the invalid TAC (4294967295) for an external neighboring cell for which the NRExternalNCellPlmn.<i>NrNetworkingOption</i> parameter is set to NSA. Set this parameter to a valid TAC (ranging from 0 to 16777215) for an external neighboring cell for which the NRExternalNCellPlmn.<i>NrNetworkingOption</i> parameter is set to SA or SA_NSA.
NR Networking Option	NRExternalNCellPlmn.<i>NrNetworkingOption</i>	Set this parameter to the secondary-operator NR networking option of an NR external neighboring cell.
Note: The preceding parameters need to be configured only when the gNodeB serving an NR external neighboring cell works in multi-operator sharing.		

Table 5-9 Parameters for configuring a relationship between NR cells

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCellRelation.NrCellId	Set this parameter based on the network plan.
Mobile Country Code	NRCellRelation.Mcc	Set these parameters to the MCC and MNC of the primary operator to which the neighboring NR cell belongs, separately.
Mobile Network Code	NRCellRelation.Mnc	
gNodeB ID	NRCellRelation.gNBId	Set this parameter based on the network plan. gNodeB IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NRCellRelation.CellId	Set this parameter based on the network plan.

Table 5-10 Parameters for configuring an external E-UTRAN cell of the gNodeB

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	gNBEutraExternal-Cell.Mcc	- If the eNodeB serving an external E-UTRAN cell works in RAN sharing with common carrier, RAN sharing with special dedicated carrier, or hybrid RAN sharing, set these parameters to the MCC and MNC of the primary operator to which the external E-UTRAN cell belongs, separately. - If the eNodeB serving an external E-UTRAN cell works in RAN sharing with dedicated carrier or does not work in RAN sharing, set these parameters to the MCC and MNC of the operator to which the external E-UTRAN cell belongs, separately.
Mobile Network Code	gNBEutraExternal-Cell.Mnc	
eNodeB ID	gNBEutraExternal-Cell.EnodebId	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	gNBEutraExternal-Cell.CelId	Set this parameter based on the network plan.
Downlink EARFCN	gNBEutraExternal-Cell.DlEarfcn	Set this parameter based on the network plan.
Physical Cell ID	gNBEutraExternal-Cell.PhysicalCellId	Set this parameter based on the network plan.
Tracking Area Code	gNBEutraExternal-Cell.Tac	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
Aggregation Attribute	gNBEutraExternal-Cell.AggregationAttribute	Select the MASTER_PLMN_RESERVED_FLAG option of the gNBEutraExternal-Cell.AggregationAttribute parameter if the eNodeB serving an external E-UTRAN cell works in RAN sharing with special dedicated carrier, and the external cell belongs to a secondary operator.

Table 5-11 Parameters for configuring the E-UTRAN external-cell PLMN list of the gNodeB

Parameter Name	Parameter ID	Setting Notes
Mobile Country Code	gNBEutraExtCellPlmn.Mcc	Set these parameters to the MCC and MNC of the primary operator to which the external E-UTRAN cell belongs, separately.
Mobile Network Code	gNBEutraExtCellPlmn.Mnc	
eNodeB ID	gNBEutraExtCellPlmn.EnodebId	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	gNBEutraExtCellPlmn.CellId	Set this parameter based on the network plan.
Shared Mobile Country Code	gNBEutraExtCellPlmn.SharedMcc	Set these parameters to the MCC and MNC of the secondary operator to which the external E-UTRAN cell belongs, separately.
Shared Mobile Network Code	gNBEutraExtCellPlmn.SharedMnc	
Note: The preceding parameters need to be configured only when the eNodeB serving an external E-UTRAN cell works in RAN sharing with common carrier or RAN sharing with special dedicated carrier.		

Table 5-12 Parameters for configuring the relationship of an NR cell with an E-UTRAN cell

Parameter Name	Parameter ID	Setting Notes
NR Cell ID	NRCeUtraNRRelation.NrCellId	Set this parameter based on the network plan.
Mobile Country Code	NRCeUtraNRRelation.Mcc	- If the eNodeB serving a neighboring E-UTRAN cell works in RAN sharing with common carrier, RAN sharing with special dedicated carrier, or hybrid RAN sharing, set these parameters to the MCC and MNC of the primary operator to which the neighboring E-UTRAN cell belongs, separately. - If the eNodeB serving a neighboring E-UTRAN cell works in RAN sharing with dedicated carrier or does not work in RAN sharing, set these parameters to the MCC and MNC of the operator to which the neighboring E-UTRAN cell belongs, separately.
Mobile Network Code	NRCeUtraNRRelation.Mnc	
eNodeB ID	NRCeUtraNRRelation.EnodebId	Set this parameter based on the network plan.
Cell ID	NRCeUtraNRRelation.CellId	Set this parameter based on the network plan.

Configuring Parameters on the eNodeB

Table 5-13 Parameters for configuring an external NR cell of the eNodeB

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NrExternalCell.GnodebId	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NrExternalCell.CellId	Set this parameter based on the network plan.
Mobile Country Code	NrExternalCell.Mcc	- If the gNodeB serving an external NR cell works in multi-operator sharing, set these parameters to the MCC and MNC of the primary operator to which the external NR cell belongs, separately. - If the gNodeB serving an external NR cell does not work in multi-operator sharing, set these parameters to the MCC and MNC of the operator to which the external NR cell belongs, separately.
Mobile Network Code	NrExternalCell.Mnc	
Physical Cell ID	NrExternalCell.PhyCellId	Set this parameter based on the network plan.
Tracking Area Code	NrExternalCell.Tac	Set this parameter based on the network plan.
Downlink ARFCN	NrExternalCell.DlArfcn	Set this parameter based on the network plan.
NR Networking Option	NrExternalCell.NrNetworkingOption	Set this parameter based on the network plan.

Table 5-14 Parameters for configuring the NR external-cell PLMN list of the eNodeB

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NrExternalCellPlmn.<i>GnodebId</i>	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NrExternalCellPlmn.<i>CellId</i>	Set this parameter based on the network plan.
Mobile Country Code	NrExternalCellPlmn.<i>Mcc</i>	Set these parameters to the MCC and MNC of the primary operator to which the external NR cell belongs, separately.
Mobile Network Code	NrExternalCellPlmn.<i>Mnc</i>	
Shared Mobile Country Code	NrExternalCellPlmn.<i>SharedMcc</i>	Set these parameters to the MCC and MNC of the secondary operator to which the external NR cell belongs, separately.
Shared Mobile Network Code	NrExternalCellPlmn.<i>SharedMnc</i>	
Tracking Area Code	NrExternalCellPlmn.<i>Tac</i>	This parameter indicates the ID of base station serving the external NR cell for a RAN sharing operator. The ID uniquely identifies the base station for the operator.
Note: The preceding parameters need to be configured only when the gNodeB serving an external NR cell works in multi-operator sharing.		

Table 5-15 Parameters for configuring a relationship of an E-UTRAN cell with an NR cell

Parameter Name	Parameter ID	Setting Notes
Local cell ID	NrNRRelationship.<i>LocalCellId</i>	Set this parameter based on the network plan.
Mobile Country Code	NrNRRelationship.<i>Mcc</i>	Set these parameters to the MCC and MNC of the primary operator to which the neighboring NR cell belongs, separately.
Mobile Network Code	NrNRRelationship.<i>Mnc</i>	

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	NrNRRelationship.Gnode bId	Set this parameter based on the network plan. Base station IDs on the live network must be planned in a unified manner and must be unique.
Cell ID	NrExternalCellPlmn.Cell Id	Set this parameter based on the network plan.

5.1.4.1.2 Using MML Commands

Before using MML commands, refer to [5.1.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

Activation Command Examples

Before activating this feature, you may need to retrieve basic operator information—for example, by running the **LST GNBOPERATOR** command. For details about basic MML commands used for multi-operator sharing, see **Man-Machine Interface Reference > 3900 & 5900 Series Base Station MML Command Reference > MML Command Reference > gNodeBFunction MML Commands > gNodeBFunction Management > gNBCU Management > Sharing Management** in *3900 & 5900 Series Base Station Product Documentation*.

In NSA and SA hybrid networking, the NG and Xn interfaces must be configured for the gNodeB serving SA UEs. For details about NG and Xn interface configurations, see *NG and Xn Self-Management*. Other configurations are as follows:

Example 1: In a new deployment for NSA and SA hybrid networking, the sharing mode of a gNodeB is changed from the independent operator mode to RAN sharing with dedicated carrier.

```
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Configuring the sharing mode, gNodeB-level operator information, and cell-level operator
information
//Configuring the sharing mode of the gNodeB
MOD GNBSHARINGMODE: gNBMultiOpSharingMode=SEPARATED_FREQ;
//Manually restarting the application referenced by the gNodeB function
RST APP: AID=1;
//Configuring the primary-operator information of the gNodeB
ADD GNBOPERATOR: OperatorId=0, OperatorName="unicom", OperatorType=PRIMARY_OPERATOR,
```

```
Mcc="460", Mnc="01", NrNetworkingOption=SA_NSA;
//Configuring the secondary-operator information of the gNodeB
ADD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,
Mcc="460", Mnc="03", NrNetworkingOption=SA_NSA;
//Configuring operator information for cell 1
ADD NRDUCELL: NrDuCellId=1, OperatorId=0;
//Configuring operator information for cell 2
ADD NRDUCELL: NrDuCellId=2, OperatorId=1;
//Configuring interfaces
//S1 and X2 interface configuration is the same as that in RAN sharing with common carrier. For details, see
the configuration examples for RAN sharing with common carrier.
//NG and Xn interface configuration is the same as that in RAN sharing with common carrier. For details,
see the configuration examples for RAN sharing with common carrier.
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
//Configuring neighboring cells
//Assuming that the neighboring base station works in RAN sharing with dedicated carrier
//Adding NR external neighboring cells, with Mcc and Mnc respectively set to the MCC and MNC of the
primary operator
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=2, CellId=3, PhysicalCellId=3, Tac=1,
SsbFreqPos=26638, NrNetworkingOption=SA_NSA;
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=2, CellId=4, PhysicalCellId=4, Tac=1,
SsbFreqPos=26639, NrNetworkingOption=SA_NSA;
//Adding external E-UTRAN cells, with Mcc and Mnc respectively set to the MCC and MNC of the operator
to which the cell belongs (not required in NSA networking)
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="01", EnodebId=0, CellId=3, DLearfcn=2200,
PhysicalCellId=3, Tac=0;
ADD GNBEUTRAEXTERNALCELL: Mcc="460", Mnc="01", EnodebId=0, CellId=4, DLearfcn=2200,
PhysicalCellId=4, Tac=0;
//Adding the secondary-operator information to the external neighboring cell PLMN list, with Mcc and Mnc
respectively set to the MCC and MNC of the primary operator
ADD NREXTERNALNCELLPLMN: Mcc="460", Mnc="01", gNBId=2, CellId=4, SharedMcc="460",
SharedMnc="03", Tac=1, NrNetworkingOption=SA_NSA;
ADD GNBEUTRAEXTCELLPLMN: Mcc="460", Mnc="01", EnodebId=0, CellId=4, SharedMcc="460",
SharedMnc="03";
//Adding the neighbor relationships with NR cells served by the neighboring base station (assuming that
NrCellId is 1 for the primary-operator cell and 2 for the secondary-operator cell), with Mcc and Mnc
respectively set to the MCC and MNC of the primary operator
ADD NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=2, CellId=3;
ADD NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=2, CellId=4;
//Adding the neighbor relationships with E-UTRAN cells served by the neighboring base station (assuming
that NrCellId is 1 for the primary-operator cell and 2 for the secondary-operator cell), with Mcc and Mnc
respectively set to the MCC and MNC of the operator to which the cell belongs (not required in NSA
networking)
ADD NRCELLEUTRANRELATION: NrCellId=1, Mcc="460", Mnc="01", EnodebId=0, CellId=3;
ADD NRCELLEUTRANRELATION: NrCellId=2, Mcc="460", Mnc="01", EnodebId=0, CellId=4;
//Configuring the TAC (not required in NSA networking)
//Configuring a valid TAC for the operator with the NR networking option set to SA or SA_NSA
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;
//Configuring a valid TAC for the cell of the operator with the NR networking option set to SA or SA_NSA
ADD NRDUCELL: NrDuCellId=1, NrDuCellName="1", DuplexMode=CELL_TDD, CellId=1, PhysicalCellId=1,
FrequencyBand=N77, DLNarfcn=620000, ULBandwidth=CELL_BW_40M, DLBandwidth=CELL_BW_40M,
SlotAssignment=4_1_DDSU, SlotStructure=SS2, TrackingAreaId=0, SsbDescMethod=SSB_DESC_TYPE_GSCN,
SsbFreqPos=7712, LogicalRootSequenceIndex=0;
```

Example 2: The sharing mode of a gNodeB that originally belonged to one operator adopting NSA networking is changed from the independent operator mode to RAN sharing with dedicated carrier, after an operator adopting SA networking is added.

```
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Configuring the sharing mode, gNodeB-level operator information, and cell-level operator
information
//Configuring the sharing mode of the gNodeB
MOD GNBSHARINGMODE: GNBMultiOpSharingMode=SEPARATED_FREQ;
//Manually restarting the application referenced by the gNodeB function
```

```
RST APP: AID=1;
//Configuring the secondary-operator information of the gNodeB
ADD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,
Mcc="460", Mnc="03", NrNetworkingOption=SA;
//Configuring operator information for cell 1
ADD NRDUCELL: NrDuCellId=1, OperatorId=0;
//Configuring operator information for cell 2
ADD NRDUCELL: NrDuCellId=2, OperatorId=1;
//Configuring neighboring cells
//Assuming that the neighboring base station works in RAN sharing with dedicated carrier
//Adding the secondary-operator information to the external-cell PLMN list, with Mcc and Mnc respectively
set to the MCC and MNC of the primary operator
ADD NREXTERNALNCELLPLMN: Mcc="460", Mnc="01", gNBId=2, CellId=4, SharedMcc="460",
SharedMnc="03", Tac=1, NrNetworkingOption=SA;
ADD GNBEUTRAEXTCELLPLMN: Mcc="460", Mnc="01", EnodebId=0, CellId=4, SharedMcc="460",
SharedMnc="03";
//Configuring the TAC
//Configuring a valid TAC for the operator with the NR networking option set to SA or SA_NSA
ADD GNBTRACKINGAREA: TrackingAreaId=0, Tac=1;
//Changing the TAC to a valid value for the cell of the operator with the NR networking option set to SA or
SA_NSA
MOD NRDUCELL: NrDuCellId=2, DuplexMode=CELL_TDD, TrackingAreaId=0;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
```

An eXn interface is configured specific to an operator, with an assumption that the IP address and related configurations of the eXn object have been added. For details on configuring an eXn interface and verifying the eXn interface setup result, see *eXn Self-Management*.

Example 1: New site deployment

```
//Configuring the eXn interface—either shared or separated, with mandatory consistency—for local
and peer base stations, based on the operator's policy
//(Case 1) Configuring a shared eXn interface (given an assumption that the IP address and related
configurations of the shared eXn interface have been added)
//Adding an eXn object on the gNodeB
ADD GNBDUEXN: gNBDuExnId=1, UpEpGroupId=1;
//(Case 2) Configuring separated eXn interfaces (specifying different eXn addresses and related
configurations for different operators)
//Adding operator information for a gNodeB DU eXn object (This operation is required for both the local
and peer base stations.)
ADD GNBDUEXNOP: gNBDuExnId=1, OperatorId=0, UpEpGroupId=1;
ADD GNBDUEXNOP: gNBDuExnId=2, OperatorId=1, UpEpGroupId=2;
//Configuring the eXn interface—whether shared or separated, with optional consistency—for local
and peer base stations, based on the operator's policy
//Enabling operator-specific eXn transmission path setup in separated mode (This operation is required for
all base stations that support coordination-based services.)
MOD GNBINTERFACEPARAM: InterfacePolicySwitch=FLEX_EXNOP_SW-1;
```

Example 2: Switching from shared eXn mode to separated eXn mode for operators that are not in an alliance on the live network

```
//Removing the shared eXn object on the gNodeB side
RMV GNBDUEXN: gNBDuExnId=1;
//Adding separated eXn interfaces (specifying different eXn addresses and related settings for different
operators)
//Adding operator information for a gNodeB DU eXn object (This operation is required for both the local
and peer base stations.)
ADD GNBDUEXNOP: gNBDuExnId=1, OperatorId=0, UpEpGroupId=1;
ADD GNBDUEXNOP: gNBDuExnId=2, OperatorId=1, UpEpGroupId=2;
```

Example 3: Switching from shared eXn mode to separated eXn mode for operators in an alliance on the live network

```
//Removing the shared eXn object on the gNodeB side
RMV GNBDUEXN: gNBDuExnId=1;
```

```
//Adding a separated eXn interface (given an assumption that two operators are in an alliance and share an
eXn user-plane path within the alliance)
//Adding operator information for a gNodeB DU eXn object (This operation is required for both the local
and peer base stations. The alliance policy must be the same among operators.)
ADD GNBDUEXNOP: gNBDuExnId=1, OperatorId=0, UpEpGroupId=1;
ADD GNBDUEXNOP: gNBDuExnId=2, OperatorId=1, UpEpGroupId=1;
ADD GNBDUEXNOP: gNBDuExnId=3, OperatorId=2, UpEpGroupId=3;
```

Optimization Command Examples

N/A

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

If the operator to be retained adopts SA networking, the deactivation command examples are the same as those in NSA networking; however, if it adopts NSA networking, the following deactivation command examples apply.

Example 1: The sharing mode of a gNodeB is changed from RAN sharing with dedicated carrier to the independent operator mode.

```
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Removing neighboring cell configurations
//Removing the neighbor relationships with NR cells served by the neighboring base station
RMV NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=2, CellId=3;
RMV NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=2, CellId=4;
//Removing the secondary-operator information from the NR external neighboring cell PLMN list
RMV NREXTERNALNCELLPLMN: CellId=4, gNBId=2, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03";
//Removing the NR external neighboring cell of the secondary operator
RMV NREXTERNALNCELL: Mcc="460", Mnc="03", gNBId=2, CellId=4;
//Deleting interface configurations
//S1 and X2 interface configuration is the same as that in RAN sharing with common carrier. For details, see
the configuration for RAN sharing with common carrier in NSA networking.
//NG and Xn interface configuration is the same as that in RAN sharing with common carrier. For details,
see the configuration for RAN sharing with common carrier in SA networking.
//Removing cell-level operator information and gNodeB-level operator information, and changing the
sharing mode
//Removing cell-level operator information
RMV NRDUCELLOP: NrDuCellId=1, OperatorId=0;
RMV NRDUCELLOP: NrDuCellId=2, OperatorId=1;
//Removing gNodeB-level secondary-operator information
RMV GNBOPERATOR: OperatorId=1;
//Changing the sharing mode of the gNodeB to the independent operator mode
MOD GNBSHARINGMODE: gNBMultiOpSharingMode=INDEPENDENT;
//Manually restarting the application referenced by the gNodeB function
RST APP: AID=1;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
```

Example 2: The sharing mode of a gNodeB is changed from RAN sharing with dedicated carrier to RAN sharing with common carrier.

```
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Adding cell-level operator information and changing the sharing mode
//Changing the sharing mode of the gNodeB to RAN sharing with common carrier
```

```
MOD GNBSHARINGMODE: gNBMultiOpSharingMode=SHARED_FREQ;  
//Manually restarting the application referenced by the gNodeB function  
RST APP: AID=1;  
//Adding the primary and secondary operators to the shared cells  
ADD NRDUCELL: NrdCellId=1, OperatorId=1;  
ADD NRDUCELL: NrdCellId=2, OperatorId=0;  
//Activating cells  
ACT NRCELL: NrdCellId=1;  
ACT NRCELL: NrdCellId=2;
```

5.1.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.1.4.2 Activation Verification

Basic Networking for RAN Sharing with Dedicated Carrier

Prerequisites:

- The gNodeB is shared based on the network plan by two operators in RAN sharing with dedicated carrier.
- The transmission links between the RAN and the core network are established and functioning normally.
- UE 1 and UE 2 separately belonging to two operators are ready.

Procedure:

Step 1 Query the cell status.

Run the MML command **DSP NRCELL** to check the cell status.

Expected result: **NR Cell State** is **Normal**.

Step 2 Check SIB1. Specifically, log in to the MAE-Access, and choose **Monitor > Signaling Trace > Signaling Trace Management > Trace Type > NR > Application Layer > Uu Interface Trace**.

Expected result: SIB1 contains the correct PLMN ID of each operator.

In SIB1 of cell 1 belonging to the primary operator, **PLMN-Identity** indicates **46001** for the primary operator. Additionally, the MCC and MNC indicated by **PLMN-Identity** are the same as those in the PLMN ID configured for the operator, showing that the PLMN ID of the operator is correctly delivered in SIB1 of cell 1.

```
systemInformationBlockType1  
cellSelectionInfo  
plmn-IdentityList  
PLMN-IdentityInfo  
plmn-IdentityList  
PLMN-Identity  
mcc  
MCC-MNC-Digit: ---- 0x4 (4)  
MCC-MNC-Digit: ---- 0x6 (6)  
MCC-MNC-Digit: ---- 0x0 (0)  
mnc
```

```
MCC-MNC-Digit: ---- 0x0 (0)
MCC-MNC-Digit: ---- 0x1 (1)
trackingAreaCode: ---- 000000000000000000000000 (00 00 00)
cellIdentity: ---- 000000000000000000000000000000010001 (00 00 00 01 01)
cellReservedForOperatorUse: ---- notReserved (1)
```

In SIB1 of cell 2 belonging to the secondary operator, **PLMN-Identity** indicates **46001** for the primary operator and **46003** for the secondary operator, and the state of the primary-operator PLMN is **reserved**. Additionally, the MCC and MNC indicated by **PLMN-Identity** are the same as those in the PLMN ID configured for the corresponding operator, showing that the PLMN ID of the corresponding operator is correctly delivered in SIB1 of cell 2.

```
systemInformationBlockType1
  cellAccessRelatedInfo
    plmn-IdentityList
      PLMN-IdentityInfo
        plmn-IdentityList
          PLMN-Identity
            mcc
              MCC-MNC-Digit: ---- 0x4 (4)
              MCC-MNC-Digit: ---- 0x6 (6)
              MCC-MNC-Digit: ---- 0x0 (0)
            mnc
              MCC-MNC-Digit: ---- 0x0 (0)
              MCC-MNC-Digit: ---- 0x1 (1)
            cellIdentity: ---- 000000000000000000000000000000010010 (00 00 00 01 02)
            cellReservedForOperatorUse: ---- reserved (0)
          PLMN-IdentityInfo
            plmn-IdentityList
              PLMN-Identity
                mcc
                  MCC-MNC-Digit: ---- 0x4 (4)
                  MCC-MNC-Digit: ---- 0x6 (6)
                  MCC-MNC-Digit: ---- 0x0 (0)
                mnc
                  MCC-MNC-Digit: ---- 0x0 (0)
                  MCC-MNC-Digit: ---- 0x3 (3)
            trackingAreaCode: ---- 000000000000000000000000 (00 00 00)
            cellIdentity: ---- 000000000000000000000000000000010010 (00 00 00 01 02)
            cellReservedForOperatorUse: ---- notReserved (1)
```

Step 3 Power on UE 1 and UE 2 and enable them to access cells.

Expected result: Service data is sent to respective core networks.

----End

NOTE

In SIB1, the *PLMN-IdentityInfo* IE contains **trackingAreaCode** for the operators with the **gNBOperator.NrNetworkingOption** parameter set to **SA** or **SA_NSA**, but not for the operators with the **gNBOperator.NrNetworkingOption** parameter set to **NSA**.

In a secondary-operator cell, the broadcast PLMN ID of the primary operator is automatically configured as **reserved**, and the *PLMN-IdentityInfo* IE does not contain **trackingAreaCode**.

5.1.4.3 Network Monitoring

In networks where RAN sharing with dedicated carrier is enabled, cells are exclusively used by individual operators. As a result, all cell-level performance counters can be monitored on a per operator basis, and can be used for preliminary fault demarcation and locating similar to that in independent operator mode.

5.2 Operator-Specific License Management

RAN sharing with dedicated carrier supports the same operator-specific license management functions as RAN sharing with common carrier. For details, see [4.2 Operator-Specific License Management](#).

5.3 Operator-Specific NCI Configuration

5.3.1 Principles

The principles of operator-specific NCI configuration in RAN sharing with dedicated carrier are the same as those in RAN sharing with common carrier. For details, see [4.6 Operator-Specific NCI Configuration](#).

After the operator-specific NCI configuration function takes effect, the system information only broadcasts the PLMN ID of the operator to which a cell belongs. Specifically, the cell belonging to the primary operator broadcasts only the primary-operator PLMN ID and the cell belonging to a secondary operator broadcasts only the secondary-operator PLMN ID.

5.3.2 Network Analysis

Network analysis for operator-specific NCI configuration in RAN sharing with dedicated carrier is the same as that in RAN sharing with common carrier. For details, see [4.6.2 Network Analysis](#).

5.3.3 Requirements

Requirements for operator-specific NCI configuration in RAN sharing with dedicated carrier are the same as those in RAN sharing with common carrier. For details, see [4.6.3 Requirements](#).

5.3.4 Operation and Maintenance

5.3.4.1 Data Configuration

5.3.4.1.1 Data Preparation

In RAN sharing with dedicated carrier, the parameters in the **NRExternalNCellPlmn** MO do not need to be configured on the gNodeB, and those in the **NrExternalCellPlmn** MO do not need to be configured on the eNodeB. Configurations of other parameters are the same as those in RAN sharing with common carrier. For details, see [4.6.4.1.1 Data Preparation](#).

5.3.4.1.2 Using MML Commands

Before using MML commands, refer to [4.6.3 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual

network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

This section uses NSA and SA hybrid networking as an example. The configurations in this networking are the same as those in NSA networking and SA networking. For details about the basic configurations of RAN sharing with dedicated carrier, see [5.1.4.1.2 Using MML Commands](#).

For this example, assume that both the local and neighboring base stations work in RAN sharing with dedicated carrier.

Activation Command Examples

Example 1: In new deployments, RAN sharing with dedicated carrier is used, and each operator has its own gNodeB ID and cell ID.

- Primary operator: For the local base station, the gNodeB ID is 111 and consists of 24 bits, and the ID of its served cell is 11. For the neighboring base station, the gNodeB ID is 211, and the ID of its served cell is 31.
- Secondary operator: For the local base station, the gNodeB ID is 112 and consists of 24 bits, and the ID of its served cell is 21. For the neighboring base station, the gNodeB ID is 212, and the ID of its served cell is 41.

S1, X2, NG, and Xn interface configuration is the same as that in RAN sharing with common carrier. For configuration examples of S1 and X2 interfaces, see [4.1.4.1.2 Using MML Commands](#). For configuration examples of NG and Xn interfaces, see [4.1.4.1.2 Using MML Commands](#).

gNodeB configurations:

```
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
//Configuring the sharing mode, gNodeB-level operator information, and cell-level operator
information
//Configuring the sharing mode of the gNodeB
MOD GNBSHARINGMODE: gNBMultiOpSharingMode=SEPARATED_FREQ;
//Manually restarting the application referenced by the gNodeB function
RST APP: AID=1;
//Configuring the gNodeB ID and gNodeB ID length for the primary operator of the gNodeB
ADD GNBOPERATOR: OperatorId=0, OperatorName="unicom", OperatorType=PRIMARY_OPERATOR,
Mcc="460", Mnc="01", NrNetworkingOption=SA_NSA, gNBId=111, gNBIdLength=24,
UserLabel="AirportSite";
//Configuring the gNodeB ID and gNodeB ID length for the secondary operator of the gNodeB
ADD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,
Mcc="460", Mnc="03", NrNetworkingOption=SA_NSA, gNBId=112, gNBIdLength=24,
UserLabel="AirportSiteTelecom";
//Configuring operator information for cell 1
ADD NRDUCELLOP: NrDuCellId=1, OperatorId=0, CellId=11, CellPrimaryOpFlag=TRUE;
//Configuring operator information for cell 2
ADD NRDUCELLOP: NrDuCellId=2, OperatorId=1, CellId=21, CellPrimaryOpFlag=TRUE;
//Configuring interfaces
//NG, S1, Xn, and X2 interface configuration is the same as that in RAN sharing with common carrier. For
details, see the configuration examples for RAN sharing with common carrier.
//Configuring neighboring cells
//Assuming that the local and neighboring base stations work in RAN sharing with dedicated carrier and
NCIs are operator-specific in the two cells served by the neighboring base station
```

```
//Adding NR external neighboring cells, with Mcc and Mnc respectively set to the MCC and MNC of the
primary operator
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=211, CellId=31, PhysicalCellId=3, Tac=1,
SsbFreqPos=26639, NrNetworkingOption=SA_NSA;
ADD NREXTERNALNCELL: Mcc="460", Mnc="03", gNBId=212, CellId=41, PhysicalCellId=4, Tac=1,
SsbFreqPos=26638, NrNetworkingOption=SA_NSA;
//Adding the neighbor relationships with NR cells served by the neighboring base station (assuming that
NrCellId of the two cells of the local base station is 1 and 2, separately), with Mcc and Mnc respectively set
to the MCC and MNC of the primary operator
ADD NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=211, CellId=31;
ADD NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="03", gNBId=212, CellId=41;
//Enabling operator-specific NCI configuration
MOD GNBSHARINGMODE: MultipleNCISwitch=ON, MultipleNciEnhanceSw=ON;
RST APP: AID=1;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
```

eNodeB configurations:

```
//Configuring the LTE-to-NR PLMN selection mode
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode= SELECTED_PLMN_ID;
//Optional) Configuring the LTE-to-NR PLMN selection mode for a specific gNodeB
ADD NBRGNODEBPOLICY: Mcc="460", Mnc="01", GnodebId=111,
LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//Configuring gNBIdLength for each involved operator
ADD NCELLPLMNLIST: Mcc="460", Mnc="01", RatType=NR, gNBIdLength=24;
ADD NCELLPLMNLIST: Mcc="460", Mnc="03", RatType=NR, gNBIdLength=24;
//Adding external NR cells with information being operator-specific
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", GnodebId=111, CellId=11, DIARfcn=636666, PhyCellId=1,
Tac=1;
ADD NREXTERNALNCELL: Mcc="460", Mnc="03", GnodebId=112, CellId=21, DIARfcn=504990, PhyCellId=2,
Tac=1;
//Configuring neighbor relationships
ADD NRNRRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=111, CellId=11;
ADD NRNRRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="03", GnodebId=112, CellId=21;
```

Example 2: Reconfiguration for the primary and secondary operators in reconstruction

- Before reconstruction:

For the local base station, the gNodeB ID is 110, and the IDs of its four served cells are 1, 2, 3, and 4 (cell IDs 1 and 2 belong to an operator, and cell IDs 3 and 4 belong to another operator), separately. For the neighboring base station, the gNodeB ID is 210, and the IDs of its four served cells are 5, 6, 7, and 8 (cell IDs 5 and 6 belong to an operator, and cell IDs 7 and 8 belong to another operator), separately.
- After reconstruction:
 - Primary operator: A different gNodeB ID is added and the cell IDs remain unchanged, and the **gNBOperator.gNBIdLength** and **gNodeBFunction.gNBIdLength** parameters are set to the same value (24).

For the local base station, the gNodeB ID is 111, and the IDs of its served cells are 1 and 2, separately. For the neighboring base station, the gNodeB ID is 211, and the IDs of its served cells are 5 and 6, separately.
 - Secondary operator: A different gNodeB ID is added and the cell IDs remain unchanged, and the **gNBOperator.gNBIdLength** and **gNodeBFunction.gNBIdLength** parameters are set to the same value (24).

For the local base station, the gNodeB ID is 112, and the IDs of its served cells are 3 and 4, separately. For the neighboring base station, the gNodeB ID is 212, and the IDs of its served cells are 7 and 8, separately.

NOTE

The **NRDUCellOp.CellId** parameter for the primary operator (specified by **NRDUCellOp.CellPrimaryOpFlag**) of a cell must be set to the same value as the **NRDUCell.CellId** parameter. As such, the cell IDs of both the primary and secondary operators remain unchanged after reconstruction.

gNodeB configurations:

```
//Configuring interfaces
//(Optional) If Xn is shared before reconstruction and independent Xn configurations are required after
reconstruction, see the configuration examples for basic networking for RAN sharing with common carrier.
//(Optional) If X2 is shared before reconstruction and independent X2 configurations are required after
reconstruction, see the configuration examples for basic networking for RAN sharing with common carrier.
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
DEA NRCELL: NrCellId=3;
DEA NRCELL: NrCellId=4;
//Configuring different gNodeB IDs and gNodeB ID lengths for different operators
//Configuring the gNodeB ID and gNodeB ID length for the primary operator of the gNodeB
MOD GNBOPERATOR: OperatorId=0, OperatorName="unicom", OperatorType=PRIMARY_OPERATOR,
Mcc="460", Mnc="01", NrNetworkingOption=SA_NSA, gNBId=111, gNBIdLength=24;
//Configuring the gNodeB ID and gNodeB ID length for the secondary operator of the gNodeB
MOD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,
Mcc="460", Mnc="03", NrNetworkingOption=SA_NSA, gNBId=112, gNBIdLength=24;
//Configuring operator information for cells 1 and 2 with cell IDs being operator-specific
MOD NRDUCELL: NrDuCellId=1, OperatorId=0, CellId=1, CellPrimaryOpFlag=TRUE;
MOD NRDUCELL: NrDuCellId=2, OperatorId=0, CellId=2, CellPrimaryOpFlag=TRUE;
//Configuring operator information for cells 3 and 4 with cell IDs being operator-specific
MOD NRDUCELL: NrDuCellId=3, OperatorId=1, CellId=3, CellPrimaryOpFlag=TRUE;
MOD NRDUCELL: NrDuCellId=4, OperatorId=1, CellId=4, CellPrimaryOpFlag=TRUE;
//Configuring neighboring cells
//(Optional) Intra-base-station neighboring cell: Removing and then reconfiguring a neighbor relationship
using the primary-operator gNodeB ID and cell ID of a neighboring cell (required when a neighboring cell
belongs to the primary operator and the gNBId parameter in the gNBOperator MO for this operator is set
to a different value from the gNBId parameter in the gNodeBFunction MO)
RMV NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=110, CellId=2;
ADD NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=111, CellId=2;
RMV NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=110, CellId=1;
ADD NRCELLRELATION: NrCellId=2, Mcc="460", Mnc="01", gNBId=111, CellId=1;
//(Optional) Intra-base-station neighboring cell: Removing and then reconfiguring a neighbor relationship
(required when a neighboring cell belongs to a secondary operator as the cell does not broadcast the
gNodeB's primary-operator information)
RMV NRCELLRELATION: NrCellId=3, Mcc="460", Mnc="01", gNBId=110, CellId=4;
ADD NRCELLRELATION: NrCellId=3, Mcc="460", Mnc="03", gNBId=112, CellId=4;
RMV NRCELLRELATION: NrCellId=4, Mcc="460", Mnc="01", gNBId=110, CellId=3;
ADD NRCELLRELATION: NrCellId=4, Mcc="460", Mnc="03", gNBId=112, CellId=3;
//(Optional) Inter-base-station neighboring cell: Removing a neighbor relationship and an external cell and
then reconfiguring an external cell and a neighbor relationship using the primary-operator gNodeB ID and
cell ID of a neighboring cell (required when a neighboring cell belongs to the primary operator and the
gNBId parameter in the gNBOperator MO for this operator is set to a different value from the gNBId
parameter in the gNodeBFunction MO)
//Cells 1 and 5 are used as examples in the following descriptions. If other neighbor relationships (those
between cells 1 and 6, between cells 2 and 5, and between cells 2 and 6) are existent, modify the related
parameter values accordingly.
RMV NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=210, CellId=5;
RMV NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=210, CellId=5;
ADD NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=211, CellId=5, PhysicalCellId=5, Tac=1,
SsbFreqPos=26639;
ADD NRCELLRELATION: NrCellId=1, Mcc="460", Mnc="01", gNBId=211, CellId=5;
//(Optional) Inter-base-station neighboring cell: Removing and then reconfiguring a neighbor relationship
(required when a neighboring cell belongs to a secondary operator as the cell does not broadcast the
gNodeB's primary-operator information)
```

```
//Cells 3 and 7 are used as examples in the following descriptions. If other neighbor relationships (those
between cells 3 and 8, between cells 4 and 7, and between cells 4 and 8) are existent, modify the related
parameter values accordingly.
RMV NREXTERNALCELLPLMN: CellId=7, gNBId=210, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03";
RMV NRCELLRELATION: NrCellId=3, Mcc="460", Mnc="01", gNBId=210, CellId=7;
RMV NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=210, CellId=7;
ADD NREXTERNALNCELL: Mcc="460", Mnc="03", gNBId=212, CellId=7, PhysicalCellId=7, Tac=1,
SsbFreqPos=26639;
ADD NRCELLRELATION: NrCellId=3, Mcc="460", Mnc="03", gNBId=212, CellId=7;
//Enabling operator-specific NCI configuration
MOD GNBSHARINGMODE: MultipleNCISwitch=ON, MultipleNciEnhanceSw=ON;
RST APP: AID=1;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
ACT NRCELL: NrCellId=3;
ACT NRCELL: NrCellId=4;
```

eNodeB configurations:

```
//Configuring the LTE-to-NR PLMN selection mode
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode= SELECTED_PLMN_ID;
//(Optional) Configuring the LTE-to-NR PLMN selection mode for a specific gNodeB
ADD NBRGNODEBPOLICY: Mcc="460", Mnc="01", GnodebId=111,
LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//Configuring gNBIdLength for each involved operator
ADD NCELLPLMNLIST: Mcc="460", Mnc="01", RatType=NR, gNBIdLength=24;
ADD NCELLPLMNLIST: Mcc="460", Mnc="03", RatType=NR, gNBIdLength=24;
//Configuring neighboring cells (ensure that pre-reconstruction neighboring cell configurations have been
removed from the eNodeB and gNodeB)
//Adding external NR cells with information being operator-specific
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=111, CellId=1, DlArfcn=636666, PhyCellId=1,
Tac=1;
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=111, CellId=2, DlArfcn=504990, PhyCellId=2,
Tac=1;
ADD NREXTERNALCELL: Mcc="460", Mnc="03", GnodebId=112, CellId=3, DlArfcn=636666, PhyCellId=3,
Tac=1;
ADD NREXTERNALCELL: Mcc="460", Mnc="03", GnodebId=112, CellId=4, DlArfcn=504990, PhyCellId=4,
Tac=1;
//Configuring neighbor relationships
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=111, CellId=1;
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=111, CellId=2;
ADD NRNRELATIONSHIP: LocalCellId=1, Mcc="460", Mnc="03", GnodebId=112, CellId=3;
ADD NRNRELATIONSHIP: LocalCellId=1, Mcc="460", Mnc="03", GnodebId=112, CellId=4;
```

Example 3: Configurations of the primary operator remain unchanged and those of the secondary operator are adjusted in reconstruction.

- Before reconstruction:

For the local base station, the gNodeB ID is 110, and the IDs of its four served cells are 1, 2, 3, and 4 (cell IDs 1 and 2 belong to an operator, and cell IDs 3 and 4 belong to another operator), separately. For the neighboring base station, the gNodeB ID is 210, and the IDs of its four served cells are 5, 6, 7, and 8 (cell IDs 5 and 6 belong to an operator, and cell IDs 7 and 8 belong to another operator), separately.
- After reconstruction:
 - Primary operator: Configurations remain unchanged.

For the local base station, the gNodeB ID is 110, and the IDs of its served cells are 1 and 2, separately. For the neighboring base station, the gNodeB ID is 210, and the IDs of its served cells are 5 and 6, separately.
 - Secondary operator: A different gNodeB ID is added with the cell IDs remaining unchanged, and the **gNBOperator.gNBIdLength** and

gNodeBFunction.gNBIdLength parameters are set to the same value (24).

For the local base station, the gNodeB ID is 112, and the IDs of its served cells are 3 and 4, separately. For the neighboring base station, the gNodeB ID is 212, and the IDs of its served cells are 7 and 8, separately.

gNodeB configurations:

```
//Configuring interfaces
//NG, S1, Xn, and X2 interface configuration is the same as that in RAN sharing with common carrier. For
details, see the configuration examples for RAN sharing with common carrier.
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
DEA NRCELL: NrCellId=3;
DEA NRCELL: NrCellId=4;
//Configuring different gNodeB IDs and gNodeB ID lengths for different operators
//Configuring the gNodeB ID and gNodeB ID length for the primary operator of the gNodeB
MOD GNBOPERATOR: OperatorId=0, OperatorName="unicom", OperatorType=PRIMARY_OPERATOR,
Mcc="460", Mnc="01", NrNetworkingOption=SA_NSA, gNBId=110, gNBIdLength=24;
//Configuring the gNodeB ID and gNodeB ID length for the secondary operator of the gNodeB
MOD GNBOPERATOR: OperatorId=1, OperatorName="telecom", OperatorType=SECONDARY_OPERATOR,
Mcc="460", Mnc="03", NrNetworkingOption=SA_NSA, gNBId=112, gNBIdLength=24;
//Configuring operator information for cells 1 and 2 with cell IDs being operator-specific
MOD NRDUCELL: NrDuCellId=1, OperatorId=0, CellId=1, CellPrimaryOpFlag=TRUE;
MOD NRDUCELL: NrDuCellId=2, OperatorId=0, CellId=2, CellPrimaryOpFlag=TRUE;
//Configuring operator information for cells 3 and 4 with cell IDs being operator-specific
MOD NRDUCELL: NrDuCellId=3, OperatorId=1, CellId=3, CellPrimaryOpFlag=TRUE;
MOD NRDUCELL: NrDuCellId=4, OperatorId=1, CellId=4, CellPrimaryOpFlag=TRUE;
//Configuring neighboring cells
//(Optional) Intra-base-station neighboring cell: Removing and then reconfiguring a neighbor relationship
using the primary-operator gNodeB ID and cell ID of a neighboring cell (required when a neighboring cell
belongs to the primary operator and the gNBId parameter in the gNBOperator MO for this operator is set
to a different value from the gNBId parameter in the gNodeBFunction MO; this operation is not required
in the given scenario)
//(Optional) Intra-base-station neighboring cell: Removing and then reconfiguring a neighbor relationship
(required when a neighboring cell belongs to a secondary operator as the cell does not broadcast the
gNodeB's primary-operator information)
RMV NRCELLRELATION: NrCellId=3, Mcc="460", Mnc="01", gNBId=110, CellId=4;
ADD NRCELLRELATION: NrCellId=3, Mcc="460", Mnc="03", gNBId=112, CellId=4;
RMV NRCELLRELATION: NrCellId=4, Mcc="460", Mnc="01", gNBId=110, CellId=3;
ADD NRCELLRELATION: NrCellId=4, Mcc="460", Mnc="03", gNBId=112, CellId=3;
//(Optional) Inter-base-station neighboring cell: Removing a neighbor relationship and an external cell and
then reconfiguring an external cell and a neighbor relationship using the primary-operator gNodeB ID and
cell ID of a neighboring cell (required when a neighboring cell belongs to the primary operator and the
gNBId parameter in the gNBOperator MO for this operator is set to a different value from the gNBId
parameter in the gNodeBFunction MO; this operation is not required in the given scenario)
//(Optional) Inter-base-station neighboring cell: Removing and then reconfiguring a neighbor relationship
(required when a neighboring cell belongs to a secondary operator as the cell does not broadcast the
gNodeB's primary-operator information)
//Cells 3 and 7 are used as examples in the following descriptions. If other neighbor relationships (those
between cells 3 and 8, between cells 4 and 7, and between cells 4 and 8) are existent, modify the related
parameter values accordingly.
RMV NREXTERNALNCELLPLMN: CellId=7, gNBId=210, Mcc="460", Mnc="01", SharedMcc="460",
SharedMnc="03";
RMV NRCELLRELATION: NrCellId=3, Mcc="460", Mnc="01", gNBId=210, CellId=7;
RMV NREXTERNALNCELL: Mcc="460", Mnc="01", gNBId=210, CellId=7;
ADD NREXTERNALNCELL: Mcc="460", Mnc="03", gNBId=212, CellId=7, PhysicalCellId=7, Tac=1,
SsbFreqPos=26639;
ADD NRCELLRELATION: NrCellId=3, Mcc="460", Mnc="03", gNBId=212, CellId=7;
//Enabling operator-specific NCI configuration
MOD GNBSHARINGMODE: MultipleNCISwitch=ON, MultipleNciEnhanceSw=ON;
RST APP: AID=1;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
ACT NRCELL: NrCellId=3;
ACT NRCELL: NrCellId=4;
```

eNodeB configurations:

```
//Configuring the LTE-to-NR PLMN selection mode
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode= SELECTED_PLMN_ID;
//Optional) Configuring the LTE-to-NR PLMN selection mode for a specific gNodeB
ADD NBRGNODEBPOLICY: Mcc="460", Mnc="01", GnodebId=110,
LteToNrPlmnSelectionMode=SELECTED_PLMN_ID;
//Configuring gNBIdLength for each involved operator
ADD NCELLPLMNLIST: Mcc="460", Mnc="01", RatType=NR, gNBIdLength=24;
ADD NCELLPLMNLIST: Mcc="460", Mnc="03", RatType=NR, gNBIdLength=24;
//Configuring neighboring cells (ensure that pre-reconstruction neighboring cell configurations have been
removed from the eNodeB and gNodeB)
//Adding external NR cells with information being operator-specific
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=110, CellId=1, DlArfcn=636666, PhyCellId=1,
Tac=1;
ADD NREXTERNALCELL: Mcc="460", Mnc="01", GnodebId=110, CellId=2, DlArfcn=504990, PhyCellId=2,
Tac=1;
ADD NREXTERNALCELL: Mcc="460", Mnc="03", GnodebId=112, CellId=3, DlArfcn=636666, PhyCellId=3,
Tac=1;
ADD NREXTERNALCELL: Mcc="460", Mnc="03", GnodebId=112, CellId=4, DlArfcn=504990, PhyCellId=4,
Tac=1;
//Configuring neighbor relationships
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=110, CellId=1;
ADD NRNRELATIONSHIP: LocalCellId=0, Mcc="460", Mnc="01", GnodebId=110, CellId=2;
ADD NRNRELATIONSHIP: LocalCellId=1, Mcc="460", Mnc="03", GnodebId=112, CellId=3;
ADD NRNRELATIONSHIP: LocalCellId=1, Mcc="460", Mnc="03", GnodebId=112, CellId=4;
```

Optimization Command Examples

Examples are the same as those in RAN sharing with common carrier. For details, see [Optimization Command Examples](#).

Deactivation Command Examples

The following provides only deactivation command examples. You can determine whether to restore the settings of other parameters based on actual network conditions.

gNodeB configurations:

```
//Deactivating cells
DEA NRCELL: NrCellId=1;
DEA NRCELL: NrCellId=2;
DEA NRCELL: NrCellId=3;
DEA NRCELL: NrCellId=4;
//Modifying the primary operator flag of cells
MOD NRDUCELL: NrDuCellId=1, OperatorId=0, CellPrimaryOpFlag=FALSE;
MOD NRDUCELL: NrDuCellId=2, OperatorId=0, CellPrimaryOpFlag=FALSE;
MOD NRDUCELL: NrDuCellId=3, OperatorId=1, CellPrimaryOpFlag=FALSE;
MOD NRDUCELL: NrDuCellId=4, OperatorId=1, CellPrimaryOpFlag=FALSE;
//Disabling operator-specific NCI configuration
MOD GNBSHARINGMODE: MultipleNCISwitch=OFF;
RST APP: AID=1;
//Activating cells
ACT NRCELL: NrCellId=1;
ACT NRCELL: NrCellId=2;
ACT NRCELL: NrCellId=3;
ACT NRCELL: NrCellId=4;
//Changing the X2 interface mode from the independent mode to the shared mode
//Removing the X2 interface binding relationship and the corresponding X2 object for the primary operator
RMV GNBCUX2OP: gNBCuX2Id=0, OperatorId=0;
RMV GNBCUX2: gNBCuX2Id=0;
//Binding the primary operator to the configured X2 interface of operator 1
ADD GNBCUX2OP: gNBCuX2Id=1, OperatorId=0;
//Optional, depending on the operator's plan) Changing the Xn interface mode from the
independent mode to the shared mode
```

```
//Removing the Xn interface binding relationship and the corresponding Xn object for the primary operator  
RMV GNBCUXNOP: gNBCuXnId=0, OperatorId=0;  
RMV GNBCUXN: gNBCuXnId=0;  
//Binding the primary operator to the configured Xn interface of operator 1  
ADD GNBCUXNOP: gNBCuXnId=1, OperatorId=0;
```

eNodeB configurations:

```
MOD ENODEBALGOEXTSWITCH: LteToNrPlmnSelectionMode=PRIMARY_PLMN_ID;
```

5.3.4.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

5.3.4.2 Activation Verification

Prerequisites

According to the network plan, the gNodeB is shared by multiple operators, and both the **gNBOperator.gNBId** and **NRDUCellOp.CellId** parameters are set to different values for each operator.

Expected Result

Separately check SIB1 of cells belonging to the primary and secondary operators. The value of **cellIdentity** broadcast in SIB1 is consistent with the settings of the **gNBOperator.gNBId** and **NRDUCellOp.CellId** parameters.

5.3.4.3 Network Monitoring

None

5.4 Frequency-based 5GC Selection

Frequency-based 5GC selection in RAN sharing with dedicated carrier is the same as that in RAN sharing with common carrier. For details, see [4.8 Frequency-based 5GC Selection](#).

5.5 Operator-Specific Paging Resource Management

Operator-specific paging resource management in RAN sharing with dedicated carrier is the same as that in RAN sharing with common carrier. For details, see [4.9 Operator-Specific Paging Resource Management](#).

6 Parameters

The following hyperlinked EXCEL files of parameter reference match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- 3900 & 5900 Series Base Station gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and the reserved parameters that fall into disuse in the current R version.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter. View its information, including the meaning, values, and impacts.

----End

7 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

- Step 1** Open the EXCEL file of performance counter reference.
- Step 2** On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID, for example, FBFD-010011.
- Step 3** Click **OK**. All counters related to the feature are displayed.

----End

8 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

9 Reference Documents

- 3GPP TS 36.423: "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 application protocol (X2AP)"
- 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification"
- 3GPP TS 38.104: "NR; Base Station (BS) radio transmission and reception"
- 3GPP TS 38.214: "NR; Physical layer procedure for data"
- 3GPP TS 23.122: "Non-Access-Stratum (NAS) functions related to Mobile Station (MS) in idle mode"
- *EPC-based NSA Basics*
- *EPC-based NSA Performance Enhancement*
- *EPC-based NSA Experience Improvement*
- *NSA-SA Selection Based on User Experience*
- *X2 and S1 Self-Management in NSA Networking*
- *X2 Self-Management in SA Networking*
- *NG and Xn Self-Management*
- *CoMP*
- *VoNR*
- *Admission Control*
- *Multi-Frequency Smart Aggregation*
- *SA Mobility Management in Connected Mode*
- *SA Mobility Management in Idle Mode*
- *SA Mobility Management in Inactive Mode*
- *Mobility Load Balancing in Connected Mode*
- *Interoperability Between E-UTRAN and NG-RAN in Connected Mode*
- *Interoperability Between E-UTRAN and NG-RAN in Idle Mode*
- *Synchronization*
- *Base Station Supporting Multi-operator PKI*
- *URLLC*
- *ANR*
- *Multi-Frequency Smart Selection*

- *Network Slicing*
- *Downlink Scheduling*
- *Multi-Frequency Smart Selection*
- *Cell Combination*
- *Carrier Aggregation*
- *Co-MM (Low-Frequency TDD)*
- *MIMO (TDD)*
- *Resource Control in NSA Networking*
- *FWA*
- *License Control Item Lists*
- *RedCap*
- *Hyper Cell*
- *UE Power Saving*
- *QCI-based Resource Control*
- *eXn Self-Management*
- *Service-differentiated Experience Guarantee*
- *Adaptive Managed Latency*
- *Technical Specifications in 3900 & 5900 Series Base Station Product Documentation*
- Feature parameter description documents in *eRAN Feature Documentation*
 - *RAN Sharing*
 - *S1 and X2 Self-Management*
 - *Transport Resource Management*
 - *ANR Management*